

Legacy Numerical Evidence for an Adaptive Chiral Wall: A Twenty-Four-Figure Working Atlas

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This working note distills the twenty-four strongest non-Bott numerical findings currently present in the repository into a single visual and interpretive record. The purpose is not to retroactively declare the planned production program complete. It is to preserve what the archive already establishes, identify what is merely suggestive, and make the best existing data immediately reusable for future plotting and comparison. A completed deterministic domain-wall audit supplies the exact target. Independent stochastic archives then show a wall-localized slow tangent sector, delayed wall purification, near- $c = 1$ logarithmic entanglement, near-unit real-space Chern response, localized feedback activity, a dynamical crossover near the exact mass transition, modular handedness, and wall-enhanced classical-record uncertainty. Exact tripartite mutual information cleanly counts one versus two walls; the corresponding stochastic pilot preserves the factor of two but has the wrong protocol and a large absolute bias. A translation-resolved flattened-parent spectrum exposes the wall branches directly. A direct reconstruction of the uniform finite-range parent shows that one shell is already spectrally gapped with unit-magnitude Chern response in its flattened-frame orientation, while the domain-wall shell hierarchy separates early agreement of entropy coefficients from slower convergence of wall correlators. Spatial maps, trajectory distributions, and a conditioning audit show both the typical wall structure and the non-negligible record tails. Two independent trajectory-resolved charge campaigns further show that the size-normalized filling distribution narrows with circumference. Two additional Colab cross-checks show that a separate pure-state covariance archive simultaneously supports near- $c = 1$ entropy and a wall-localized algebraic correlator, and that feedback sharply reshapes the finite particle-Choi transfer mode in its reduced slab basis. A completed correct-feedback comparison also confirms that the explicit interface survives when domain-wall support truncation is disabled and the controller projectors overlap it. Finally, a direct reanalysis of the pure-state covariance snapshots finds logarithmic subsystem charge fluctuations with a trajectory-resolved current level near $k = 1$ at both available circumferences and shell depths, matched to the independently extracted entropy coefficient on the same ensembles. An independent deterministic form-factor reconstruction further shows that the practical $n_{\text{shell}} = 1$ three-by-three OW mode retains 99.2155% of its unwindowed norm and preserves the target-band overlap winding, even though the associated smooth compact-support spinor is necessarily Chern trivial. A four-point mass sweep then verifies the bundle statement directly: unwindowed and $n_{\text{shell}} = 0, 1, 2$ overlaps all have total winding -1 in the $C = 1$ phase at $\alpha = 1, 1.5$, while all are gapped with zero winding in the trivial phase at $\alpha = 3, 30$. An exact deterministic CPU reconstruction finally separates the autonomous finite-success mean channel from its continuous weak-success generator. The untruncated controller has paired wall-localized occupation branches through half filling, while all three shell choices retain an extensively mixed interface sector and show second-order fixed-time channel-to-generator convergence. Every figure is generated by one documented Jupyter notebook, and every panel is tied to an explicit source manifest and a machine-readable diagnostic ledger.

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visual hierarchy also intentionally gives no weight to Bott-only results: the topology panels use the real-space tripartition Chern estimator, while the wall evidence comes from entropy, tangent, purification, activity, modular, and record observables.

A. One-notebook figure contract

The sole plotting source is `experiment_review/legacy_evidence_figure_atlas/legacy_evidence_figure_atlas.ipynb`. It begins with an editable CPU-affinity cell; states the estimators and evidence class before each result; reads metadata and archived tables or arrays; places each figure's creation commands in the cell immediately above its output; and ends with raw diagnostics and integrity assertions. Execution writes exactly twenty-four PDF/PNG pairs to the local `figures/` directory. It also writes `figure_manifest.json`, which records every input path and claim, and `figure_diagnostics.json`, which records the numerical values used below. The small `build_notebook.py` helper only serializes the notebook cells; it neither reads scientific data nor produces a figure.

The rendering follows the Bhuiyan–Pan–Jian (BPJ) grammar used in the repository paper: Times-compatible serif text, Computer-Modern mathematics, red upward triangles with dotted lines, green squares with dashed lines, blue circles with solid lines, black theory or transition references, boxed axes, inward ticks, compact legends, and panel letters outside the upper-left corner. Sequential blue scales are reserved for genuine spatial or space-time densities rather than used decoratively.

B. Formatting and source-figure audit

Every numbered figure was inspected at its compiled 7.05-inch width rather than only as a notebook thumbnail. Existing publication-ready legacy renders are preserved in their native visual language: Fig. 11 is resized and cropped only to remove the unrelated upper spectrum row, and Fig. 13(a) embeds the archived local-Chern render without recoloring or re-marking it. These rendered exceptions are identified in their captions because their plotted numerical arrays were not exported. All other panels are rebuilt from machine-readable tables or arrays and therefore use the common BPJ grammar. Where an identifiable generating or characterization notebook exists, `figure_manifest.json` records that notebook alongside the exact data or render; the manifest never substitutes a notebook path for the underlying evidence artifact.

The audit also checks that panel letters, legends, color bars, and axis labels remain readable after two-column compilation; that native rendered figures are not silently restyled; and that every displayed file is produced or republished by the single truth notebook. Thus “single plotting source” means one reproducible assembly

notebook, not that visually strong legacy plots must be reformatted merely for uniformity.

II. MODEL, OBJECTS, CLOCKS, AND ESTIMATOR CONVENTIONS

This section fixes the notation needed to read every panel without consulting the source code. More detailed reductions, including the tripartite-information derivation and the mean-channel block algorithm, are collected in Appendix A; the compact panel-to-equation index is Table III.

A. Lattice Hamiltonian and two-wall geometry

There are two complex-fermion orbitals per unit cell on an $N_x \times N_y$ square lattice. With zero-based coordinates, the atlas uses the one-body ordering

$$i(\mu, x, y) = \mu + 2x + 2N_x y, \quad \mu \in \{0, 1\}. \quad (1)$$

Unless a caption says otherwise, both directions are periodic. A periodic topological slab consequently has two oppositely oriented interfaces. Writing $\Psi_{x,y} = (c_{x,y,0}, c_{x,y,1})^T$, the exact benchmark Hamiltonian is

$$H_{\text{DW}} = \sum_{x,y} \Psi_{x,y}^\dagger \alpha(x) \sigma_z \Psi_{x,y} + \sum_{x,y} (\Psi_{x,y}^\dagger t_x \Psi_{x+1,y} + \Psi_{x,y}^\dagger t_y \Psi_{x,y+1} + \text{H.c.}), \quad (2)$$

$$t_x = -\frac{1}{2}\sigma_z - \frac{i}{2}\sigma_x, \quad t_y = -\frac{1}{2}\sigma_z - \frac{i}{2}\sigma_y. \quad (3)$$

The mass profile is

$$\alpha(x) = \begin{cases} \alpha_{\text{in}}, & x \in \mathcal{I}_{\text{top}}, \\ \alpha_{\text{out}}, & x \notin \mathcal{I}_{\text{top}}, \end{cases} \quad (4)$$

with the wall columns equal to the two endpoints of $\mathcal{I}_{\text{top}}$. The standard topological/trivial choice is $(\alpha_{\text{in}}, \alpha_{\text{out}}) = (1, 30)$; the precise interval and geometry used by each result are stated in its caption or manifest. In a uniform region, Fourier transformation gives

$$h(\mathbf{k}) = \sin k_x \sigma_x + \sin k_y \sigma_y + (\alpha - \cos k_x - \cos k_y) \sigma_z. \quad (5)$$

The gap closes at $\alpha = 0, \pm 2$, and $0 < \alpha < 2$ is the nontrivial phase while $\alpha > 2$ is trivial. With the displayed (k_x, k_y) orientation and the standard spectral-projector ordering defined below, the lower band has $C = +1$ at $\alpha = 1, 1.5$. Reversing one momentum axis reverses this oriented integer. Complex conjugation also reverses it; in this repository that conjugation arises because the production covariance stores creation and annihilation indices in the transpose ordering. The form-factor sweep

always quotes the spectral-projector convention. The separate flattened-parent calculation quotes both its raw lower-band FHS integer and the output of the real-space routine, so those two signs are never silently identified.

B. Covariances, controller modes, and support flags

Two matrix orderings for the same physical two-point function occur in the archive:

$$\begin{aligned} \Gamma_{ij} &= \text{Tr}(\rho c_j^\dagger c_i), & C_{ij} &= \text{Tr}(\rho c_i^\dagger c_j), \\ C &= \Gamma^\top = \Gamma^*, & G &= 2C - \mathbf{1}. \end{aligned} \quad (6)$$

Here Γ is the one-body operator (spectral-projector) ordering, whereas C and G are the production-storage ordering used by the circuit code. Both are Hermitian, have the same spectrum, and describe the same state; transposing the index convention does not change entropy or filling, and $0 \leq C \leq \mathbf{1}$. A pure number-conserving Gaussian state obeys $C^2 = C$, and half filling means $\text{Tr } C = N_x N_y$. For the exact ground state,

$$\Gamma_{\text{ex}} = P_-(H_{\text{DW}}), \quad C_{\text{ex}} = P_-^*(H_{\text{DW}}). \quad (7)$$

The archived real-space Chern routines explicitly reconstruct $\Gamma = [(\mathbf{1} + G)/2]^*$ before applying an oriented projector formula. If the stored matrix $C = \Gamma^*$ were instead interpreted directly as a Bloch projector, complex conjugation would reverse the reported Chern sign; this is the source of the otherwise apparent sign flip, not a second physical state. Concretely, the X -trial OW builder stores the components of the row $\langle \tau | P_\pm$ as a production ket, i.e., the conjugate ordering of $P_\pm | \tau \rangle$. The raw lower band of the resulting flattened-frame matrix therefore has $C = -1$ relative to the $C = +1$ exact-projector convention. Result 12 passes its covariance so that the real-space routine reconstructs this same occupied projector; its FHS and real-space answers consequently agree at -1 rather than being made to agree by filling the complementary band.

For trial spinors $|\tau_A\rangle = (1, 1)^\top / \sqrt{2}$ and $|\tau_B\rangle = (1, -1)^\top / \sqrt{2}$, projected overcomplete-Wannier (OW) modes are obtained from

$$P_\pm(\mathbf{k}) = \frac{1}{2} \left[\mathbf{1} \pm \frac{h(\mathbf{k})}{|h(\mathbf{k})|} \right], \quad |\chi_{\mathbf{r}, \nu, \pm}\rangle \propto P_\pm | \mathbf{r}, \tau_\nu \rangle. \quad (8)$$

The desired local outcomes occupy the lower-band modes and empty the upper-band modes. For the nonnegative integer values used here, $n_{\text{shell}} = n$ multiplies each OW mode by the wrapped square mask $g_n(\Delta x, \Delta y) = \mathbf{1}_{|\Delta x| \leq n} \mathbf{1}_{|\Delta y| \leq n}$ and then renormalizes it; $n = 0, 1, 2$ therefore mean one-by-one, three-by-three, and five-by-five unit-cell support. The value $n_{\text{shell}} = \text{None}$ means the OW mode is unwindowed. This flag is logically independent of `dw.truncation`: when the latter is true, a mode is additionally cut at the domain-wall sector containing its center; when false, its support may overlap the interface. Thus “unwindowed” never means “domain-wall truncation off.”

The one-body projector of a controller channel is $P_j = |\chi_j\rangle\langle\chi_j|$. A circuit cycle is one complete ordered sweep of the declared controller channels. A trajectory ξ is one complete Born-sampled measurement-and-feedback history. The symbol S in a caption denotes the number of independent trajectories, while C_{cyc} denotes cycle count; the covariance itself is always the italic matrix C . Partial post-selection is not used anywhere in this atlas.

C. Five distinct numerical objects

The following objects answer different questions and are not interchangeable.

1. $\Gamma_{\text{ex}} = P_-(H_{\text{DW}})$, equivalently the stored $C_{\text{ex}} = P_-^*(H_{\text{DW}})$, is the pure exact ground state. It calibrates the intended wall CFT.
2. $C_\xi(C_{\text{cyc}})$ is a nonlinear, conditioned Gaussian trajectory produced by Born measurement and feedback. A trajectory average is formed only after each recordwise observable is evaluated unless explicitly stated otherwise.
3. P_{flat} fills the lowest fixed number of eigenvectors of a static flattened controller parent. It characterizes the target frame, not the distribution generated at finite circuit depth.
4. A particle–Choi transfer mode is an eigenvector of a reduced, conditioned transfer construction. It is not a covariance-tangent Oseledec vector.
5. C_{ss} is the mixed stationary covariance of the record-discarded affine mean channel or its continuous weak-success generator. It is deterministic and is not the covariance of a typical conditioned trajectory.

Likewise, five clocks occur. The circuit clock C_{cyc} counts complete controller sweeps; the tangent time T is normalized exactly as declared by its campaign; physical Hamiltonian time t_{phys} generates $e^{-iH_{\text{DW}}t_{\text{phys}}}$; modular time t_{mod} generates the state-dependent entanglement Hamiltonian; and the continuous mean-channel time t is the semigroup parameter of Eq. (47). A velocity is meaningful only with its own clock. Throughout, positive displacement means increasing periodic y ; reversing wall orientation must reverse the sign, but no equality of the physical, modular, and dissipative-response magnitudes is implied.

D. Core estimators and averaging order

For a periodic interval of length A_y , define the chord and log-chord variables

$$\ell(A_y) = \frac{N_y}{\pi} \sin\left(\frac{\pi A_y}{N_y}\right), \quad x_{\text{ch}}(A_y) = \log \ell(A_y). \quad (9)$$

The Calabrese–Cardy form [1] gives, for a full- x interval that intersects two independent chiral walls,

$$\overline{S}(A_y) = s_0 + \frac{c}{3}x_{\text{ch}}(A_y) + \cdots, \quad c_{\text{eff}} = 3m_S. \quad (10)$$

A contour sum isolating one wall carries $c/6$, hence $c_{\text{eff}} = 6m_S$. The chiral fermion correlator correspondingly has $|G(r_y)|^2 \propto \ell(r_y)^{-2}$.

For any covariance, the global Gaussian entropy and intrinsic quantum charge variance are

$$\begin{aligned} S(C) &= -\text{Tr}[C \log C + (\mathbf{1} - C) \log(\mathbf{1} - C)], \\ V_Q(C) &= \text{Tr}[C(\mathbf{1} - C)]. \end{aligned} \quad (11)$$

The restricted versions and contour decomposition are defined in Eqs. (A1)–(A2). The saved finite-time tangent gap is an operational separation between conditional co-cycle rates; only matched clocks and normalizations may be compared.

For premeasurement Born probabilities s_j , the retained classical-record estimator is

$$\begin{aligned} \hat{H}_{\text{chain}} &= \sum_j h_2(s_j), \\ h_2(s) &= -s \log s - (1 - s) \log(1 - s). \end{aligned} \quad (12)$$

It is the mean conditional uncertainty, not the realized-record self-information $-\log p_\xi$, because the outcome bits were not retained.

Modular handedness uses an unwrapped displacement $D_w(t_{\text{mod}})$ and

$$v_{\text{mod},w} = \left. \frac{dD_w}{dt_{\text{mod}}} \right|_{\text{fit}}; \quad (13)$$

physical and dissipative-response velocities use the analogous derivative with their own clocks. Finally, the static controller parent and its additive wrong-outcome loss are

$$H_{\text{flat}} = \sum_j (P_{j,+} - P_{j,-}), \quad (14)$$

$$\mathcal{L}_{\text{wrong}}(P) = \text{const} + \text{Tr}(PH_{\text{flat}}). \quad (15)$$

Ky Fan’s minimum principle selects the N_{occ} lowest eigenvectors of H_{flat} . It minimizes a sum of marginal wrong-outcome probabilities, not the joint success probability of overlapping noncommuting tests.

For a quantity O_{ξ,y_0} evaluated at translated origin y_0 , translated origins are variance-reduction samples within one record:

$$O_\xi = \frac{1}{N_{y_0}} \sum_{y_0} O_{\xi,y_0}, \quad \overline{O} = \frac{1}{S} \sum_{\xi=1}^S O_\xi. \quad (16)$$

They are never counted as additional independent trajectories. Unless a caption says otherwise, stochastic bars and bands are the trajectory standard error, deterministic curves have no sampling bars, and regression errors refer only to the declared fit points.

III. GROUP I: DETERMINISTIC TARGET AND ADAPTIVE TOPOLOGY

A. Result 1: the exact wall realizes the clean chiral target

Figure 1 consolidates the accepted version-2 B0 audit. At $N_x = 20, N_y = 48$, the full-strip fit gives $c = 1.000092$ for both the coupled interface and hard-exterior construction. The wall-correlator exponents are 2.07447 and 2.07437, respectively. The two physical velocities have opposite signs and magnitudes close to 0.872, the primary coupled-wall modular fits give $(-2.930, +4.536)$, and the four construction/wall combinations carry signed twist flows $(-1, +1, -1, +1)$. The transverse gate accepts $N_x = 20$ and larger tested widths; $N_x = 12, 16$ remain visibly underconverged for the coupled local checks.

Evidence class: completed deterministic audit. This is the clean reference against which the stochastic estimators should be judged. **What it does not establish:** It tests the exact ground-state and analysis pipeline, not preparation by the adaptive trajectory ensemble.

B. Result 2: a sharply localized slow tangent sector

Figure 2 places the two finite-time implementations and their selected-mode localization on the same visual footing. Two independent implementations give the same contrast. In the CPU $T = 2N_y$ campaign, the wall gaps fall from 0.1125 to 0.1014 over $N_y = 20, 22, 24$, while the uniform values remain 0.894–0.896. The selected wall-sector weight is 0.992–0.996. In the GPU $T = N_y$ campaign, the wall gaps are 0.02698(313), 0.02431(287), and 0.01968(225) at $N_y = 30, 40, 50$, compared with uniform gaps 0.7217(110), 0.7483(86), and 0.7676(87). The corresponding slow-vector wall weights are 0.933, 0.935, and 0.944.

Thus the important repeatable statement is not an absolute gap exponent: it is a roughly eightfold CPU suppression, a 27–39-fold GPU suppression, and strong spatial localization in both data sets. **Evidence class:** canonical CPU and GPU finite-time precursors. **What it does not establish:** The campaigns use different clocks and trajectory suites, and neither is the long-time $S = 100$ production tangent family.

C. Result 3: purification leaves a wall-localized residual

Figure 3 collects the entropy, intrinsic charge variance, mass control, and spatial-contour evidence for the surviving wall sector. Starting from $C = 1/2$, the canonical $S = 100$ GPU campaign rapidly removes the extensive mixed-state entropy but leaves a finite residual at $2N_y$. For $N_y = 30, 40, 50$, the final entropies are 0.2444(233), 0.2907(284), and 0.3313(271); the charge variances are

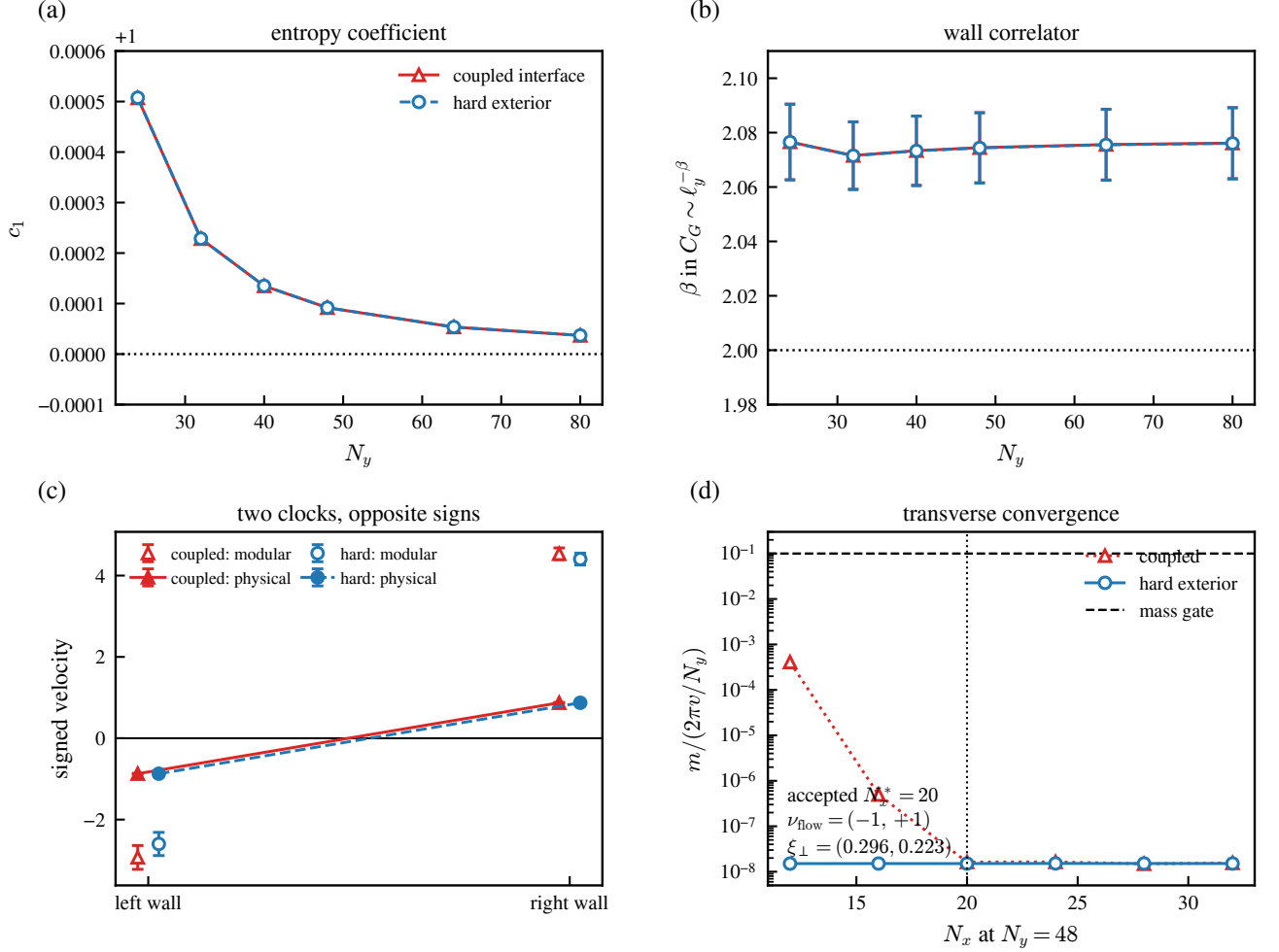


FIG. 1. Exact domain-wall calibration. (a) Full-strip entropy coefficient versus circumference, shown as c near the $c = 1$ target. (b) Squared wall-correlator exponent near the r_y^{-2} target. (c) Opposite physical and modular velocities on the two walls; the magnitudes use different clocks and are not equated. (d) Transverse convergence and the accepted $N_x = 20$ gate. Bars in (b) and (c) are one-standard-error regression uncertainties; where absent visually, they are smaller than the plotting symbols. Panels (a) and (d) are deterministic diagnostics with no sampling error bars. The 20×48 accepted spectral projector is $\Gamma_{\text{ex}} = P_-(H_{\text{DW}})$, equivalently the production covariance $C_{\text{ex}} = P_+^*(H_{\text{DW}})$, from Eqs. (2)–(7); panels (a)–(c) use Eqs. (10), (A3), and (13). This calibrates the target and estimators, not adaptive preparation.

0.06647(802), 0.08464(991), and 0.09733(952); and the simultaneous Chern estimates remain 0.9891, 0.9908, and 0.9879. An independent 16×32 contour archive places 94.25% of the support-terminated final entropy on the two wall columns. The same smaller CPU sweep shows that a topological interior $\alpha_{\text{in}} = 1$ retains orders of magnitude more entropy than the trivial $\alpha_{\text{in}} = 3$ control.

Evidence class: canonical ensemble dynamics plus a smaller spatial precursor. **What it does not establish:** No saved artifact contains the same record’s full tangent cocycle and purification spectrum, so the proposed recordwise tangent–purification bridge and a universal purification exponent remain untested. The principal ensemble stops at $2N_y$.

D. Result 4: logarithmic entropy is close to the two-wall $c = 1$ value

Figure 4 shows the raw log-chord fit, its residuals, the size comparison, and the cycle-resolved fitted coefficient. The strongest common-depth entropy sequence uses $N_x = 20$, $S = 100$, $C = 50$, and $N_y = 30, 40, 50$. Fits over $A_y \geq 8$ give the slopes

$$m_1 = 0.35572(60), \quad 0.35092(58), \quad 0.34920(43), \quad (17)$$

or, directly, $c_{\text{eff}} = \frac{3m_1}{1.0672(18), 1.0528(17), 1.0476(13)}$, with $R^2 \geq 0.99997$. The discrepancy from $c = 1$ decreases from 6.72% to 4.76% over this common sequence. A separate time-scaled $S = 10$, $C = N_y/2$ sequence

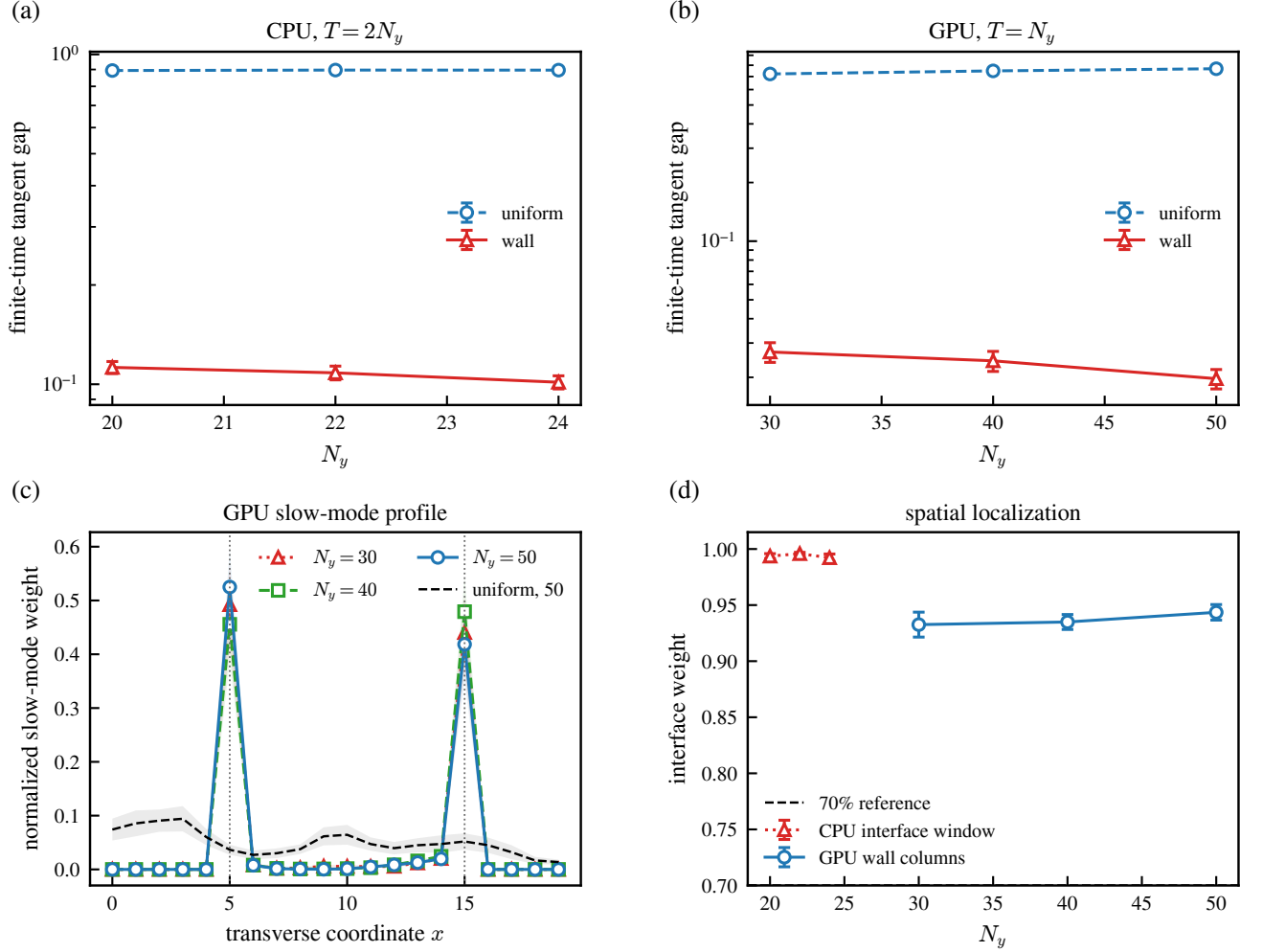


FIG. 2. Wall-localized tangent sector. (a) CPU wall and uniform finite-time gaps. (b) Independent GPU gaps on larger circumferences. (c) GPU transverse slow-mode profiles, with two peaks at the two interfaces. (d) Integrated interface weight for both campaigns; the black dashed line is a 70% localization reference. Bars in (a), (b), and (d), and shaded bands in (c), are trajectory standard errors for $S = 25$; several are smaller than the symbols or line width. The CPU horizon is $T = 2N_y$ for $N_y = 20, 22, 24$, whereas the GPU horizon is $T = N_y$ for $N_y = 30, 40, 50$. Both are $N_x = 20$, $n_{\text{shell}} = 1$, perfect-correction campaigns; the wall arm has $(\alpha_{\text{in}}, \alpha_{\text{out}}) = (1, 30)$, $\text{DW}=\text{True}$, and $\text{dw.truncation}=\text{True}$, while the uniform control has $\alpha = 1$, $\text{DW}=\text{False}$, and no wall truncation. Both use the QR reduction in Eq. (A23). Accordingly, the robust comparison is wall suppression and localization within each campaign, not equality of their absolute finite-time rates.

gives $c_{\text{eff}} = 1.08460(96), 1.0665(13), 1.05947(54)$ at $N_y = 40, 60, 80$, and $1.04098(42), 1.0708(14)$ at $N_y = 100, 120$. Because its depth and sample count differ, it is displayed as a separate series rather than folded into the controlled extrapolation. At 20×40 , $c_{\text{eff}}(C)$ settles near 1.05 after the early transient.

Evidence class: canonical GPU $S = 100$ state-entropy precursor plus the broader time-scaled $S = 10$ sequence. **What it does not establish:** The controlled $N_y = 30$ – 50 trend is consistent with a $c = 1$ two-wall state, but it has a persistent positive finite-size/protocol bias and does not by itself establish the current-algebra charge sector.

E. Result 5: tripartite information counts one wall versus two

Figure 5 presents the raw exact and stochastic fits together with their converted coefficients and symmetry-preserving fit-window sensitivity. For full-width von Neumann entropies, the positive convention is

$$\begin{aligned} \Delta I^{(\text{full})} &\equiv I_2 - I_3 = I(A : C \mid B) \geq 0, \\ I_3 - I_2 &= -\Delta I^{(\text{full})}. \end{aligned} \quad (18)$$

Here $I_2 = I(A : C)$ and $I_3 = I_3(A : B : C)$. A one-wall datum is different: every subsystem still spans the full transverse width, and only the final entanglement-contour

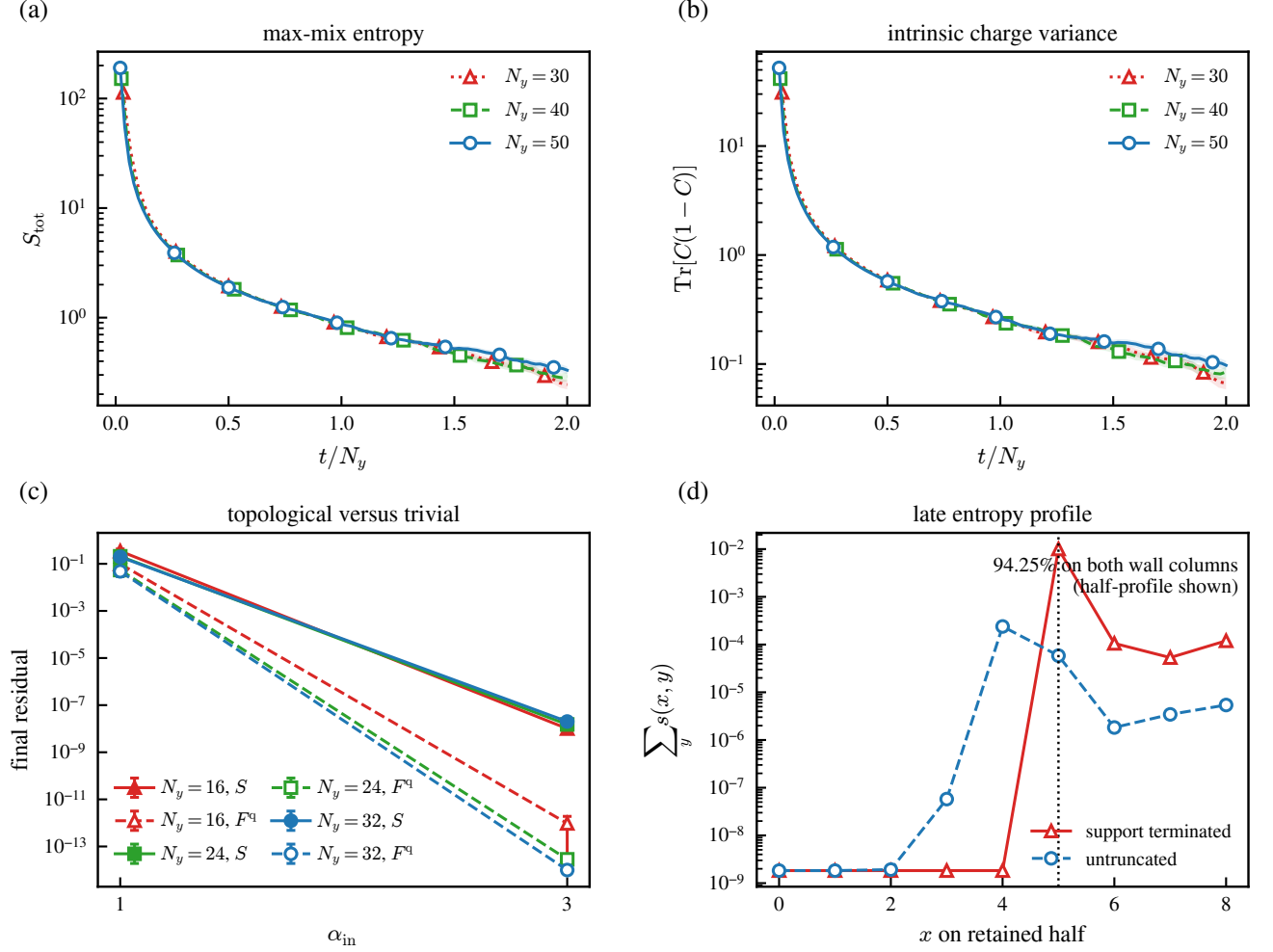


FIG. 3. Max-mix purification. (a) Total entropy and (b) intrinsic charge variance collapse approximately with t/N_y but retain a late residual. (c) The residual strongly distinguishes the topological and trivial interiors in the CPU mass sweep. (d) A smaller contour archive localizes the late entropy to the wall; only one half of the periodic transverse profile is displayed. Bands in (a) and (b) are $S = 100$ trajectory standard errors, and bars in (c) are $S = 10$ standard errors; many are smaller than the curves or markers. The contour export in (d) contains only its mean, so no defensible recordwise error bar is available there. The canonical panels use $N_x = 20$, $N_y = 30, 40, 50$, $S = 100$, and stop at $2N_y$; panel (d) is the separate 16×32 archive. Entropy, intrinsic variance, and contours are defined by Eqs. (11), (A1), and (A2); the figure does not establish a purification exponent.

sum is restricted to a wall window $\mathcal{X}$. Its plotted quantity is the algebraic contour contribution

$$\Delta I^{(\mathcal{X})} = S_{AB}^{(\mathcal{X})} + S_{BC}^{(\mathcal{X})} - S_B^{(\mathcal{X})} - S_{ABC}^{(\mathcal{X})}. \quad (19)$$

It is not the entropy of a separately cut one-wall Hilbert space and is not, by itself, a von Neumann conditional mutual information; strong subadditivity therefore constrains only Eq. (18). The wall curves are empirically positive, so the atlas uses the same positive plotting orientation for both quantities. Equations (A7)–(A14) give the precise distinction and show why both CFT reductions are linear in $z = \log[1/(1-x)]$.

The gray fit bands in Figs. 5(a) and 5(b) do not mark ranges selected afterward by fit quality. Both source

calculations used the conservative periodic-geometry rule

$$L_A + L_C \leq L_B \leq N_y - 2(L_A + L_C), \quad (20)$$

which requires both the intervening interval B and the complementary interval $D = N_y - (L_A + L_B + L_C)$ outside ABC to have length at least $L_A + L_C$. The rule avoids either separation becoming the shortest scale and treats B and D symmetrically; it was fixed by the generating notebooks, not optimized against these data. Consequently, panel (a) fits every stored $L_B = 6, \dots, 28$ row and panel (b) every stored $L_B = 8, \dots, 15$ row. Since the chord cross ratio is invariant under $L_B \leftrightarrow D$ when $L_A = L_C$, symmetry-related rows can occupy the same displayed z coordinate. The all-row fit remains the provenance-preserving baseline rather than a window chosen to mini-

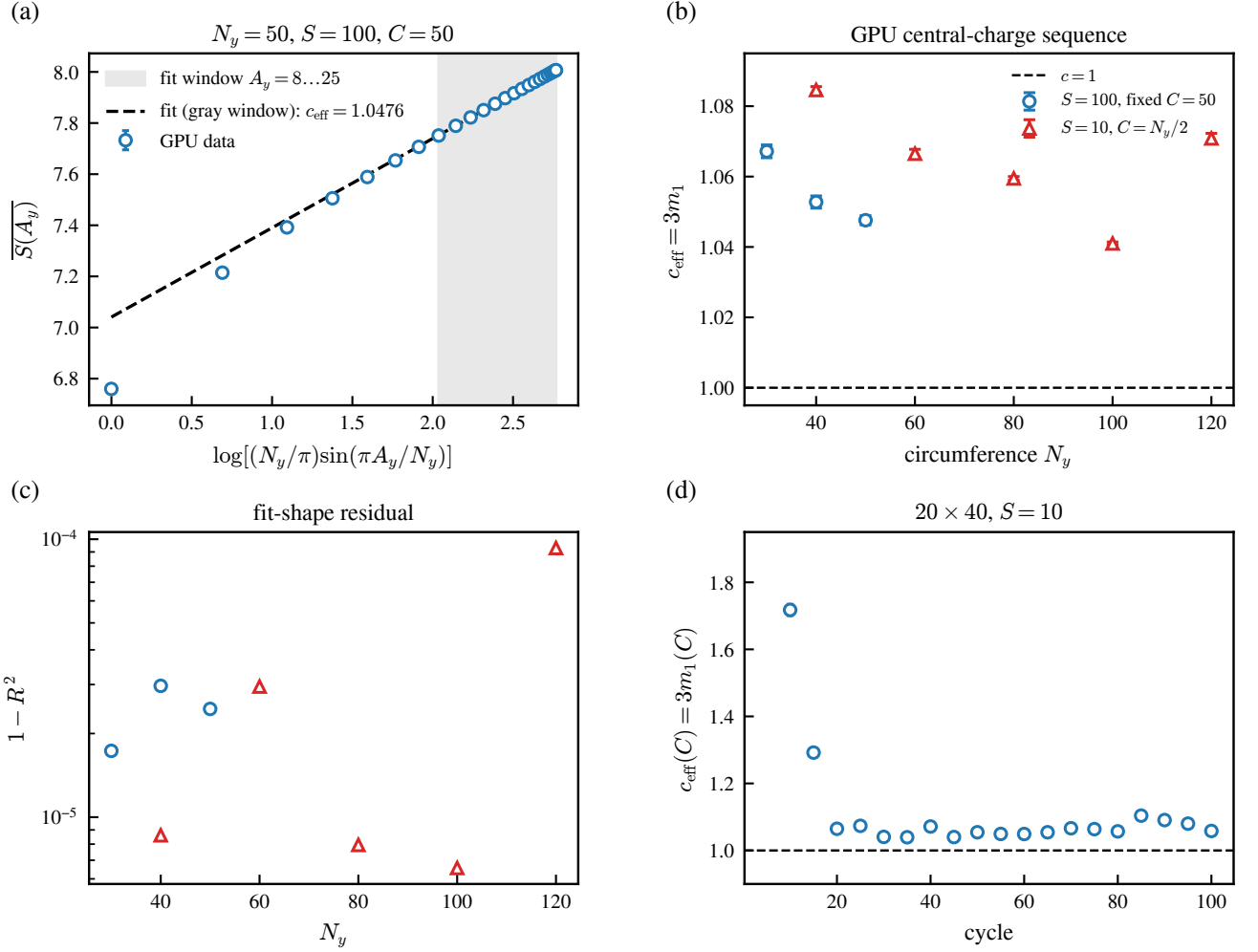


FIG. 4. Trajectory-averaged full-strip entropy. (a) Raw log-chord data for the largest controlled size. The gray region is the actual fit window; empirical points are unconnected, and the black dashed line extrapolates the fit across the displayed range even though only $8 \leq A_y \leq N_y/2 = 25$ points in the gray window enter the regression. (b) Direct GPU $c_{\text{eff}} = 3m_1$ versus circumference for the fixed- C , $S = 100$ sequence and the separate time-scaled $S = 10$ sequence. Empirical points are not connected; the dashed line is the $c = 1$ target, not a fit. (c) Fit residuals. (d) $c_{\text{eff}}(C)$ at 20×40 , again without connecting empirical points. Bars in (a) are stored entropy standard errors; bars in (b) and (d) are one-standard-error fit uncertainties. Most are smaller than the symbols or line width. The fixed-depth arm is $N_x = 20, S = 100, C_{\text{cyc}} = 50, N_y = 30, 40, 50$; the time-scaled arm has $S = 10$ and $C_{\text{cyc}} = N_y/2$. Equations (A1) and (10) define the reduction. The two arms are not combined into one finite-size extrapolation.

mize $|c_{mTMI} - 1|$.

A symmetry-preserving sensitivity scan imposes the stronger rule $L_B, D \geq q$ and keeps only fits with at least three distinct z values. For the exact data, increasing q from 6 to 8 changes (c_{mwall}, c_{mfull}) from $(1.001806, 1.002697)$ to $(1.000840, 1.000922)$; at $q = 11$ both are 1.000464 to the shown precision and the full/one-wall slope ratio is 2.0000003. This is the expected small short-distance correction, although the leverage and number of distinct coordinates decrease as q grows. The historical stochastic result does not behave as though its roughly 50% excess came only from the endpoint rows. It permits only one nontrivial trim before fewer

than three distinct coordinates remain: $q = 9$ gives $(c_{mwall}, c_{mfull}) = (1.4748, 1.5135)$ and ratio 2.0525, compared with $(1.4920, 1.5335)$ and 2.0556 at $q = 8$. The separate $L_A = L_C = 3$ full-width fit falls from 1.5754 at $q = 6$ to 1.4547 at $q = 10$, where only three distinct z values remain. Thus trimming improves the exact calibration and modestly lowers the stochastic coefficients, but no defensible window makes the wrong-protocol stochastic pilot a $c = 1$ result. The notebook diagnostic ledger records every accepted window and its regression and whole-trajectory bootstrap uncertainties.

For the exact 20×40 ground state with $L_A = L_C = 3$ and fixed $y_0 = 0$, the raw contour-decomposed

curves give slopes 0.166967668(27005) on each wall and 0.334232464(95377) across the full width. The full-to-one-wall ratio is 2.00178, an especially clean calibration of the 1/6 and 1/3 coefficients. These fits imply

$$\begin{aligned} c_{\text{TMI}}^{\text{wall}} &= 6m = 1.001806(162), \\ c_{\text{TMI}}^{\text{full}} &= 3m = 1.002697(286), \end{aligned} \quad (21)$$

which is one of the strongest numerical $c = 1$ results in the repository. The available historical stochastic pilot uses 12×31 , $C_{\text{cyc}} = 100$, $S = 100$, $L_A = L_C = 4$, unwindowed OW modes, the legacy nonperfect `dw_symmetric` schedule, and the same full-subsystem/post-contour estimator. Its trajectory-first curves give 0.2487(14) for one wall and 0.5112(14) for the matched full window, a ratio 2.05549, or $c_{\text{TMI}} = 1.4922(84)$ and 1.5336(42). It therefore retains one-versus-two-wall counting while overshooting the central charge by roughly 49% and 53%; the separately archived $L_A = L_C = 3$ full-window fit gives $c_{\text{TMI}} = 1.5753(48)$.

For the stochastic raw curves, translated origins are averaged inside each trajectory before the $S = 100$ trajectory mean is formed, as in Eq. (16). Pointwise bars are trajectory SEMs. The quoted slope parentheses are ordinary regression errors on the mean curve; a fixed-seed whole-trajectory bootstrap, which resamples the 100 records and repeats both the mean and fit, is retained separately as a correlated-fit sensitivity check. These three notions of uncertainty are not interchanged.

A separate block in `notebooks/analysis.ipynb` contains stored trajectory-fit outputs labeled $S = 250$, but its present source sets `MAX_SAMPLES_MI=2`; its embedded output is therefore stale relative to the executable cell. It also uses a 12×16 , infinite-shell, old random-sequence reduced-strip estimator rather than the full-width contour restriction above. This is a source/output and estimator mismatch, not superior production TMI evidence, so its coefficients are not promoted here.

Evidence class: deterministic calibration and a provenance-recovered wrong-protocol stochastic pilot. The exact and stochastic curve rows, geometry, averaging order, fit rows, and source fingerprints are now machine-readable in the atlas-local cache rebuilt from their generating calculation and covariance history. **What it does not establish:** Recovering the raw curves does not repair the stochastic protocol: it remains a one-size historical pilot with unwindowed controller modes and nonperfect feedback, and is not production evidence for the universal coefficient.

F. Result 6: uniform trajectories approach the Chern-one target

Figure 6 displays both the trajectory means and the individual final-state spread of the real-space Chern estimator. For uniform perfect-correction trajectories at $N_x = 20$, $S = 25$, and depths $C = N_y$, the final real-space

tripartition Chern means are

$$\begin{aligned} \bar{C}_R &= 0.9999950(23), & N_y = 30, \\ &= 0.999687(298), & N_y = 40, \\ &= 0.999927(60), & N_y = 50. \end{aligned} \quad (22)$$

where the parenthetical values are standard errors. The lowest individual final value is 0.992548, occurring in the $N_y = 40$ ensemble; the rest of the distributions cluster tightly near unity. This is the clearest adaptive bulk-topology precursor in the archive and uses a Chern observable rather than a Bott index.

Evidence class: canonical GPU trajectory precursor. **What it does not establish:** It is one radius-eight real-space estimator on three finite rectangles and lacks the complete production suite of marker profiles, purity-gap gates, and independent estimator cross-checks.

IV. GROUP II: FEEDBACK ACTIVITY AND THE TRANSITION AXIS

A. Result 7: feedback activity is concentrated at the wall

Figure 7 compares the spatial activity profile, integrated rates, and the independently measured tangent-gap suppression. The exact branchwise identity between the saved signed and absolute wrong-outcome indicators is satisfied in the underlying activity archive. At $N_y = 24$, the transverse profile has two sharp peaks at the wall locations, of order 0.09 per update, while the uniform background is a few 10^{-3} . Across $N_y = 20, 22, 24$, the all-domain activity is enhanced by factors 4.50, 4.47, and 4.84 in the wall geometry. Separately measured tangent gaps are suppressed by factors 7.94, 8.28, and 8.83 over the same sizes.

Evidence class: canonical CPU finite-size precursor. **What it does not establish:** The activity and spectral observables were obtained from separate trajectory suites. Their coexistence supports a mechanism hypothesis, but no recordwise activity-gap covariance, profile overlap, matched purification residual, or causal statement is available.

B. Result 8: independent diagnostics change near $\alpha = 2$

Figure 8 places the particle-Choi and max-mix purification precursors on the same mass axis while keeping their meanings separate. The exact positive-mass transition lies at $\alpha_c = 2$. In the no-feedback GPU particle-Choi proxy, the $N_y = 20, 30, 40$ finite gaps at $\alpha = 1.9$ are 0.01215(229), 0.00808(182), and 0.00249(90), and all rise on the trivial side by $\alpha = 2.1$. The perfect-correction proxy rises more sharply as $\alpha \rightarrow 2^-$ and loses finite samples at or above the transition; open points mark this censoring and

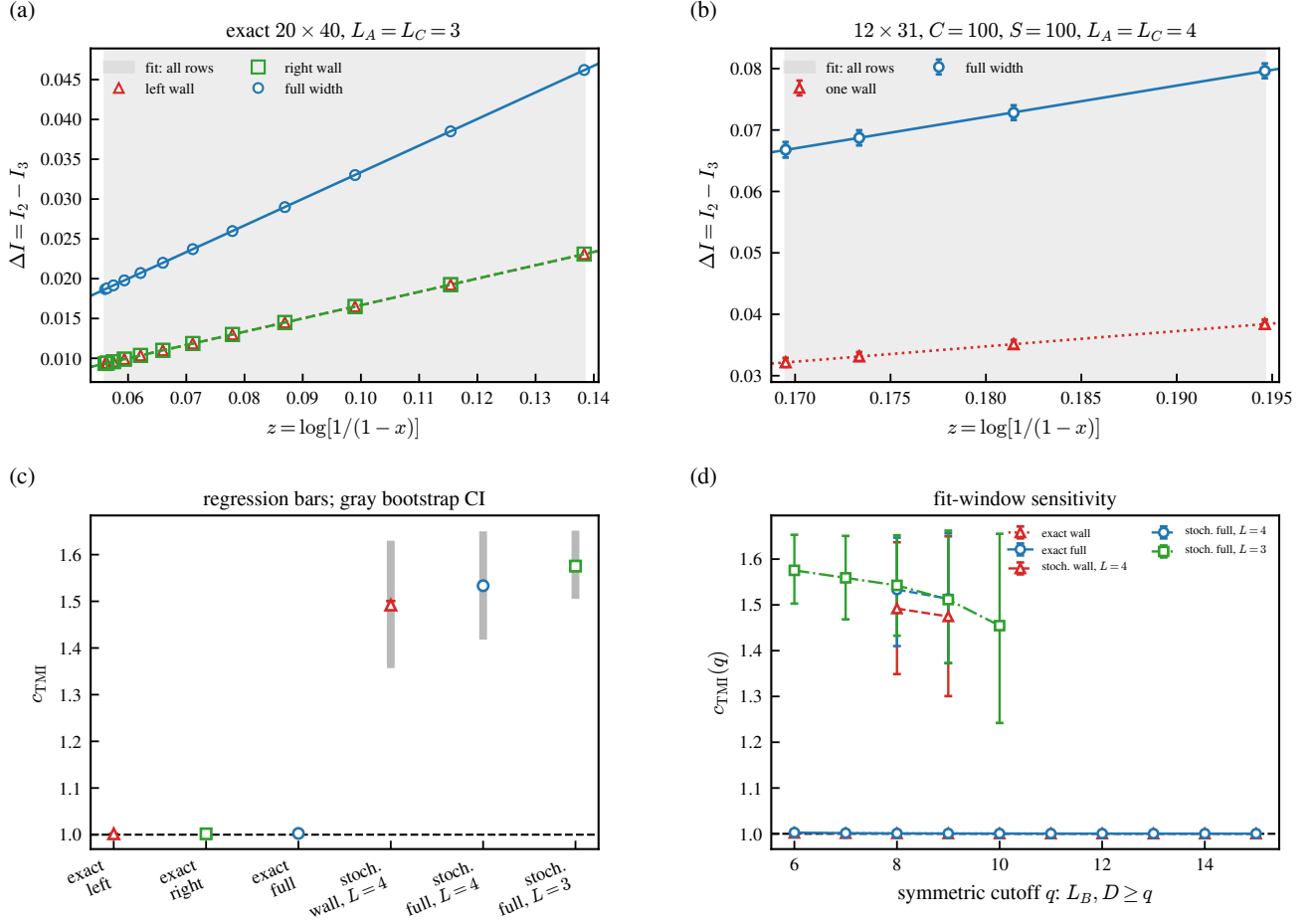


FIG. 5. Raw full-width conditional mutual information and one-wall contour contributions. The rigorous full-width quantity is $\Delta I^{(\text{full})} = I_2 - I_3 = I(A : C | B) \geq 0$, so the requested $I_3 - I_2$ is its negative. A wall curve is instead the algebraic post-contour $\Delta I^{(\mathcal{X})}$ in Eq. (19); it uses the same positive plotting convention but is not itself a conditional mutual information. (a) Exact 20×40 ground-state curves at $L_A = L_C = 3$, $y_0 = 0$, for the left-wall, right-wall, and full- x post-contour sums. Unconnected markers are measured values; the gray band contains all stored $L_B = 6, \dots, 28$ rows used in the regression, and each fitted line is extended across the full displayed $z = \log[1/(1-x)]$ range. There is no trajectory sampling error. This complete range is the source-era geometry rule in Eq. (20), not a residual- or fit-quality cut. (b) Historical 12×31 , $C_{\text{cyc}} = 100$, $S = 100$, $L_A = L_C = 4$ stochastic curves for the matched wall and full-width estimator. Origins are averaged within a trajectory before the record mean; vertical bars are trajectory SEMs. Lines fit all stored $L_B = 8, \dots, 15$ rows in the gray window and are not connections between points. The same Eq. (20) fixes this complete stored range; symmetry-related L_B and complementary- D rows may coincide in z . This arm uses $n_{\text{shell}} = \text{None}$, nonperfect feedback, and the old `dw_symmetric` schedule. (c) Exact and stochastic c_{TMI} , using $6m$ for one wall and $3m$ for two walls; the additional $L_A = L_C = 3$ stochastic full-width fit is kept distinct. Bars are one-standard-error regression uncertainties. The thick gray segments on the stochastic points are fixed-seed 95% whole-trajectory bootstrap intervals; they are also recorded in the diagnostic ledger and are not substituted for the regression bars. (d) Symmetry-preserving window scan imposing $L_B, D \geq q$ and retaining at least three distinct z values. Exact bars are regression standard errors and are generally smaller than the markers; stochastic bars are fixed-seed 95% whole-trajectory bootstrap intervals. Trimming drives the exact coefficients still closer to one but only modestly lowers the stochastic values. The corresponding full/one-wall slope ratios remain $2.0018 \rightarrow 2.0000$ exact and $2.0556 \rightarrow 2.0525$ stochastic over their accepted scans. The stochastic result is an estimator pilot, not a production-protocol $c = 1$ measurement.

a missing finite sector is not converted into an infinite gap. Independently, the CPU max-mix scan at $N_y = 32$ changes from entropy/variance $0.1244/0.03447$ at $\alpha = 1.8$ to $0.002492/0.000309$ at 1.9 and $2.0 \times 10^{-8}/7.9 \times 10^{-12}$ at 2.0 .

Evidence class: canonical GPU transfer and CPU

purification precursors. **What it does not establish:** The observables are not interchangeable: the transfer quantity is a particle-Choi proxy and the CPU quantities describe transient max-mix purification. There is no single saved stationary pure-state sweep containing topology, entropy, charge, and tangent gates together.

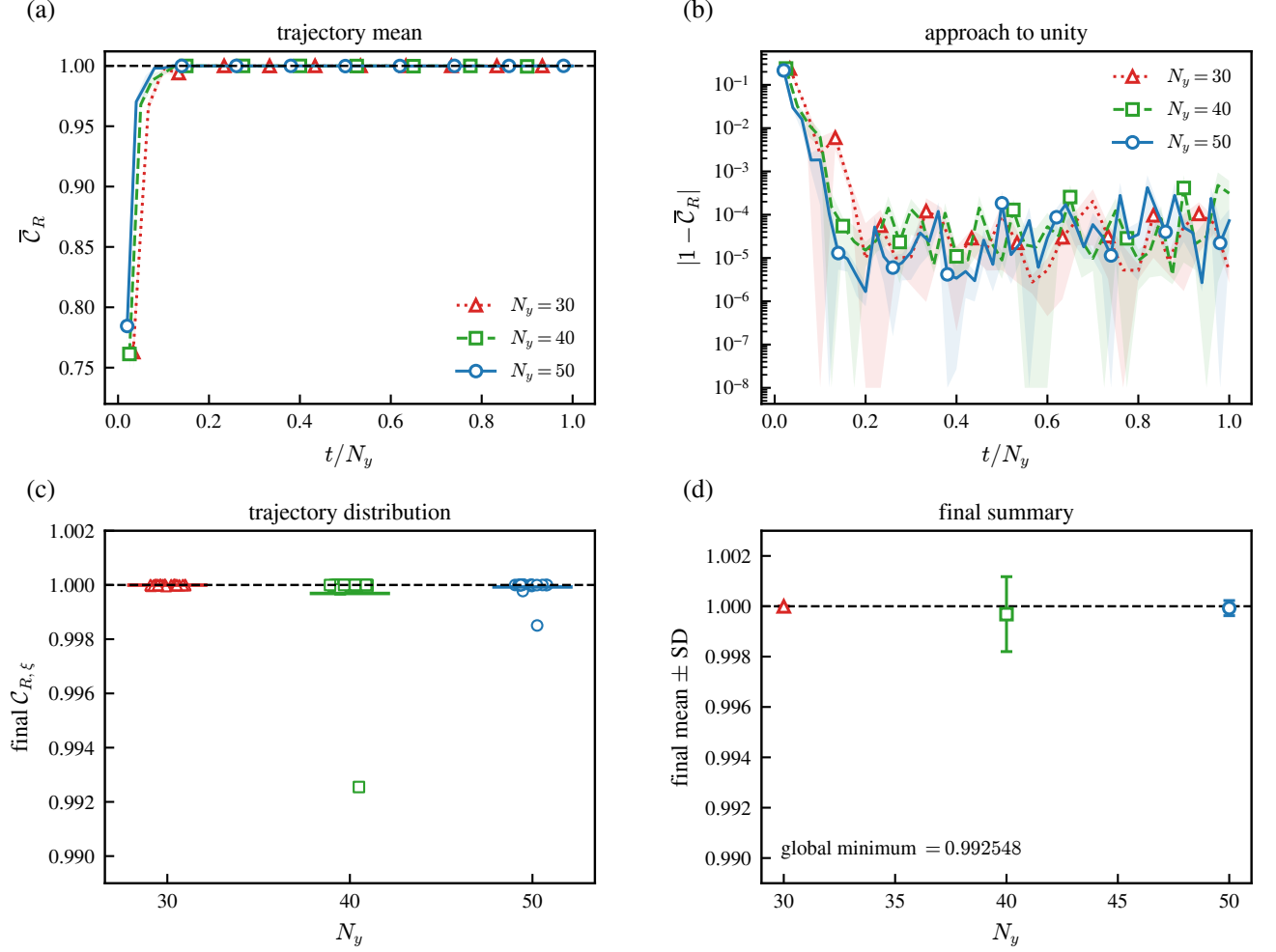


FIG. 6. Uniform adaptive topology. (a) Trajectory-mean real-space Chern estimator. (b) Absolute deviation from unity. (c) Individual final-trajectory values and ensemble means. (d) Final means with standard-deviation bars, exposing the single lower $N_y = 40$ realization rather than hiding it in the mean. Shaded bands in (a) and (b) are trajectory standard errors for $S = 25$ and are mostly smaller than the lines; (d) deliberately uses standard deviations to display ensemble spread. All panels use uniform $N_x = 20$, perfect-correction runs at $C_{\text{cyc}} = N_y$ for $N_y = 30, 40, 50$ and the radius-eight estimator in Eq. (A17). This is one finite-radius topology precursor, not a complete estimator cross-check.

V. GROUP III: HANDEDNESS AND RECORD OBSERVABLES

A. Result 9: modular flow has the orientation-dependent sign pattern

Figure 9 shows the exact signed velocities and the corresponding endpoint-resolved stochastic packet displacements. The accepted B0 state gives modular velocities $(-2.930, +4.536)$ for the two walls, with the same orientation reversal as the physical velocities. A distinct observer applied to stochastic trajectory covariances also gives the required endpoint-resolved signs. At $n_{\text{shell}} = 2$, the four final packet displacements are

$$(8.821, -8.813, 8.489, -8.520), \quad (23)$$

where the sign alternates with packet endpoint and wall orientation according to the stored labels. The $n_{\text{shell}} = 1$ and 2 observers agree on this pattern.

Evidence class: completed exact handedness calibration plus a canonical stochastic precursor. **What it does not establish:** The stochastic data use translated cuts from a small number of trajectories; those cuts are correlated variance-reduction samples, not independent records. No matched trivial modular controller or final $S = 100$ accepted-size family exists, and modular motion is not physical monitored-time transport.

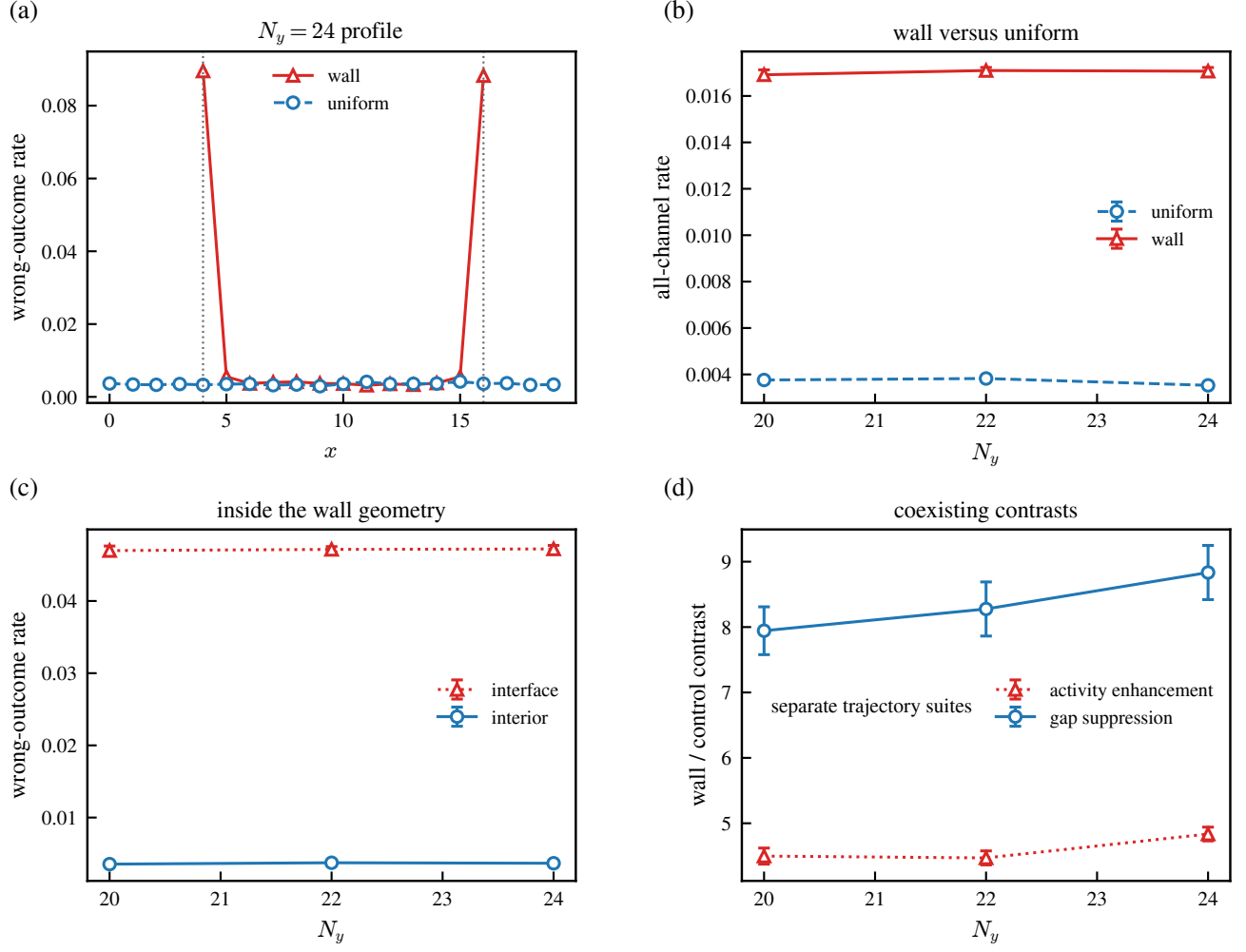


FIG. 7. Born feedback activity. (a) Two interface peaks in the $N_y = 24$ wall profile. (b) Wall versus uniform all-domain rate. (c) Interface and interior rates within the wall geometry. (d) Activity enhancement and independently measured tangent-gap suppression; the annotation emphasizes that these are separate trajectory suites. The bands in (a) and bars in (b),(c) are trajectory standard errors for $S = 25$; (d) uses the corresponding propagated standard errors for each ratio. Some are smaller than the markers. The $N_y = 20, 22, 24$ CPU activity and tangent observables use $N_x = 20$, $S = 25$, $C_{\text{cyc}} = 2N_y$, $n_{\text{shell}} = 1$, perfect correction, and a raster- y schedule. The wall arm has $(\alpha_{\text{in}}, \alpha_{\text{out}}) = (1, 30)$ and `dw.truncation=True`; the uniform control has $\alpha = 1$ and no wall truncation. They are computed by Eqs. (A22) and (A23), but come from separate trajectory suites. Panel (d) therefore supports coexistence, not a recordwise activity-gap correlation.

B. Result 10: the classical probability record detects an active wall sector

Figure 10 displays the extensive chain entropy and its wall-resolved event entropy density. The probability maps retained by the streaming GPU campaigns determine Eq. (12) even though the realized bits were not saved. At late time, the chain entropies per cycle are 89.6613(941), 119.7817(1013), and 149.7625(1005) for $N_y = 30, 40, 50$. Dividing by N_y gives 2.98871(314), 2.99454(253), and 2.99525(201), showing a stable extensive background. At 20×40 , the late binary entropy per event is 0.10129 in the distance-one wall neighborhoods and 0.01007 elsewhere—a factor of about ten.

Evidence class: canonical GPU chain-entropy feasibility result. **What it does not establish:** This is not an individual record self-information, replica partition function, aspect-ratio scaling, or record-CFT extraction. It establishes that the archived classical probabilities contain a sharply wall-enhanced uncertainty signal and motivates retaining complete record bits in future runs.

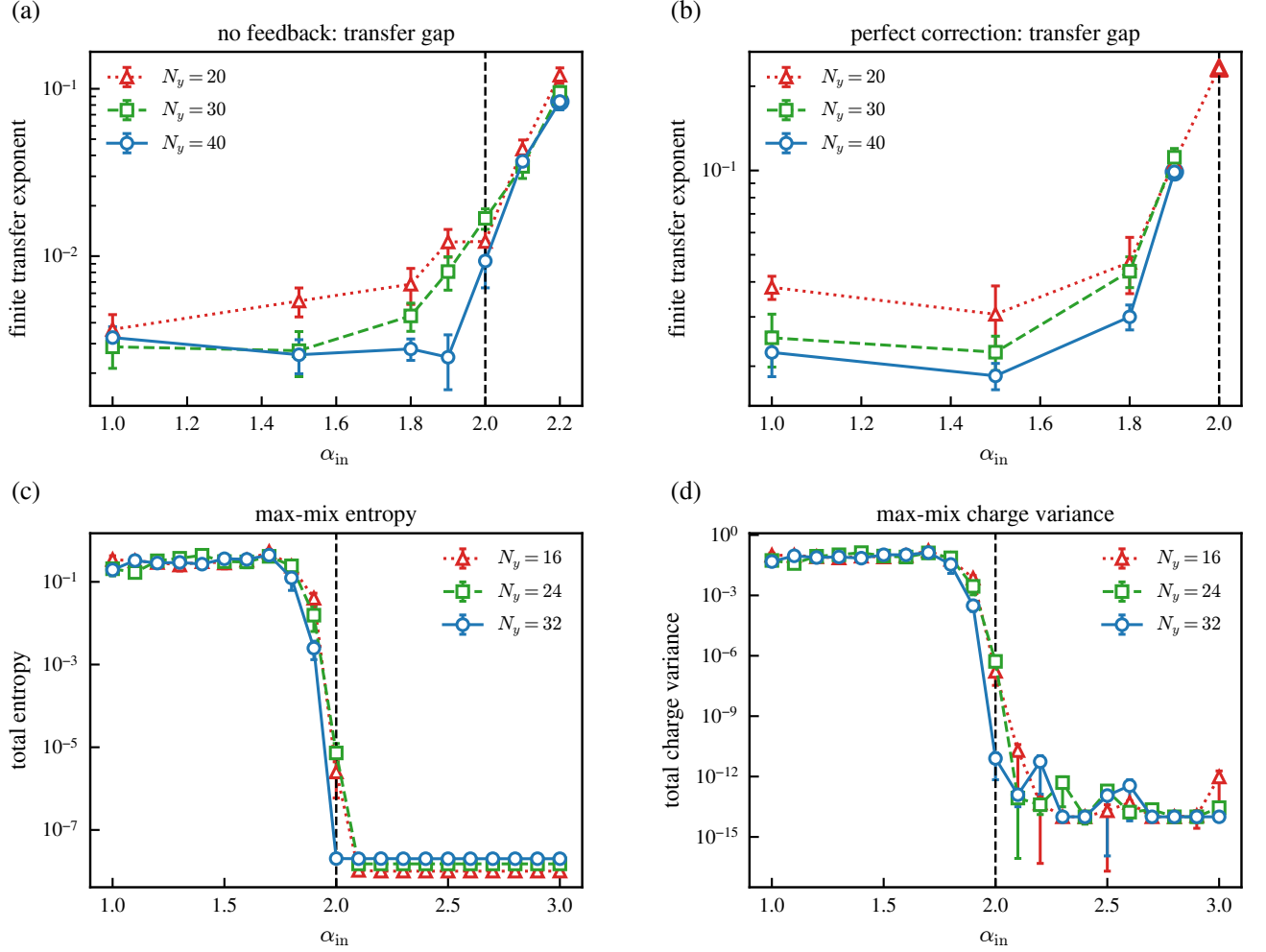


FIG. 8. Mass-transition precursors. (a) No-feedback and (b) perfect-correction finite transfer exponents. Open overplotted markers denote censored sample sets. (c) Max-mix entropy and (d) max-mix charge variance from an independent CPU sweep. The vertical dashed line is the exact $\alpha_c = 2$ transition. Bars in (a),(b) are standard errors over the finite transfer samples; bars in (c),(d) are $S = 10$ trajectory standard errors. Where not visible, they are smaller than the markers or line width. Panels (a,b) use $N_x = 20$, $N_y = 20, 30, 40$, $S = 10$, $C_{\text{cyc}} = 2N_y$, $n_{\text{shell}} = 1$, and `dw.truncation=True`, comparing no feedback with perfect correction. Their reduced particle–Choi exponent and gap are Eqs. (A24) and (A25). Panels (c,d) use the independent $N_x = 16$, $N_y = 32$, $S = 10$, $C_{\text{cyc}} = 2N_y$, $n_{\text{shell}} = 1$, `dw.truncation=True` perfect-correction max-mix covariance and Eq. (11). The two diagnostics are not identified, and missing finite sectors are censoring rather than divergent gaps.

VI. GROUP IV: EFFECTIVE PARENTS, SPATIAL STRUCTURE, AND ENSEMBLE SPREAD

A. Result 11: translation-resolved flattened spectra expose the wall modes

The flattened-parent source notebook preserves translation symmetry along the periodic direction and therefore decomposes the one-body problem into k_y blocks. The archived spectrum shows the corresponding wall physics without first reducing it to a fitted coefficient: the X - and Y -trial parents have especially clean counter-propagating in-gap branches crossing zero, while the Z trial retains the

crossing within a more structured bulk spectrum. This is the spectral version of having two oppositely oriented interfaces in a periodic transverse geometry.

Evidence class: rendered deterministic flattened-Hamiltonian benchmark. The source notebook and high-resolution PNG are both archived. **What it does not establish:** The eigenvalue array itself was not exported separately, so Fig. 11 is a full-width crop of the legacy render rather than a reconstruction from raw numeric rows. It demonstrates visible spectral flow but does not supply a new finite-size fit.

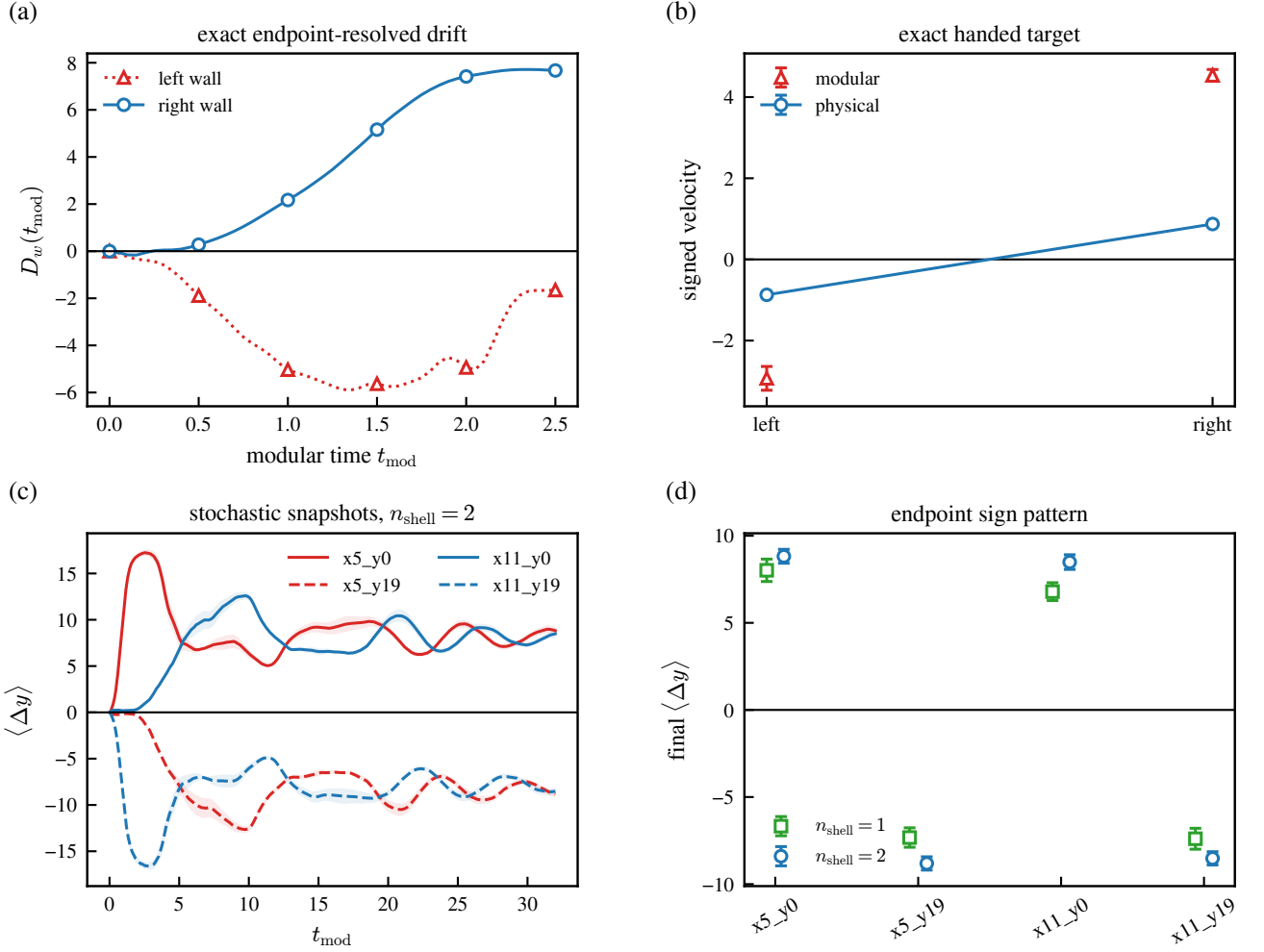


FIG. 9. Modular handedness. (a) Exact endpoint-resolved displacement. (b) Exact modular and physical velocity signs. (c) Stochastic packet trajectories for $n_{\text{shell}} = 2$. (d) Final endpoint sign pattern for two shell choices. Bars in (b) are regression standard errors; bands in (c) and bars in (d) are trajectory standard errors after first averaging 40 translated origins within each of $S = 10$ independent records. The superseded pooled-cut SEM is not plotted. Several bars are smaller than the curves or symbols; (a) is deterministic. The stochastic panels use the cycle-50 16×40 observer and the averaging order in Eq. (16). Equations (A27) and (13) define the motion; modular time is not the monitored circuit clock.

B. Result 12: the finite-range uniform parent is already a gapped Chern insulator

Figure 12 compares the corrected uniform-parent Chern response, gap, and the separate domain-wall entropy and correlator diagnostics across shell depth. The flattened-parent bulk topology must be tested in a uniform gapped geometry, not with a tripartition disk that intersects both sides of a domain wall. Direct reconstruction at 20×40

gives

n_{shell}	$\mathcal{C}_R(\alpha = 1)$	$\mathcal{C}_R(\alpha = 3)$	$\min E (\alpha = 1)$
0	0	0	1.582825
1	-0.999999829	-2.82×10^{-8}	1.802517
2	-0.999997053	-5.46×10^{-7}	1.902787
full	-0.999942295	-1.30×10^{-5}	2.000000

(24)

for the X -trial construction. An independent momentum-space FHS evaluation of the $n_{\text{shell}} = 1$ lower band gives $\mathcal{C}_{\text{FHS}} = -1$ at $\alpha = 1$ and zero at $\alpha = 3$; the real-space calculation now evaluates that same occupied projector and agrees, $\mathcal{C}_R = -0.999999829$ and -2.82×10^{-8} , respectively. Thus its sign is not produced by a complementary-band substitution. It is the raw flattened-frame orientation, which is opposite the $C = +1$ spectral-projector orienta-

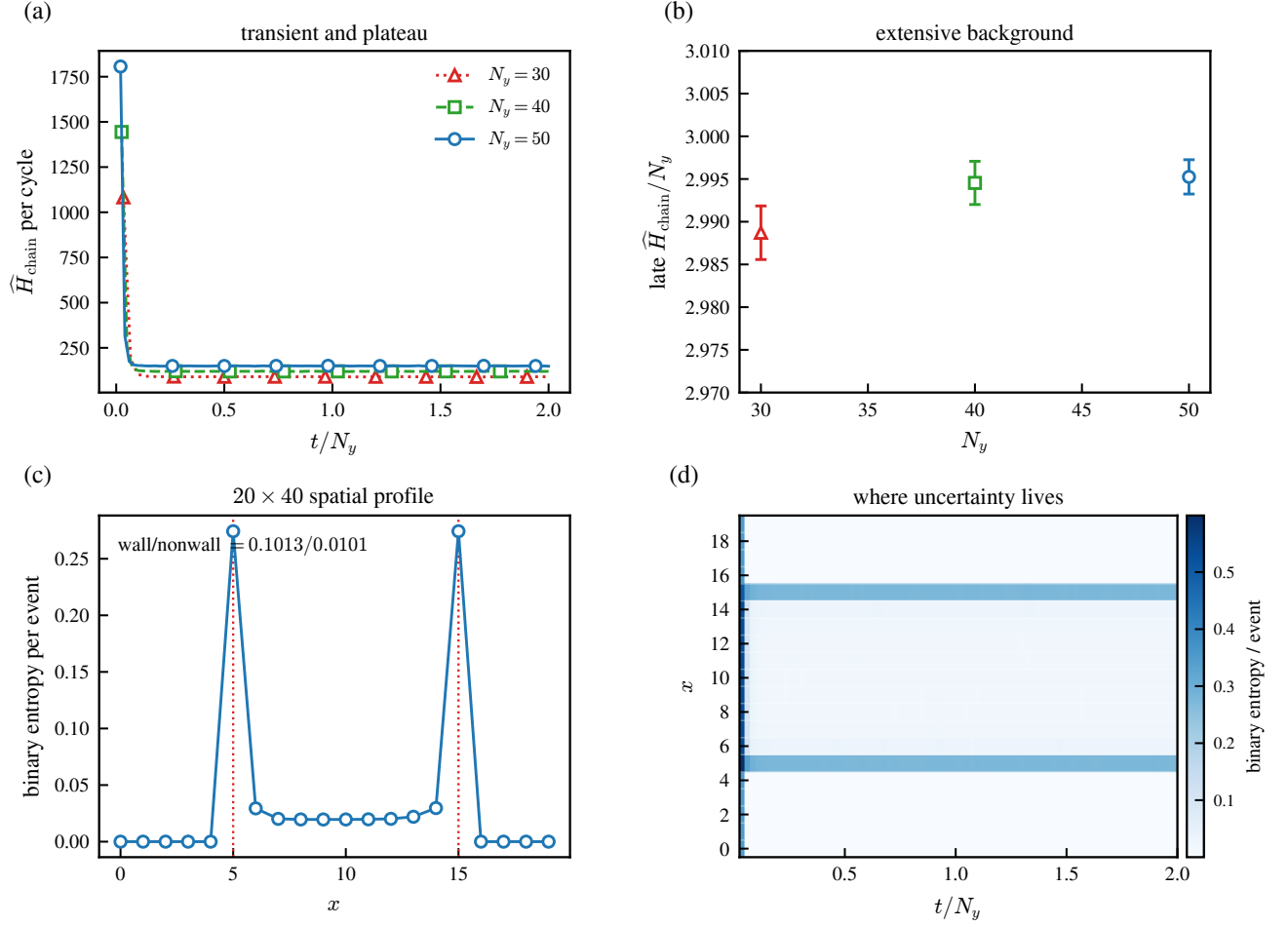


FIG. 10. Classical probability-record uncertainty. (a) Chain-entropy transient and plateau. (b) Late entropy divided by N_y . (c) Late transverse profile at 20×40 . (d) Space-time map using the BPJ-compatible sequential blue scale; the two persistent high-uncertainty bands coincide with the walls. The band in (a) and bars in (b),(c) are trajectory standard errors for $S = 100$; most are smaller than the line or symbols. Panel (d) is the ensemble-mean field and has no scalar error-bar overlay; its fine white mesh marks individual transverse cells, with integer centers labeled every two sites. These $N_x = 20$, $S = 100$, $N_y = 30, 40, 50$ streaming runs use the chain estimator in Eq. (12); panel (c) is the 20×40 late profile. Because outcome bits were not saved, none of the panels is a realized-record self-information or a record-CFT coefficient.

tion used for the exact Hamiltonian and form-factor sweep. The $n_{\text{shell}} = 1$ occupied band has width 0.386523, so “flattened parent” does not mean an exactly flat finite-range Chern band.

The separate domain-wall tables answer a different convergence question. With domain-wall truncation, one shell already gives the two-wall and one-wall entropy slopes 0.333983 and 0.166750, close to $1/3$ and $1/6$, while the wall-correlator exponent is 3.20326. Two shells give 3.15179, and full support gives 2.00421. The formerly quoted finite-domain-wall Chern scalar is excluded here because its radius-eight tripartition disk, centered at $x = 10$, crosses the interfaces at $x = 6, 14$ and is not a uniform bulk invariant. Thus finite support already preserves the uniform bulk topology; the slower shell convergence belongs to the domain-wall correlator tail.

Equations (14) and (15) also make precise the relation to Ky Fan optimization: filling the lowest parent eigenstates minimizes the simultaneous *additive* penalty for undesired local outcomes. It does not maximize a single joint probability for mutually overlapping measurements.

Evidence class: direct deterministic reconstruction of the uniform flattened parent from the canonical CPU implementation, independently checked by real-space tripartition and momentum-space FHS topology; machine-readable domain-wall entropy and correlator tables. **What it does not establish:** This benchmark characterizes an effective static parent and its ground state; it does not prove that finite-depth adaptive dynamics samples that state exactly. The uniform FHS integer is not assigned to the globally gapless domain-wall geometry.

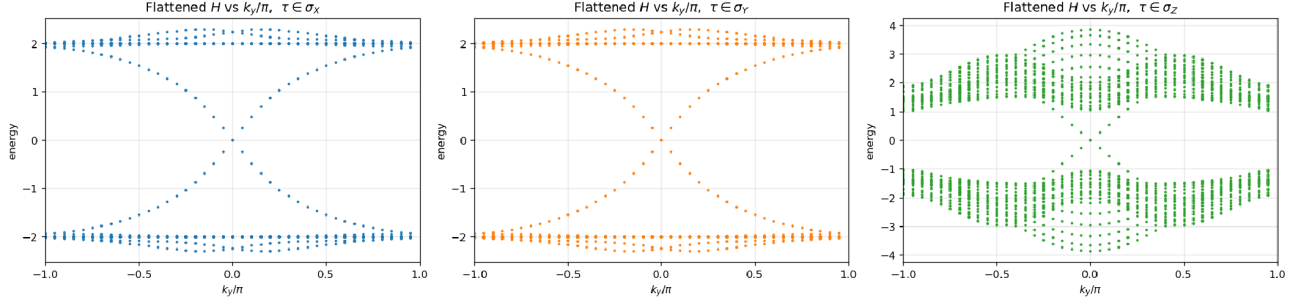


FIG. 11. Translation-resolved flattened-parent spectra for the X, Y, and Z trial orbitals. The k_y -resolved lower row of the archived benchmark is given the full text width so the two in-gap wall branches and their zero-energy crossing remain legible. The source geometry is periodic 20×40 with $(\alpha_{\text{in}}, \alpha_{\text{out}}) = (1, 30)$ and $n_{\text{shell}} = \text{None}$; each trial-orbital panel overlays `dw.truncation=False` and true. It is the deterministic spectrum of Eq. (14) after exact y -translation reduction, not a trajectory average; the plotted lower row is a crop of the archived render because eigenvalue rows were not exported. Visible spectral flow is therefore preserved, but no new size fit or uncertainty is inferred.

C. Result 13: spatial maps resolve the wall rather than a featureless bulk

Four complementary maps in Fig. 13 make the geometry visible. The exact local Chern marker is uniform in the topological strip and switches across its interfaces. The $S = 25$ slow-vector density is concentrated at the two wall columns. An independent $S = 10$ half-strip entanglement contour at circuit cycle 50 is similarly wall-centered, and the $S = 100$ max-mix charge-variance map at $2N_y$ shows where residual purification noise lives. These fields are particularly useful guards against a scalar average being produced by a diffuse bulk background.

Evidence class: one rendered exact calibration and three machine-readable adaptive maps. **What it does not establish:** The adaptive panels use different accepted precursor campaigns and sizes. They show mutually compatible localization, not a recordwise equality between tangent, entanglement, and charge fluctuations.

D. Result 14: modular propagation is visible before center-of-mass reduction

The signed displacements in Result 9 are reductions of a much richer density movie. Figure 14 shows four snapshots from the cycle-50, sample- and origin-averaged modular evolution at 16×40 . The initially localized charge packet redistributes through the half-strip while retaining pronounced structure tied to the two wall columns. This provides a direct spatial check on the endpoint-resolved sign analysis without confusing modular time with monitored circuit time.

Evidence class: canonical CPU observer applied to $S = 10$ GPU covariance snapshots. **What it does not establish:** The origin translations are correlated variance-reduction samples, and this is not a real-time charge response or a matched trivial control.

E. Result 15: typical trajectories cluster near the target, but tails remain

The shared final-cycle features used in Results 15 and 16 are defined from the local cell occupation $q_\xi(x, y) = \sum_\mu C_{\xi, (x, y, \mu), (x, y, \mu)}$ and $Q_\xi = \sum_{x, y} q_\xi(x, y)$ by

$$d_{Q, \xi} = 100 \frac{|Q_\xi - N_x N_y|}{N_x N_y}, \quad d_{C, \xi} = |C_{R, \xi} - 1|, \quad r_{\text{wall}, \xi} = \left[\frac{1}{|\mathcal{W}|} \sum_{(x, y) \in \mathcal{W}} q_\xi(x, y)^2 \right]^{1/2}. \quad (25)$$

Here $\mathcal{W}$ contains all y and the five columns within radius two of either wall, exactly as in the saved feature table. Thus “wall-charge RMS” is an RMS of the local cell occupation, not an RMS of the globally centered filling.

The $C_{\text{cyc}} = 2N_y$ distributions in Fig. 15 replace a vague claim of ensemble cleanliness with a record-resolved statement. Across $S = 100$ trajectories, the means of $c_{\text{eff}} = 3m_s$ are 1.0794, 1.0476, and 1.0436 at $N_y = 30, 40, 50$; the corresponding standard deviations are 0.0894, 0.0350, and 0.0364. The real-space Chern medians are 0.9993, 0.9978, and 0.9983, although rare records extend down to 0.433, 0.623, and 0.617. Charge drift remains discrete and decreases with circumference. Independently, final max-mix V_Q is broad and right-skewed: its medians are 0.0304, 0.0357, and 0.0573, well below its means 0.0665, 0.0846, and 0.0973.

Evidence class: canonical $S = 100$ trajectory-resolved pure-state and purification diagnostics. **What it does not establish:** The typical records support near- $c = 1$, near-unit-Chern behavior, but the tails rule out saying that all records are equivalent. The pure-state and max-mix histograms come from separate ensembles and are not paired trajectory by trajectory.

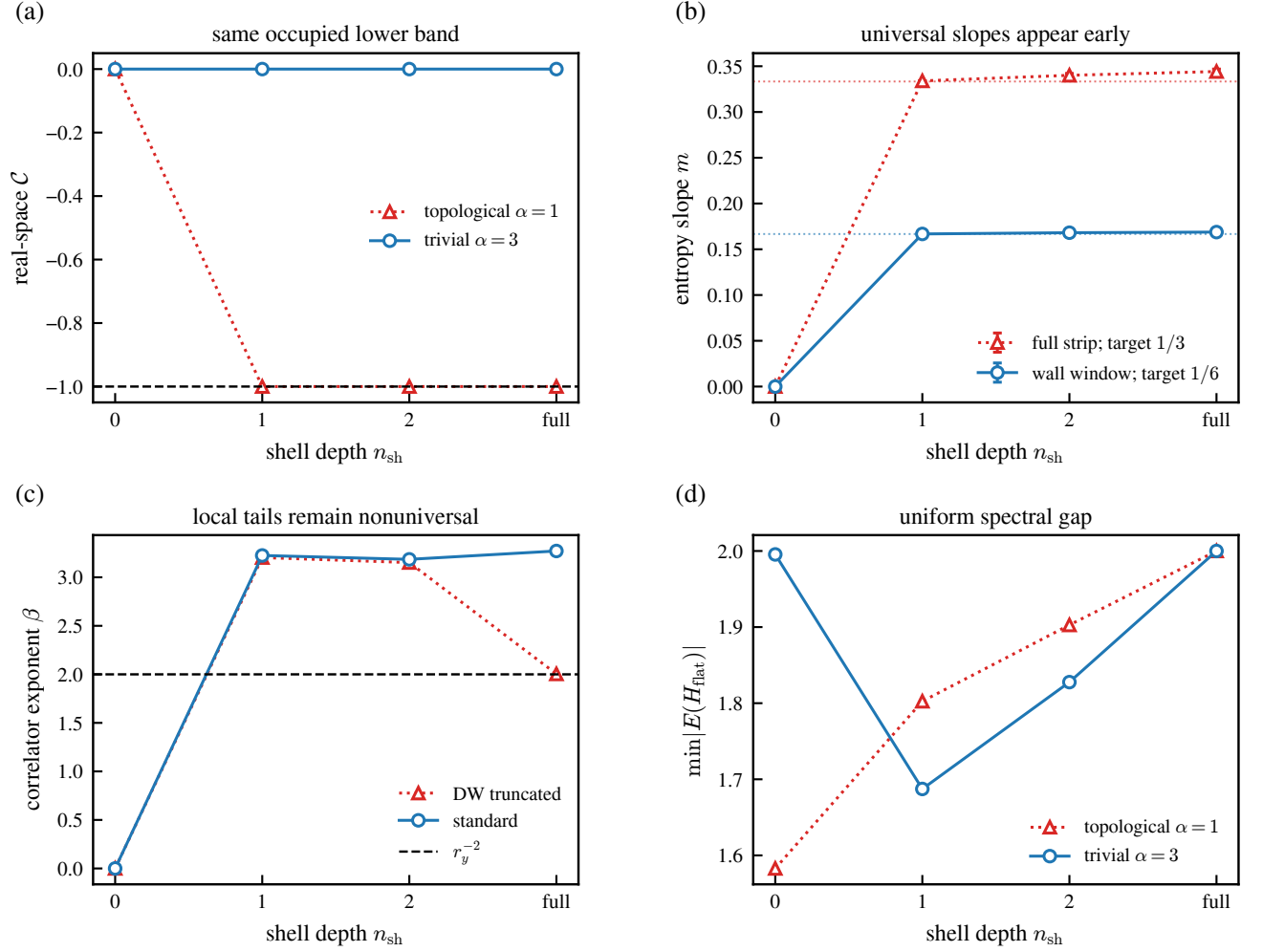


FIG. 12. Corrected flattened-parent audit. (a) Uniform real-space Chern response for the topological $\alpha = 1$ parent and trivial $\alpha = 3$ control. The independent $n_{\text{shell}} = 1$ lower-band FHS integers are -1 and 0 , respectively; the real-space tripartition routine evaluates the same occupied projector and agrees in sign and magnitude. (b) Full-strip and wall-window entropy coefficients in the separate domain-wall construction. (c) Its wall-correlator exponent. (d) Uniform distance of the flattened-parent spectrum from zero. “Full” means the untruncated shell sum. Bars in (b) are the RMS one-standard-error regression uncertainty across the contributing interval fits and are generally smaller than the markers. The remaining panels are deterministic, so sampling error bars do not apply. The uniform panels use 20×40 and Eqs. (A17), (A19), and (A20); the entropy/correlator panels belong to a separate domain-wall construction governed by Eqs. (10) and (A3). Static-parent topology is not finite-depth adaptive preparation.

F. Result 16: ensemble cleaning is observable-dependent

Figure 16 shows how the inferred coefficient changes under each declared trajectory-ranking functional and retained fraction. Post hoc rank conditioning does not produce a unique “clean” ensemble. Retaining the smallest global charge drifts can leave the central-charge excess unchanged or even worsen it, especially at $N_y = 30$. Conditioning on wall-charge RMS or Chern error reduces the excess for some circumferences and retained fractions, but the behavior is not uniform. The aggressive 10% points contain only ten records and carry correspondingly larger

uncertainty. Any future cleaning rule must therefore be preregistered, report its retained fraction, and be checked against multiple independent observables.

Evidence class: post hoc diagnostic on the canonical $S = 100$ final-cycle feature table. **What it does not establish:** This is a sensitivity analysis, not a production reweighted ensemble and not evidence that unfavorable trajectories should be discarded.

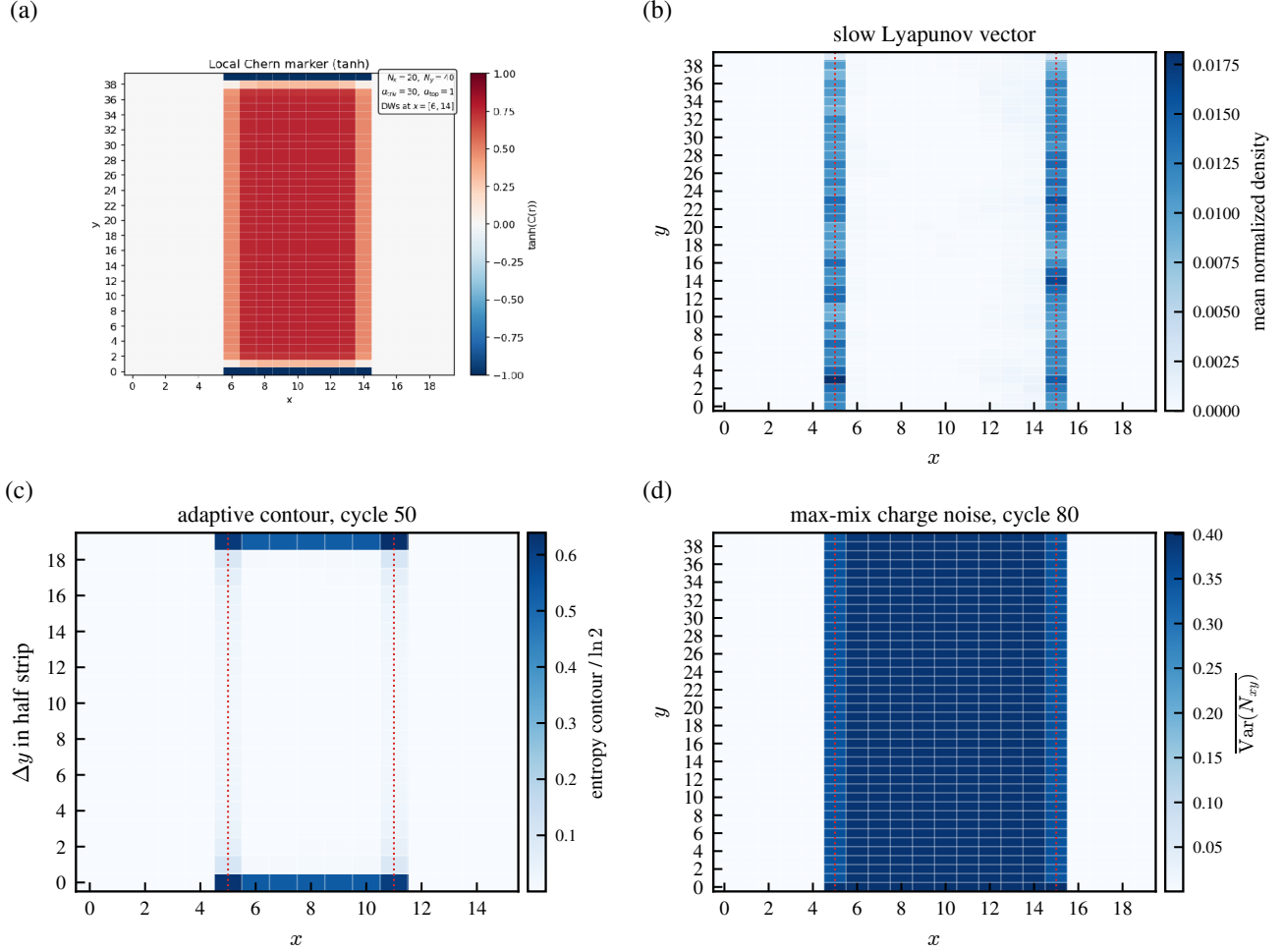


FIG. 13. Spatial evidence. (a) Archived exact local-Chern render; the numeric field was not separately exported. (b) Ensemble-averaged density of the minimum-absolute Lyapunov vector at 20×40 . (c) Final 16×40 half-strip entanglement contour. (d) Final 20×40 max-mix local charge variance. Dotted red lines mark the wall columns in the machine-readable adaptive panels. The fine white mesh delineates every unit cell, with integer centers labeled every two sites. Panels (a)–(d) respectively use Eqs. (A18), (A23), and (A2); the adaptive inputs have $S = 25$, $S = 10$ at cycle 50, and $S = 100$ at $2N_y$, respectively. Their compatible localization does not make them recordwise paired observables.

G. Result 17: the filling distribution narrows with circumference

Figure 17 gives the cycle-resolved population variance and its final-size comparison for both saved preparations. The final-cycle charge-drift histogram in Result 15 hides a useful finite-size fact. For each record ξ , define the centered filling deviation in percentage points by

$$\delta\nu_\xi(C_{\text{cyc}}) = 100 \frac{Q_\xi(C_{\text{cyc}}) - N_x N_y}{N_x N_y}, \quad (26)$$

$$\sigma_\nu^2(C_{\text{cyc}}) = \text{Var}_\xi[\delta\nu_\xi(C_{\text{cyc}})].$$

The raw cell-charge histories show the same narrowing in two independent $S = 100$ campaigns. At $C_{\text{cyc}} = 2N_y$, the pure-state variances are 0.29684, 0.21614, and 0.16534 squared percentage points for $N_y = 30, 40, 50$;

the max-mix purification values are 0.28533, 0.20166, and 0.17023. Thus the $N_y = 30 \rightarrow 50$ reduction is 44.3% and 40.3%, respectively. The corresponding standard deviations narrow from 0.545 to 0.407 and from 0.534 to 0.413 percentage points. Agreement between the two preparations makes the trend more credible than either archive alone, although three sizes do not determine an asymptotic exponent.

Evidence class: two canonical $S = 100$, $N_x = 20$ trajectory-resolved charge campaigns. **What it does not establish:** σ_ν^2 is sample-to-sample variance of normalized total filling, not the within-trajectory quantum variance $V_Q = \langle Q^2 \rangle - \langle Q \rangle^2$ and not proof that every trajectory is identical.

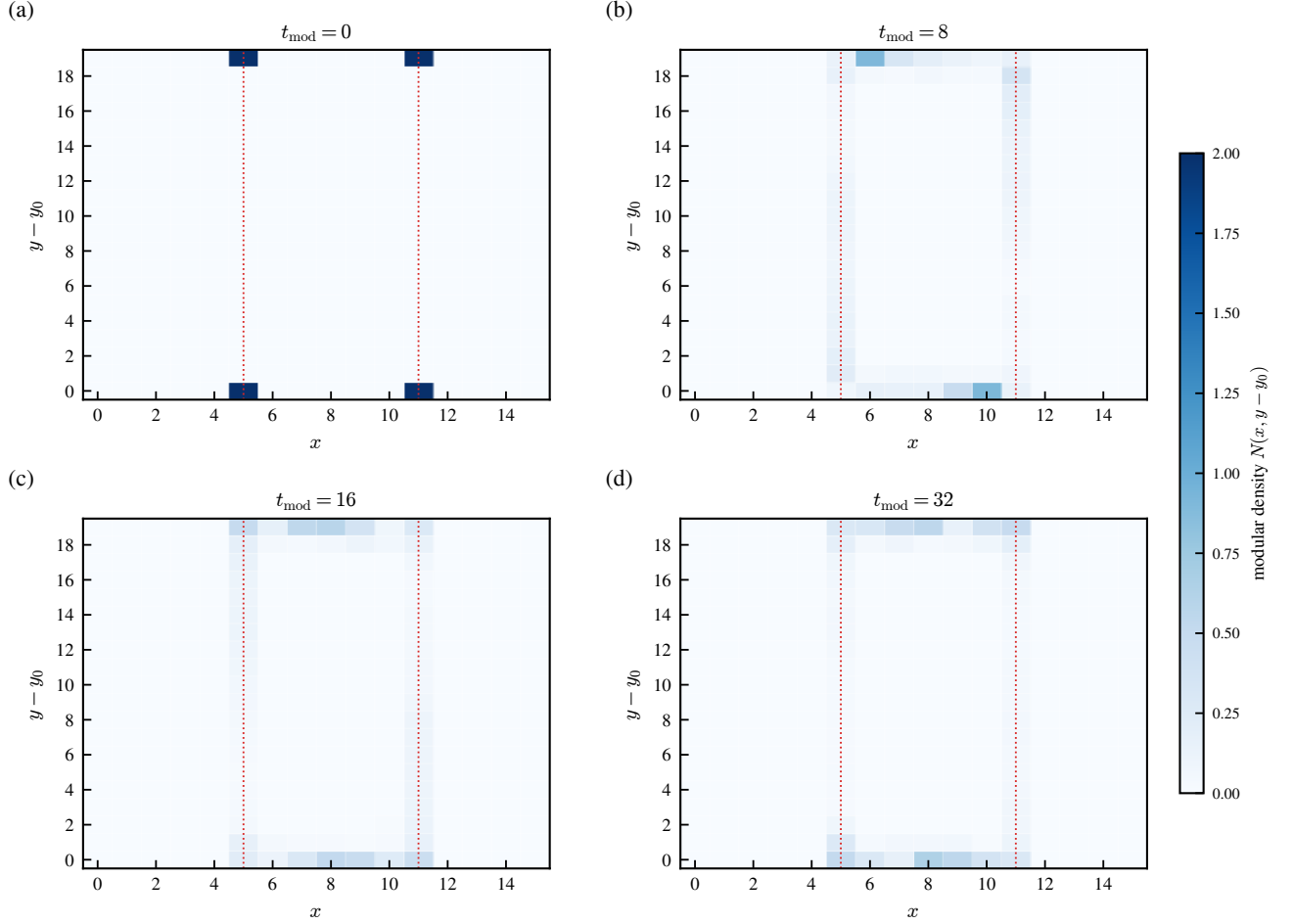


FIG. 14. Modular charge-density snapshots at $t_{\text{mod}} = 0, 8, 16, 32$. The horizontal coordinate is the physical transverse direction; the vertical coordinate is the periodic position relative to the half-strip origin. Red dotted lines mark the two domain walls; the fine white mesh delineates individual unit cells, with integer centers labeled every two sites. These are cycle-50, 16×40 , $S = 10$ covariance snapshots propagated by Eq. (A27), with translated origins averaged within each record before the ensemble mean. They visualize modular, not physical-time, density propagation and carry no independent scalar error overlay.

H. Result 18: an independent pure-state archive checks entropy and correlations

The Colab audit found a second physical-state archive that had not been represented in the atlas. Canonical perfect-correction covariance snapshots at $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, $S = 10$, and $C = 50$ support both entropy and correlation analyses. The late-window full-width fits give

	$n_{\text{shell}} = 1$	$n_{\text{shell}} = 2$
$N_y = 30$	$c_{\text{eff}} = 1.04583(13)$	$1.05513(16)$
$N_y = 40$	$c_{\text{eff}} = 1.03244(22)$	$1.08650(21)$

(27)

with $R^2 \geq 0.99995$. The quoted entropy uncertainties are regression standard errors over the fit points, not trajectory errors. On the declared correlator window $2 \leq r_y \leq N_y/4$, the two-wall-window exponents are 2.154(36), 2.126(33), 2.191(36), and 2.092(34) in the same order;

the nonwall controls instead lie between 4.11 and 6.37. Thus the archive independently supports a wall-localized algebraic sector at both shell depths. The endpoint map in Fig. 18(c) also makes the limitation visible: the numerical exponent moves substantially under arbitrary endpoint choices, so this remains a selected-window cross-check rather than a precision universal-exponent determination.

Evidence class: canonical $S = 10$ perfect-correction covariance snapshots with independent entropy and correlator analyses. **What it does not establish:** Only two circumferences are present. A high R^2 at one declared window does not remove the explicit fit-window sensitivity, and the regression errors are not a whole-trajectory finite-size uncertainty.

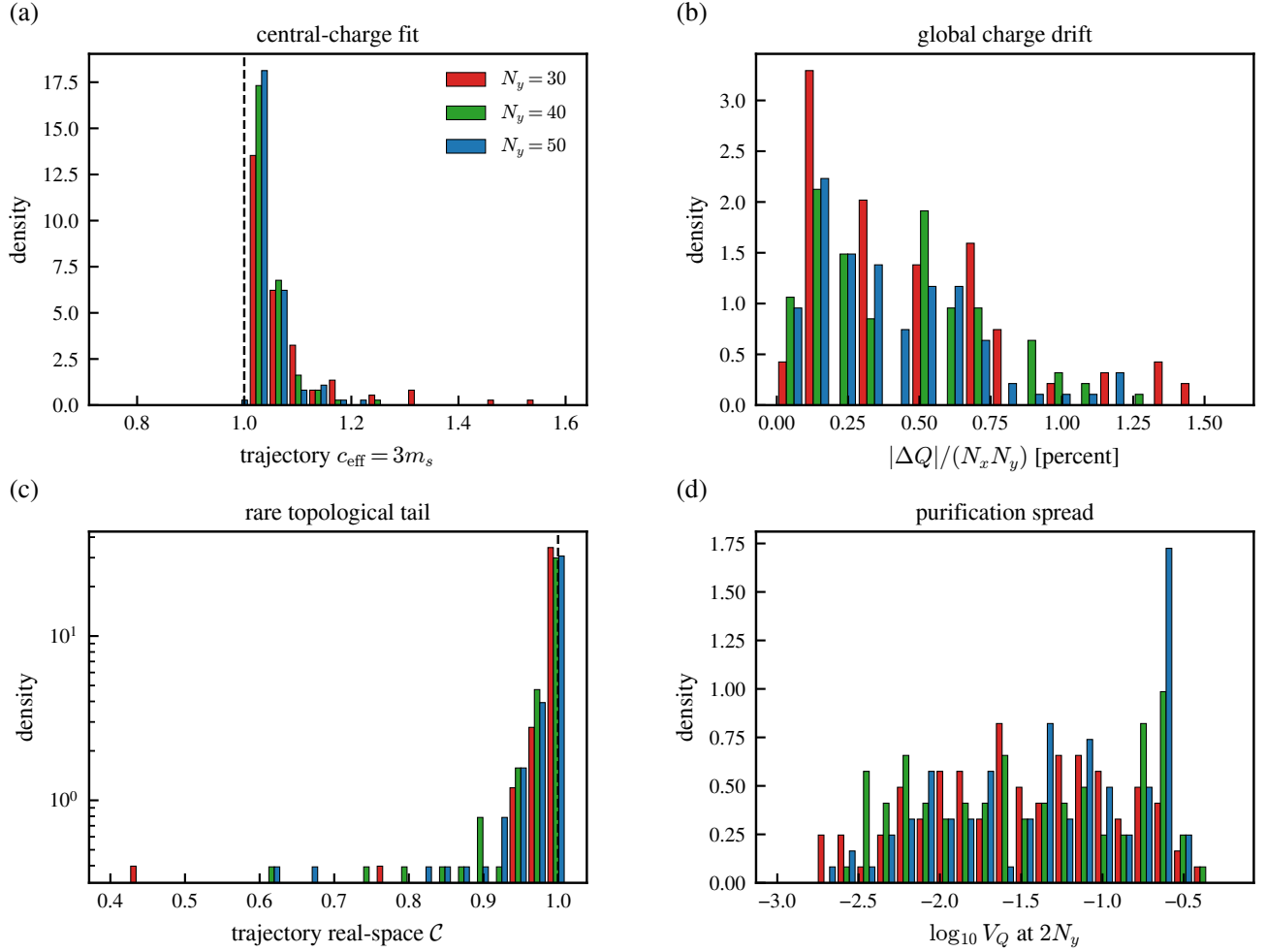


FIG. 15. Trajectory distributions at $C_{\text{cyc}} = 2N_y$: cycles $C_{\text{cyc}} = 60, 80, 100$ for $N_y = 30, 40, 50$, respectively. (a) Effective central charge from the per-trajectory entropy slope. (b) Absolute global charge drift. (c) Real-space Chern response on a logarithmic density axis to expose rare low-Chern records. (d) Final intrinsic charge variance during max-mix purification. All panels use side-by-side filled histogram bars with narrow inter-bin spacing and black outlines; colors identify circumference as in the legend. Each bar sample is one trajectory scalar from $S = 100$ at $N_x = 20$, with cycles 60, 80, 100 for $N_y = 30, 40, 50$. Panels (a–d) use Eqs. (10), (25), (A17), and (11). The pure-state arm is perfect correction with $n_{\text{shell}} = 1$ and `dw_truncation=True`; pure-state and max-mix records are separate ensembles. The histograms carry no separate error bars; their recordwise spread is the displayed distribution. The tails preclude an all-records-equivalent claim.

I. Result 19: feedback reshapes the finite particle–Choi transfer mode

The mass-sweep Colab folder contains more than the scalar gaps in Result 8. Its executed mode analysis maps each saved reduced-basis eigenvector back to physical unit cells. At $N_y = 40$ and $\alpha_{\text{in}} = 1$, perfect correction raises the mean probability within one column of either reduced-slab edge from 0.45187 to 0.87933 and lowers the mean edge distance from 1.9944 to 0.5745 cells. The site participation ratio simultaneously falls from 156.24(728) to 56.72(278). Along the perfect-correction branch the edge weight decreases to 0.65428 at $\alpha = 1.5$ and 0.46673 at 1.9, while the mean edge distance grows to 1.2694 and

1.9432. The common-scale unit-cell maps in Fig. 19(c,d) show the same feedback-induced localization directly.

The terminology matters. The legacy arrays call these “finite Lyapunov modes,” but they are eigenmodes of the conditioned particle–Choi transfer construction in a reduced topological slab basis. They are not the covariance-tangent Oseledec modes used in Result 2. This classification is why the result strengthens the spatial interpretation of the S2 transfer precursor without being promoted into a thermodynamic tangent claim.

Evidence class: canonical GPU $S = 10$ finite particle–Choi transfer eigenvectors; 7,151 parsed finite-mode rows with no parse failures. **What it does not establish:** The basis is a conditioned reduced Choi slab, and finite-

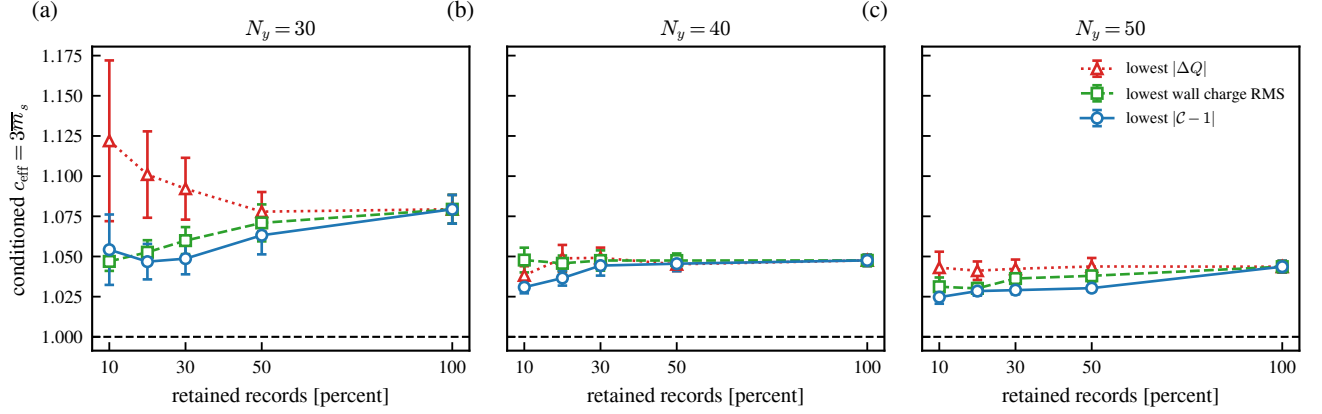


FIG. 16. Conditioned effective central charge versus retained record fraction. Each panel fixes a circumference; curves rank trajectory by global charge drift, wall-charge RMS, or real-space Chern error. Error bars are standard errors of the retained subset (some are smaller than the markers), and the dashed line is $c = 1$. Every point is an unweighted recomputation from the $S = 100$ final-cycle feature table after retaining the displayed rank fraction, with uncertainty defined by Eqs. (25) and (A32). The table uses $N_x = 20$, $N_y = 30, 40, 50$, $C_{\text{cyc}} = 2N_y$, $n_{\text{shell}} = 1$, `dw.truncation=True`, and perfect correction. This is a post hoc sensitivity analysis, not partial post-selection or a redefined physical ensemble.

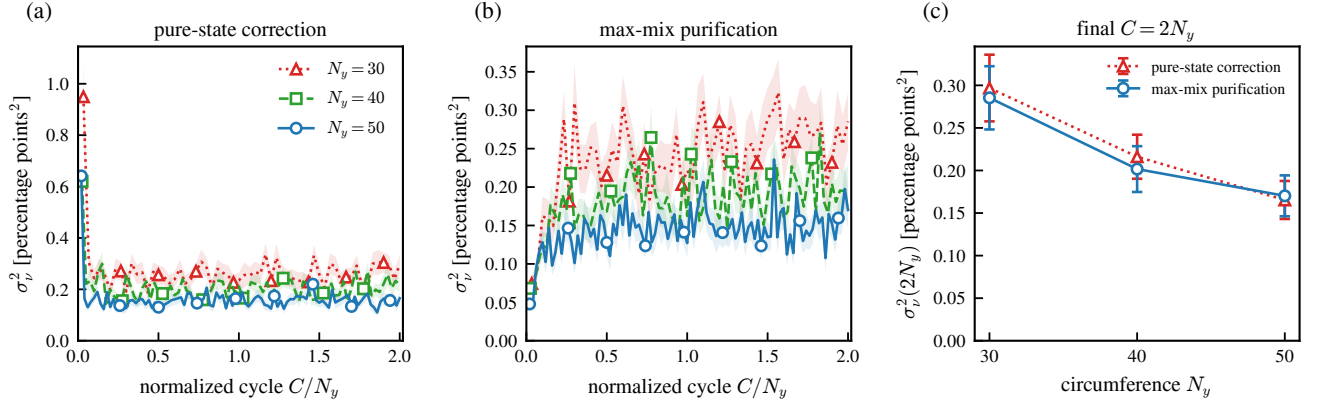


FIG. 17. Trajectory-to-trajectory filling variance. (a) Pure-state perfect correction and (b) maximally mixed purification versus normalized cycle C_{cyc}/N_y . Curves show the population variance across $S = 100$ records; translucent bands are leave-one-record-out jackknife standard errors and can be narrower than the lines. (c) Final values at $C_{\text{cyc}} = 2N_y$, i.e., cycles 60, 80, 100 for $N_y = 30, 40, 50$. Error bars are the same jackknife standard errors from Eqs. (26) and (A33). Both preparations use $N_x = 20$, $S = 100$, $n_{\text{shell}} = 1$, `dw.truncation=True`, and perfect correction; the plotted quantity is the population variance of normalized trajectory filling, not the intrinsic quantum variance in Eq. (A5), and three sizes do not fix an asymptotic exponent.

sector loss near the transition is censoring. The maps do not establish a physical tangent eigenmode or a new bulk topological invariant.

J. Result 20: correct feedback was run on the untruncated overlapping interface

The Colab charge-sharpening archive contains a completed construction that was missed in the earlier W1 inventory. Every untruncated run keeps `DW=True`, fixes the outer controller at $\alpha_{\text{out}} = 30$, and sets `dw.truncation=False`; the inner parameter is swept

through $\alpha_{\text{in}} = 1, 1.5, 1.8, 1.9, 2, 2.1, 2.2, 2.5, 3$. Thus the two spatial interfaces remain present while the controller projectors are allowed to overlap them. This is not a homogeneous `DW=False` control. The completed perfect-correction arm has $N_x = 16$, $N_y = 16, 24, 32$, ten trajectories per parameter point, and $T = 2N_y$: 27 configurations and 270 trajectories in total.

Mathematically, both constructions retain the same piecewise controller,

$$\alpha(x) = \begin{cases} \alpha_{\text{in}}, & x \in \mathcal{I}_{\text{top}}, \\ \alpha_{\text{out}} = 30, & x \in \mathcal{I}_{\text{triv}}, \end{cases} \quad (28)$$

and differ only in whether controller support is terminated

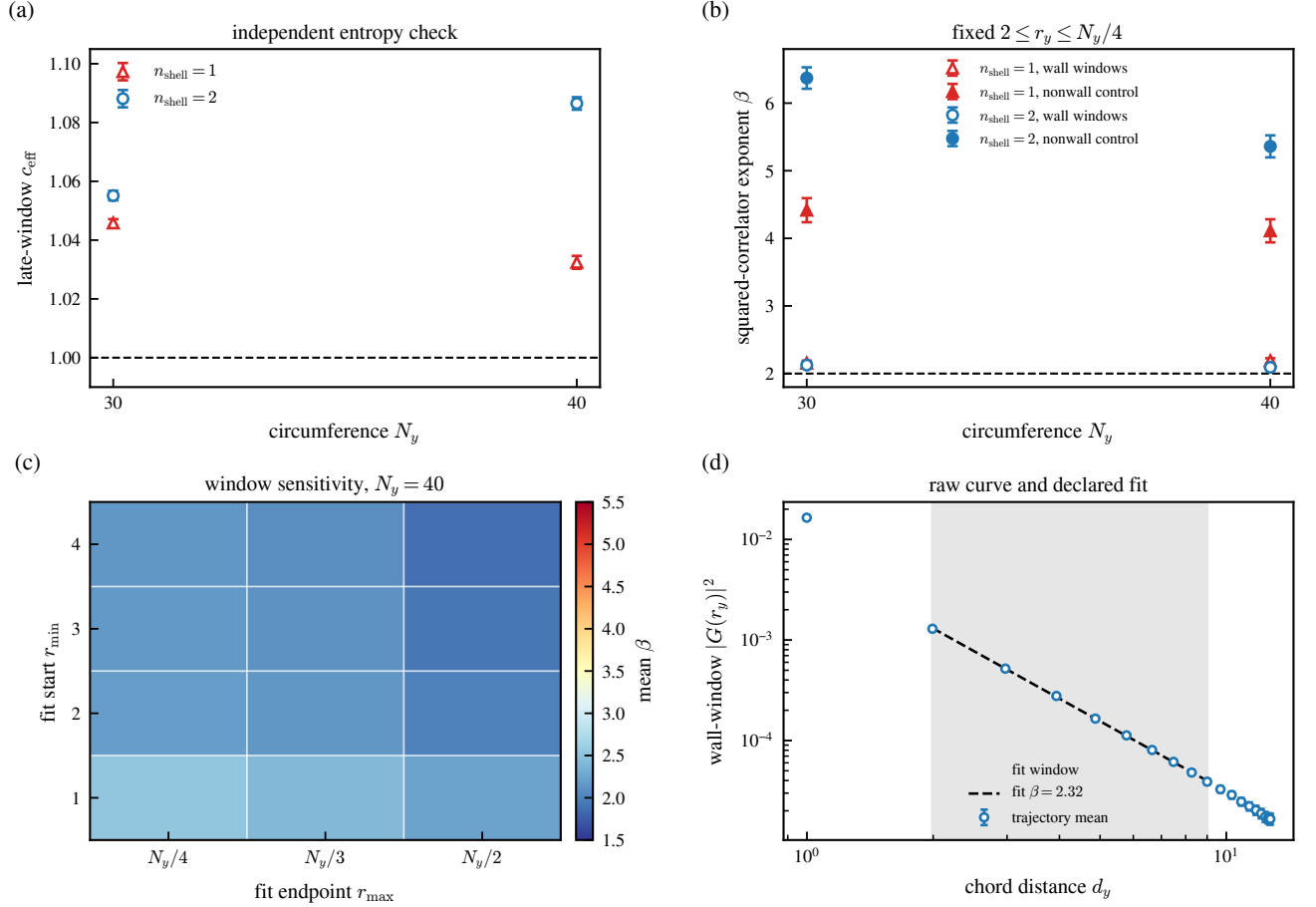


FIG. 18. Independent pure-state cross-check. (a) Late-window effective central charge at two shell depths; empirical points are not connected and the dashed line is $c = 1$. (b) Squared-correlator exponents for the two wall windows and a matched nonwall control on the fixed $2 \leq r_y \leq N_y/4$ interval. (c) Mean wall exponent as the lower and upper fit endpoints are varied; each rectangular cell is one explicitly labeled endpoint pair. (d) The $N_y = 40$, $n_{\text{shell}} = 1$ trajectory-mean wall curve. Points are unconnected, the gray region is the declared fit window, and the dashed line is the fit. Error bars in (a) are regression standard errors; those in (b) and (d) are trajectory standard errors and can be smaller than the markers. All inputs are cycle-50, $N_x = 16$, $S = 10$ perfect-correction covariances at $N_y = 30, 40$ and $n_{\text{shell}} = 1, 2$; Eqs. (A1), (10), and (A3) define the reductions. Only two sizes are present, and panel (c) exposes rather than hides fit-window sensitivity.

at the interfaces. To avoid mistaking the larger full-layer support for a physical enhancement, the construction comparison uses

$$\begin{aligned} \overline{O}_\gamma &= O_\gamma / N_{\text{mode}}^\gamma, \\ R_O(\alpha, N_y) &= \overline{O}_{\text{untruncated}} / \overline{O}_{\text{terminated}}, \\ O &\in \{S_{\text{tot}}, \text{Tr}[C(\mathbf{1} - C)]\}. \end{aligned} \quad (29)$$

At $N_y = 32$, the untruncated interface has final entropy 0.10488, 0.26286, and 6.06×10^{-5} at $\alpha_{\text{in}} = 1, 1.5, 1.9$, respectively, and is at the numerical purity floor by $\alpha_{\text{in}} = 2$. Its corresponding intrinsic charge variances are 0.03279, 0.07807, and 4.98×10^{-6} . The matched support-terminated values at the first three parameters are entropy 0.18509, 0.45161, and 0.06416, and charge variance 0.04626, 0.13334, and 0.02241. Figure 20 therefore shows both the common crossover near $\alpha = 2$ and the

generally faster subcritical purification of the overlapping construction. Because the active supports differ, panels (c,d) also report ratios per measured mode rather than comparing raw totals alone.

Evidence class: canonical CPU perfect-correction interface comparison; no post-selection arm is used. **What it does not establish:** The campaign saves purification and charge-sharpening observables on the full top layer, but not Chern profiles, log-chord entropy, tangent localization, fitted interface leakage, or opposite-wall overlap. It is a strong W1 construction precursor, not the preregistered W1 equivalence gate.

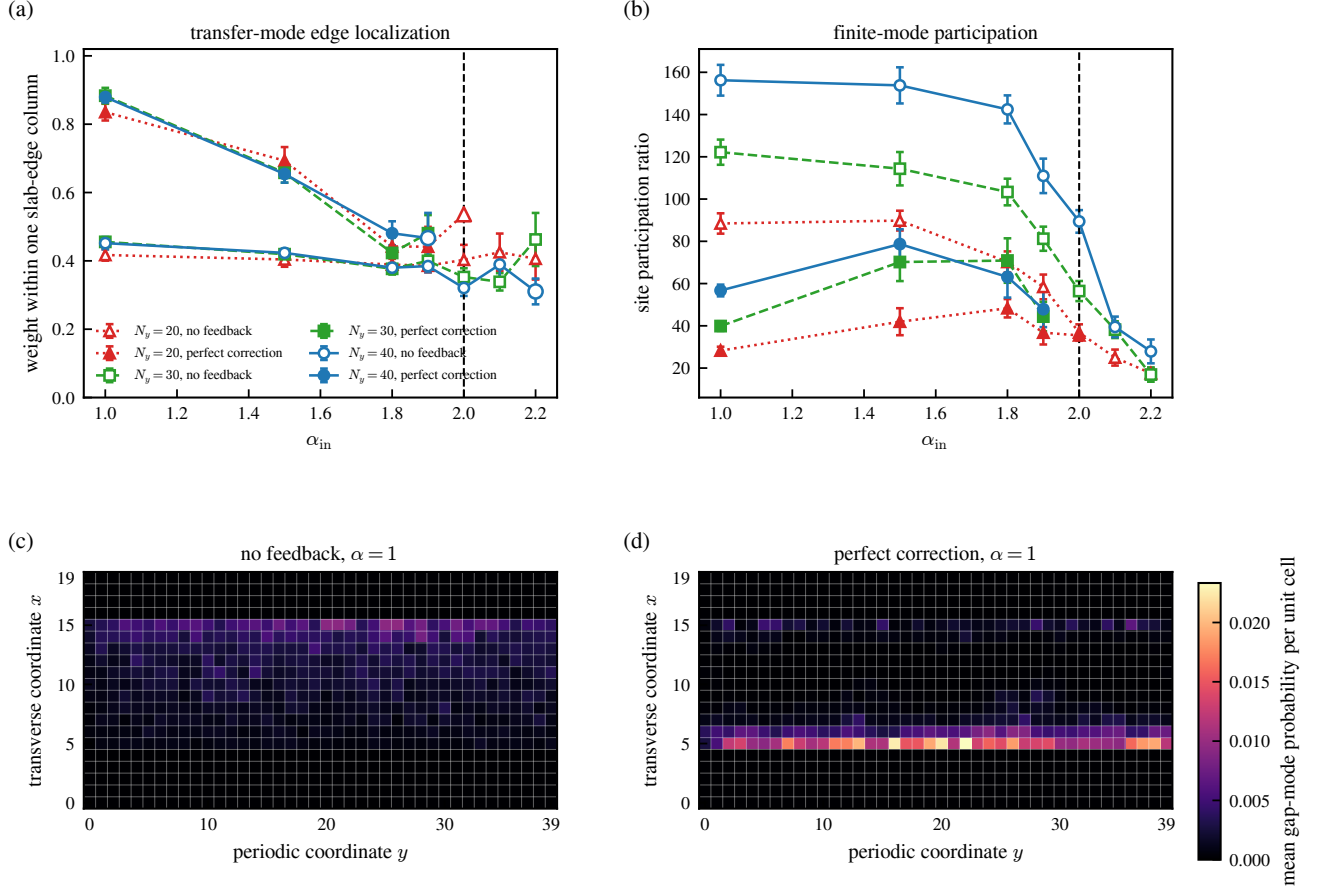


FIG. 19. Finite particle-Choi transfer-mode structure. (a) Mean probability within one column of either reduced-slab edge and (b) site participation ratio versus α_{in} . Filled markers denote perfect correction; open markers denote no feedback. Bars are trajectory standard errors and can be smaller than the markers; open overplots at censored points denote fewer than ten finite samples. (c,d) Ten-trajectory mean gap-mode probability per physical unit cell at $N_y = 40$ and $\alpha = 1$, plotted with one common color scale. Thin white lines mark every unit-cell boundary exactly; the horizontal coordinate is periodic y and the vertical coordinate is transverse x . All arms use $N_x = 20$, $N_y = 20, 30, 40$, $S = 10$, $C_{\text{cyc}} = 2N_y$, $n_{\text{shell}} = 1$, and `dw_truncation=True`; filled and open series distinguish perfect correction and no feedback. The saved reduced slab is $x = 5, \dots, 15$ (11 columns). Equations (A24), (A25), and (A26) define the reduced-basis mode and its statistics. These are conditioned particle-Choi modes; censored finite sectors are not infinite gaps, and the maps are not covariance-tangent Oseledec vectors.

K. Result 21: critical subsystem charge fluctuations resolve the current sector

Figure 21 presents the raw interval fit, trajectory-resolved coefficients, matched entropy comparison, and coefficient distributions. The raw covariance snapshots underlying Result 18 also determine an observable that had not been reduced in the archived characterization notebooks: the quantum fluctuation of charge in a full- x periodic interval. For a conditioned number-conserving Gaussian trajectory with $C = (G + \mathbf{1})/2$ and interval A ,

$$F_{A,\xi}^q = \text{Var}_\xi(Q_A) = \text{Tr}[C_{A,\xi}(\mathbf{1}_A - C_{A,\xi})]. \quad (30)$$

This is evaluated separately for every trajectory and averaged over all periodic origins only within that trajectory. The calculation does not use the classical variance of the trajectory means, and it is distinct from the full-system

intrinsic variance used to diagnose max-mix purification.

The origin average can be evaluated directly from covariance blocks. With $d = 2N_x$, $\bar{n} = N_y^{-1} \text{Tr } C$, and $K_r = N_y^{-1} \sum_y \|C_{y,y+r}\|_F^2$,

$$\bar{F}_\xi^q(A_y) = A_y(\bar{n} - K_0) - \sum_{r=1}^{A_y-1} (A_y - r)(K_r + K_{-r}). \quad (31)$$

The notebook checks Eq. (31) against explicit restricted submatrices at three interval lengths in every configuration. For a two-wall charged CFT,

$$\bar{F}^q(A_y) = \frac{k}{\pi^2} \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + b + \dots, \quad (32)$$

$$k_{\text{eff}} = \pi^2 m_F.$$

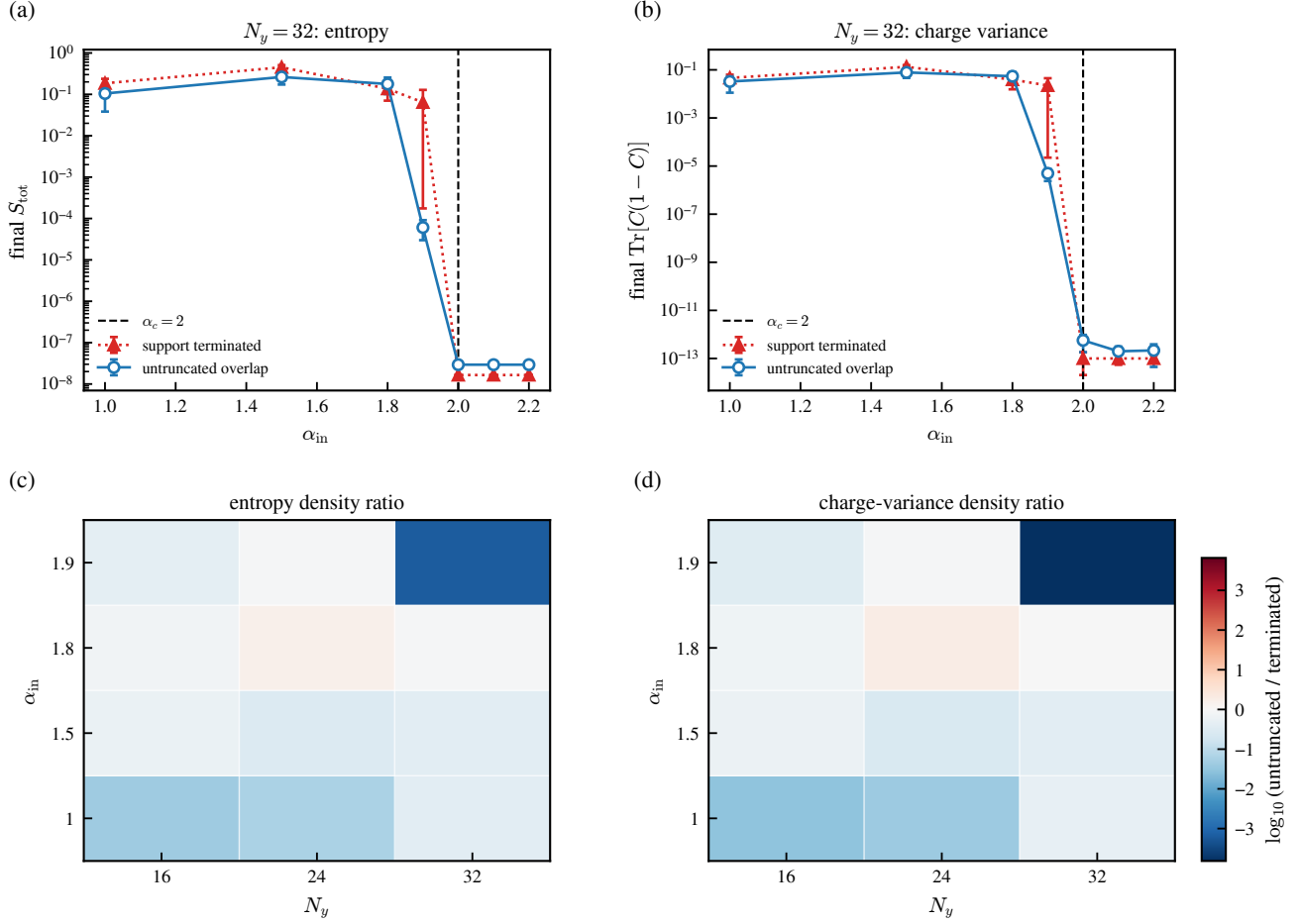


FIG. 20. Perfect-correction comparison of the support-terminated and untruncated overlapping interfaces. (a,b) Final entropy and intrinsic charge variance at $N_y = 32$; bars are trajectory standard errors and are often smaller than the markers. Lines only guide the eye between the discrete simulated α_{in} values, the black dashed line marks $\alpha_c = 2$, and values at the right edge reach the numerical floor. (c,d) Base-ten logarithm of the untruncated to support-terminated ratio for entropy and charge variance per measured mode over the subcritical sweep. White grid lines mark individual $(\alpha_{\text{in}}, N_y)$ configurations exactly, and both heat maps share one color scale. Both constructions keep `DW=True`, use perfect correction for $2N_y$ cycles and $S = 10$, and differ only in `dw.truncation`; Eqs. (28) and (29) define the comparison. It tests purification of an overlapping interface, not topological equivalence of the two constructions.

Fitting $8 \leq A_y \leq N_y/2$ at cycle 50 gives

	$n_{\text{shell}} = 1$	$n_{\text{shell}} = 2$
$N_y = 30$	$k_{\text{eff}} = 1.0409(166)$	$1.0468(106)$
$N_y = 40$	$k_{\text{eff}} = 1.0328(346)$	$1.0853(449)$

(33)

where parentheses are trajectory standard errors from ten independently fitted records. The ensemble-mean curves have $R^2 > 0.99996$. These levels track the matched entropy values $c_{\text{eff}} = 1.0458, 1.0551, 1.0324, 1.0865$ in the same order. Thus the saved states contain a direct, near-level-one charge-sector signal rather than only an entropy coefficient compatible with $c = 1$.

Evidence class: canonical GPU perfect-correction raw covariance snapshots; $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, ten trajectories, cycle 50; no partial post-selection. **What it does not establish:** The charge and entropy esti-

mates use the same four trajectory ensembles, so their agreement is a matched-observable test rather than an independent replication. Only two circumferences and ten trajectories per configuration are available; there is no thermodynamic extrapolation, one-wall charge decomposition, higher-cumulant analysis, or independent chirality test in this result.

L. Result 22: a three-by-three OW window preserves the target-band overlap obstruction

Figure 22 displays the retained norm, localization scales, overlap zeros, and the distinction between overlap winding and compact-spinor topology. The circuit measures compact overcomplete-Wannier (OW) modes, so the finite

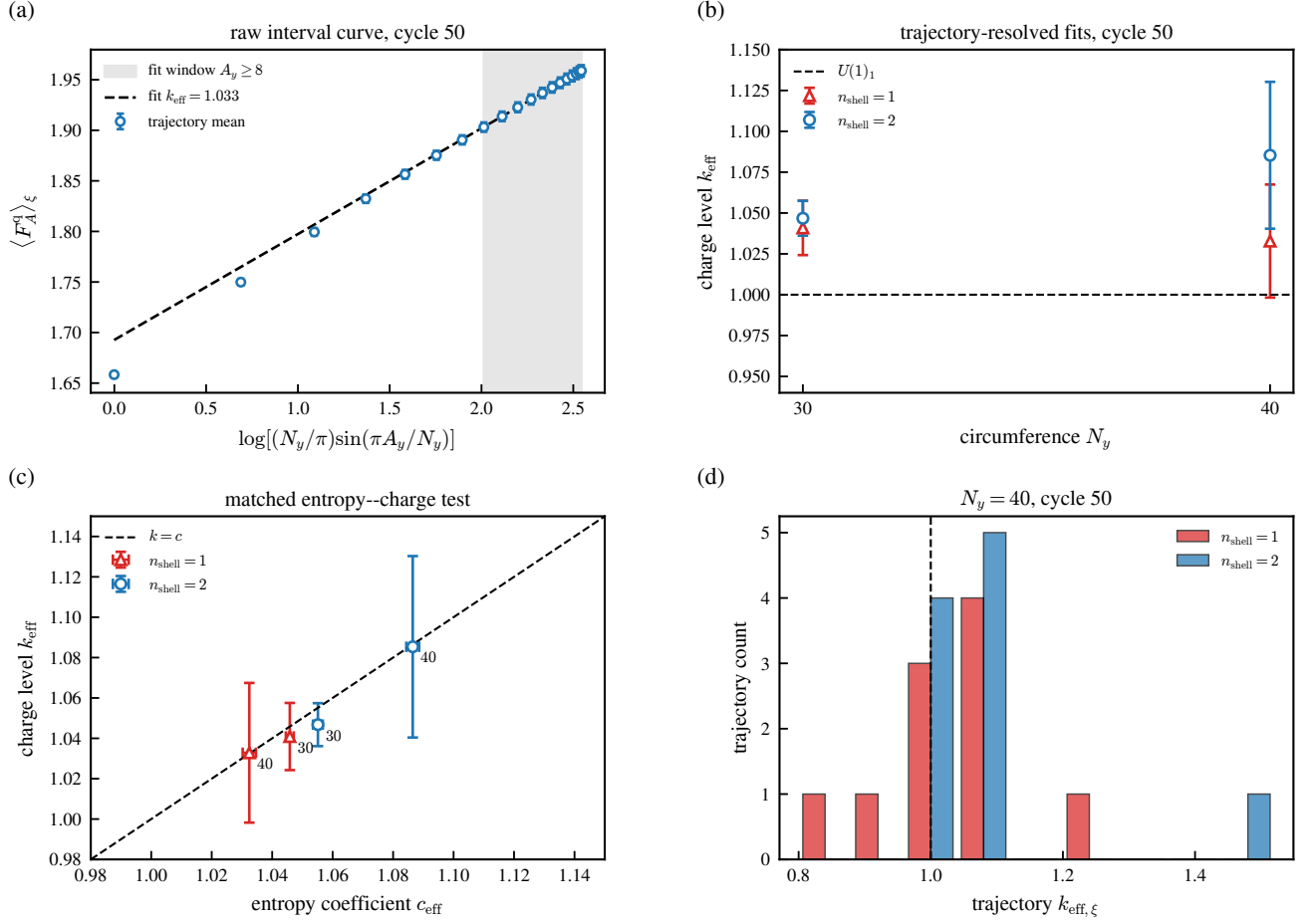


FIG. 21. Critical subsystem charge fluctuations from canonical perfect-correction covariance snapshots. Every panel uses cycle 50. (a) Mean full- x interval variance for $N_y = 40$, $n_{\text{shell}} = 1$; points are not connected, the gray region is the declared $A_y \geq 8$ fit window, and the dashed fitted line is extended across the full measured range. Bars are trajectory standard errors and are mostly smaller than the markers. (b) Trajectory-resolved k_{eff} versus size; bars are standard errors over ten trajectory fit coefficients. (c) Matched charge and entropy coefficients from the same ensembles; horizontal bars are regression errors over entropy fit points and vertical bars are trajectory standard errors. Numerals label N_y . (d) Solid, black-outlined histograms of the ten $N_y = 40$ trajectory coefficients; the dashed references in all applicable panels mark the level-one or $k = c$ target. Equations (A5)–(32) define the cycle-50, $N_x = 16$, $S = 10$ reductions for $N_y = 30, 40$ and $n_{\text{shell}} = 1, 2$. Origins are averaged inside each record before fitting; the charge/entropy match uses the same ensembles and is not an independent replication.

support must be checked directly against the target Chern band rather than inferred from a single localization length. For trial spinor $|\tau\rangle$, band form factor $f(\mathbf{k}) = \langle u_-(\mathbf{k}) | \tau \rangle$, and real-space mask g_n , the windowed mode has the twisted-convolution overlap

$$f_n^{\text{eff}}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} \tilde{g}_n(\mathbf{k}' - \mathbf{k}) f(\mathbf{k}) \langle u_-(\mathbf{k}') | u_-(\mathbf{k}) \rangle. \quad (34)$$

Its zero and local phase winding test whether the finite measurement mode still encounters the target-band obstruction. This diagnostic must be distinguished from the Chern number of the compact-support spinor

$$|\psi_n(\mathbf{k})\rangle = \sum_{\mathbf{r} \in \text{supp}(g_n)} e^{-i\mathbf{k} \cdot \mathbf{r}} C_{\mathbf{r}} |\tau\rangle. \quad (35)$$

When $|\psi_n(\mathbf{k})\rangle$ is smooth, periodic, and nowhere zero, it is a global section and therefore has $C_{\psi_n} = 0$; its topology is not the overlap winding.

For a normalized unwrapped OW mode centered at $\mathbf{R}$, the RMS spread, its topological lower bound, and the asymptotic tail length are distinct quantities:

$$\begin{aligned} \xi_{\text{rms}}^2 &= \frac{\sum_{\mathbf{r}} |\mathbf{r} - \mathbf{R}|^2 \|\chi_{\mathbf{R}}(\mathbf{r})\|^2}{\sum_{\mathbf{r}} \|\chi_{\mathbf{R}}(\mathbf{r})\|^2}, \\ \xi_{\text{rms}} &\geq \xi_{\text{top}}, \\ \xi_{\text{top}} &= \sqrt{2|C|} \ell, \quad 2\pi\ell^2 = A_{\text{uc}}. \end{aligned} \quad (36)$$

The inequality is the coherent-state spread bound of Li, Dong, Ledwith, and Khalaf [2]. Independently,

$$\|\chi_{\mathbf{0}}(\mathbf{r})\| \sim e^{-|\mathbf{r}|/\xi_{\text{exp}}} \quad (|\mathbf{r}| \rightarrow \infty), \quad (37)$$

up to algebraic prefactors, defines the exponential tail length. Thus ξ_{top} is a lower bound on the unwound RMS spread, not a radius that certifies a hard truncation, and ξ_{exp} controls the far tail rather than the second moment. For the documented uniform $\alpha = 1$ two-band projector and X trial spinor, $A_{\text{uc}} = a^2$, $|C| = 1$, and the three scales are $\xi_{\text{top}} = 0.56419a$, $\xi_{\text{rms}} = 0.67352a$, and $\xi_{\text{exp}} = 1.44270a$. The measured RMS spread is therefore about 19.4% above, rather than at, the rigorous lower bound. Direct Fourier reconstruction gives

n_{shell}	retained norm	$k_x^{(0)}/\pi$	ν_n
0	0.620624	0.209346	-1
1	0.992155	0.492186	-1
2	0.999137	0.492488	-1

(38)

compared with the unwound zero at $k_x/\pi = 1/2$ on $k_y = 0$. Thus the practical $n_{\text{shell}} = 1$ three-by-three mask removes less than one percent of the OW norm and only weakly displaces the obstruction zero. A stable Fukui–Hatsugai–Suzuki evaluation gives $C_{\psi_n} = 0$ for $n_{\text{shell}} \leq 2$, as required. Apparent nonzero values for larger masks on a coarse grid are not used: the compact-spinor minimum norm is already 2.05×10^{-3} at $n_{\text{shell}} = 4$ and 3.36×10^{-5} at $n_{\text{shell}} = 8$, so that calculation does not resolve the near-singular limit.

Evidence class: deterministic reconstruction of the uniform- $\alpha = 1$ form-factor model; projector Fourier grid, real-space norm, continuum overlap zeros, local winding, and the stable compact-spinor FHS check are all recomputed in the sole atlas notebook. **What it does not establish:** This is a controller-design check for one model, trial spinor, gauge, and mask family. It contains no domain wall, disorder, conditioned trajectory, or dynamical observable, and it does not establish the B1 static-frame prediction criterion.

M. Result 23: overlap winding follows the target Bloch-bundle phase

Figure 23 shows the resolved overlap zeros and winding through the topological and trivial mass values in the sweep. The winding in Result 22 is not merely the visual circulation around a convenient dip. Let $L \rightarrow T^2$ be the occupied-band line bundle and choose local frames related on patch overlaps by $|u_j\rangle = e^{i\chi_{ij}}|u_i\rangle$. For any smooth periodic trial field $|\phi_n(\mathbf{k})\rangle$, including the unwound trial spinor and either finite OW mask,

$$f_{n,i}(\mathbf{k}) = \langle u_i(\mathbf{k}) | \phi_n(\mathbf{k}) \rangle, \quad f_{n,j} = e^{-i\chi_{ij}} f_{n,i}. \quad (39)$$

Thus f_n is a section of the dual bundle L^* . If its zeros are isolated, their local indices may be related to C directly, without taking a zero-index theorem as a black box. Delete counterclockwise disks D_a around the zeros and set $M = T^2 \setminus \bigcup_a D_a$. On M the normalized projected trial state

$$|v_n\rangle = \frac{P_- |\phi_n\rangle}{\|P_- |\phi_n\rangle\|} = e^{i \arg f_{n,i}} |u_i\rangle \quad (40)$$

is a gauge-independent, single-valued occupied-band frame. If $A_i = -i\langle u_i | du_i \rangle$, its connection is $A_v = A_i + d \arg f_{n,i}$ and has curvature $F = dA_i$. Stokes' theorem on the punctured torus therefore gives

$$\begin{aligned} 2\pi C &= \lim_{D_a \rightarrow 0} \int_M F = \lim_{D_a \rightarrow 0} \oint_{\partial M} A_v \\ &= - \sum_a \lim_{D_a \rightarrow 0} \oint_{\gamma_a} (A_i + d \arg f_{n,i}) = -2\pi \sum_a \nu_a. \end{aligned} \quad (41)$$

Each hole occurs clockwise in ∂M , producing the minus sign, while the integral of smooth A_i vanishes as its disk shrinks. Hence

$$\sum_a \nu_a = \sum_a \frac{1}{2\pi} \oint_{\gamma_a} d \arg f_n = -C. \quad (42)$$

A nonzero target-band Chern number therefore forces overlap zeros even though a smooth, nonvanishing compact spinor $|\phi_n\rangle$ defines a trivial bundle of its own. In particular, the $n_{\text{shell}} = 0$ compact spinor has Chern number zero, but its overlap has winding -1 when the target band has $C = +1$. When $C = 0$, accidental zero–antizero pairs remain possible, but their total index must vanish.

The numerical implementation first evaluates the gauge-invariant magnitude

$$|f_n(\mathbf{k})|^2 = \langle \phi_n(\mathbf{k}) | P_- (\mathbf{k}) | \phi_n(\mathbf{k}) \rangle \quad (43)$$

on a periodic 181×181 grid. It identifies all discrete local minima, refines the lowest 32 candidates by periodic two-dimensional Nelder–Mead minimization, and merges coincident solutions. A refined residual below 10^{-6} is classified as a zero. For each zero, one reference spinor $|r_a\rangle$ is chosen at its center and held fixed on a radius-0.02 loop, defining the smooth local frame

$$\begin{aligned} |u_a^{\text{loc}}(\mathbf{k})\rangle &= \frac{P_- (\mathbf{k}) |r_a\rangle}{\|P_- (\mathbf{k}) |r_a\rangle\|}, \\ \nu_a &= \text{round} \left[\frac{1}{2\pi} \left\{ \text{unwrap } \arg f_n(\mathbf{k}(\theta)) \right\}_{\theta=0}^{2\pi} \right]. \end{aligned} \quad (44)$$

The loop uses 721 angular samples. An independent FHS surface integral supplies C for the target lower band. A positive refined global minimum is reported as a gapped overlap, not assigned a spurious winding around an arbitrary minimum.

For the requested sweep the result is

α	C	ν_{none}	ν_0	ν_1	ν_2
1	+1	-1	-1	-1	-1
1.5	+1	-1	-1	-1	-1
3	0	0	0	0	0
30	0	0	0	0	0

(45)

Here “none” means no real-space truncation. The zero locations $k_x^{(0)}/\pi$ for (none, 0, 1, 2) are (0.500000, 0.209346, 0.492186, 0.492488) at $\alpha = 1$ and (0.333333, 0.064606, 0.346511, 0.345054) at $\alpha = 1.5$,

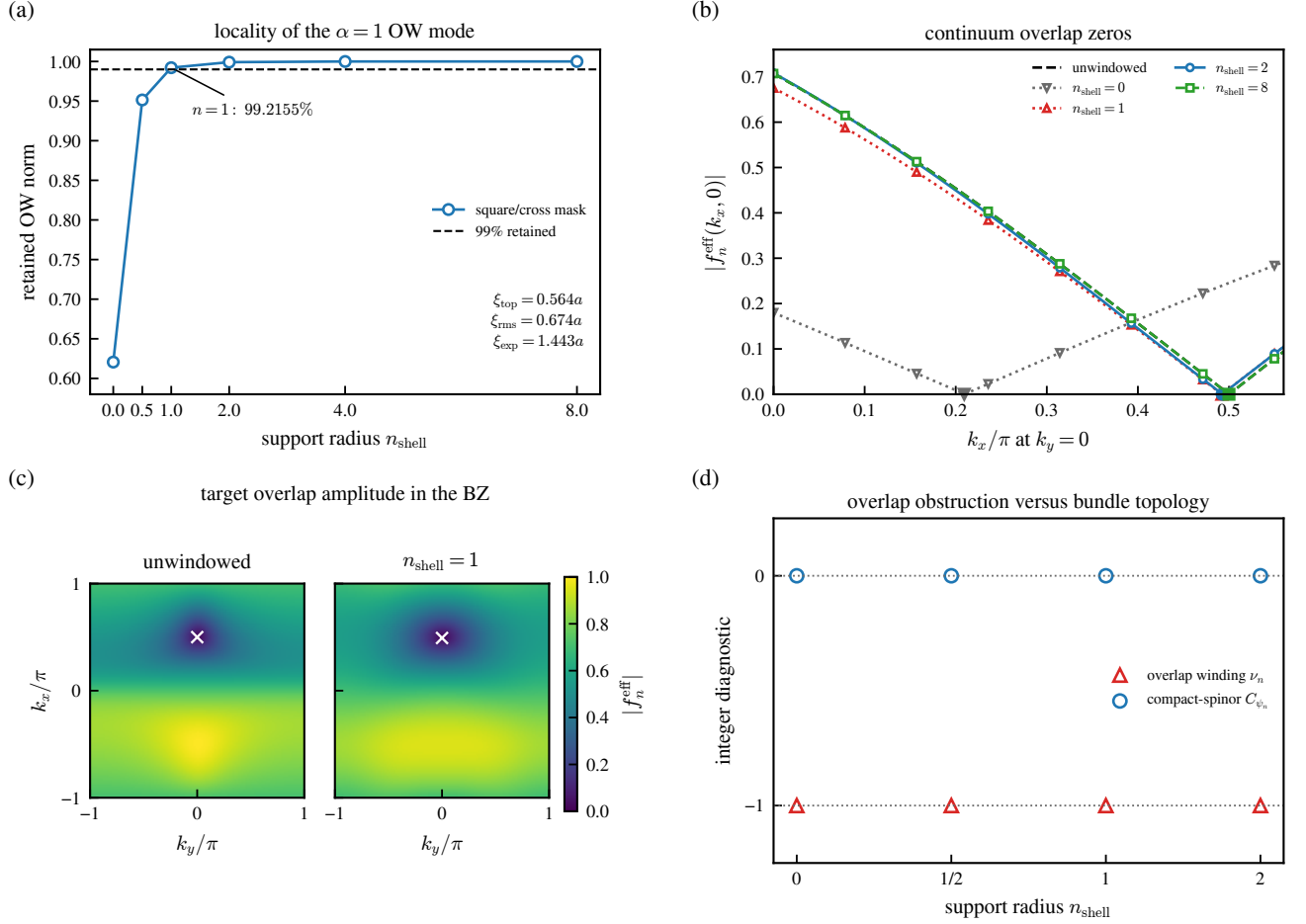


FIG. 22. Finite-shell OW form-factor audit for the uniform $\alpha = 1$ projector and X trial spinor. (a) Fraction of projected-mode norm retained by each square or cross mask, together with the topological, RMS, and exponential localization scales defined in Eqs. (36) and (37). These are properties of the unwound mode, not hard window radii. (b) $k_y = 0$ cuts of the target-band overlap; symbols mark the continuum-minimized zeros, and lines show the corresponding overlap functions rather than fits. (c) Unwound and $n_{\text{shell}} = 1$ overlap-amplitude maps over the Brillouin zone; white crosses mark the zeros. (d) The target-band overlap winding remains $\nu_n = -1$ while the smooth compact-support spinor has $C_{\psi_n} = 0$. Coarse-grid FHS values at $n_{\text{shell}} = 4, 8$ are deliberately omitted because the spinor is too close to singular for a stable bundle diagnostic. The deterministic uniform- $\alpha = 1$ calculation uses Eqs. (36), (34), (A21), and (42); lines in (b) are functions, not fits. The result tests one X trial spinor and mask family, not a domain-wall trajectory.

with $|k_y^{(0)}|/\pi < 4 \times 10^{-9}$. All eight phase-circulation calculations give one isolated zero with $\nu = -1$. The $n_{\text{shell}} = 0$ compact spinor therefore has $C_{\psi_0} = 0$ while its overlap has turn number -1 : the two invariants are not the same. At $\alpha = 3, 30$, the respective global overlap minima for (none, 0, 1, 2) are (0.500000, 0.652414, 0.525837, 0.499765) and (0.694808, 0.706710, 0.694824, 0.694800). These positive gaps exclude zeros for all four trial fields, so the total winding is zero, exactly tracking the independently computed change from $C = 1$ to $C = 0$.

Evidence class: deterministic global-zero search, local phase circulation, and independent target-band FHS check for the two-band X -trial model. **What it does not establish:** The sweep tests four mass values and three square masks. It is not a finite-size or trajectory

calculation and does not resolve the critical point $\alpha = 2$, where the target-band gap closes and C is undefined.

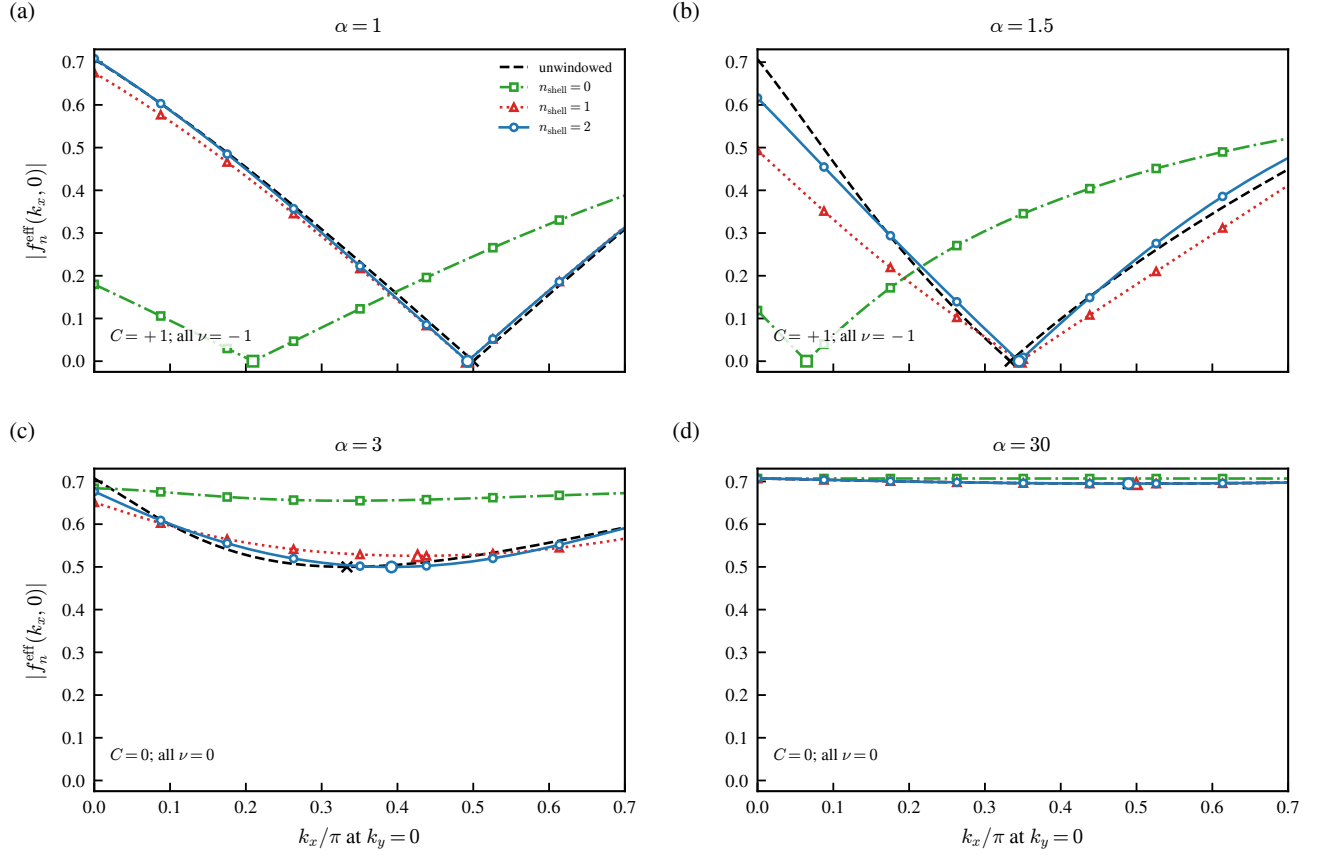


FIG. 23. Overlap-winding sweep for the unwound and $n_{\text{shell}} = 0, 1, 2$ X -trial modes. (a,b) In the $C = +1$ phase at $\alpha = 1, 1.5$, every overlap has one resolved zero on $k_y = 0$ and total winding -1 ; large symbols mark the refined zeros. (c,d) In the trivial phase at $\alpha = 3, 30$, every curve remains strictly positive and has total winding zero. Lines are deterministic $k_y = 0$ overlap cuts, not fits; positive global minima are quoted in the text, and the displayed C values come from an independent full-BZ FHS calculation. The zero search, local circulation, and bundle relation are Eqs. (43)–(42); there are no sampling bars. The four-point sweep excludes $\alpha = 2$, where the isolated-band Chern number is undefined.

VII. GROUP V: RECORD-DISCARDED MEAN CHANNEL AND CONTINUOUS LINDBLADIAN

A. Result 24: stationary interface occupations and deterministic density response

The autonomous trajectory-averaged channel is analytically closed at the one-body level, so neither Born trajectories nor random schedule descendants are required. We evaluate the exact completely positive affine Gaussian channel and its continuous weak-success generator at fixed physical time. The hybrid reconstruction uses $N_x = 20$. The eight-point static mass scan is restricted to $N_y = 64$, while the $(\alpha_{\text{in}}, \alpha_{\text{out}}) = (1, 30)$ response endpoint uses $N_y = 32, 48, 64$, $n_{\text{shell}} = 1, 2, \text{None}$, and domain-wall-sector truncation both on and off. The canonical arm keeps that truncation on. It starts from $C(0) = \frac{1}{2}\mathbf{1}$, uses no burn-in, and observes state relaxation through $2N_y$.

Here “untruncated controller” means $n_{\text{shell}} = \text{None}$, not `dw_truncation=False`. Because the unwindowed controller remains exactly translation invariant in y , its gain and loss frame operators are inverse-Fourier transformed on the physical- y index into independent $2N_x \times 2N_x$ matrices $V_{\pm}(k_y)$. Before this block reduction, these positive frame operators are

$$V_{\pm} = \sum_{\mathbf{r}, \nu=A,B} |\chi_{\mathbf{r}, \nu, \pm}\rangle \langle \chi_{\mathbf{r}, \nu, \pm}|, \quad (46)$$

with the normalized modes of Eq. (8). The scalar $n_a \in [0, 1]$ is the prescribed ancilla filling, equivalently the gain weight; every displayed Result-24 arm uses $n_a = 1/2$. For every momentum, the deterministic covariance satisfies

$$\dot{C}(k_y) = n_a V_{-}(k_y) - \frac{1}{2} \{D(k_y), C(k_y)\}, \quad (47)$$

$$D(k_y) = n_a V_{-}(k_y) + (1 - n_a) V_{+}(k_y), \quad (48)$$

$$D(k_y) C_{\text{ss}}(k_y) + C_{\text{ss}}(k_y) D(k_y) = 2n_a V_{-}(k_y). \quad (49)$$

Equation (49) is solved separately in each momentum block and each $C_{\text{ss}}(k_y)$ is then diagonalized. No sliding window, convolution, or average over y is used to create Fig. 24(a): its k_y resolution is an exact symmetry decomposition. Both `dw_truncation=True` and false curves in that panel are k_y resolved because both use $n_{\text{shell}} = \text{None}$; finite-shell products are deliberately not assigned a momentum spectrum.

At each wall x_w , the mode weight is the sum over both orbitals and the three columns $x_w - 1, x_w, x_w + 1$. Modes with weight at least 0.25 are eligible, and the selected mode minimizes

$$\mathcal{J}_a(k_y) = |\nu_a(k_y) - \frac{1}{2}| + 0.02 [1 - w_a(k_y)]. \quad (50)$$

If no mode reaches the threshold at a momentum, the maximum-weight mode is used as an explicit fallback. A straight line is fit only for $|k_y| \leq \max(2.5 \cdot 2\pi/N_y, 0.22)$.

At $N_y = 64$ the two wall-selected branches cross half occupation with fitted slopes ± 0.949605 . This is a stationary mixed-state occupation-band statement, not physical-time or twist spectral flow and not preparation of a pure chiral critical state. Finite-shell products remain strictly real-space or integrated.

For the dynamical probe, exact local $\pm\epsilon$ density-phase unitaries with $\epsilon = 10^{-3}$ are central-differenced about C_{ss} . If $\delta n_w(y, t)$ is the response summed over the same wall window, its positive-lobe center and fitted velocity are

$$y_{+,w}(t) = \frac{\sum_y \tilde{y} [\delta n_w(y, t)]_+}{\sum_y [\delta n_w(y, t)]_+},$$

$$v_{+,w} = \frac{dy_{+,w}}{dt}, \quad \mathcal{D}_w = \left\langle \frac{\sum_y \tilde{y} \delta n_w(y, t)}{\sum_y |\delta n_w(y, t)|} \right\rangle_{t \in \mathcal{W}_{\text{fit}}}, \quad (51)$$

where $\tilde{y}$ is the signed periodic displacement from the source and $[q]_+ = \max(q, 0)$; $\mathcal{D}_w$ is the panel’s signed directionality averaged over the same active fit-time window. The signed response has positive-lobe velocities $(+0.077455, -0.077455)$ on the two canonical walls at $N_y = 64$, with $R^2 > 0.9975$ and stable signs at all three circumferences. Turning off domain-wall-sector truncation preserves the signs while reducing the magnitudes to $(+0.038030, -0.038030)$. Uniform topological and trivial controls return the numerical null. This is wall-resolved directional density response, not quantized conductance; gain/loss dynamics do not provide a conserved-charge continuity equation.

For $p = 1, 1/2, 1/4, 1/8, 1/16$, exact gain/loss semigroups are composed with Strang splitting and the response is compared with the continuous solution at fixed time $N_y/2$. The fitted response-convergence order is 2.0008. Neither a covariance matrix nor a covariance response history is archived.

Evidence class: deterministic complex128 CPU reconstruction of the exact affine mean channel and continuous gain/loss generator, with compact shell-validated outputs. **What it does not establish:** The calculation is an averaged channel-level statement. It does not replace conditioned-trajectory observables, and only $n_{\text{shell}} = \text{None}$ supports a k_y -resolved spectral claim.

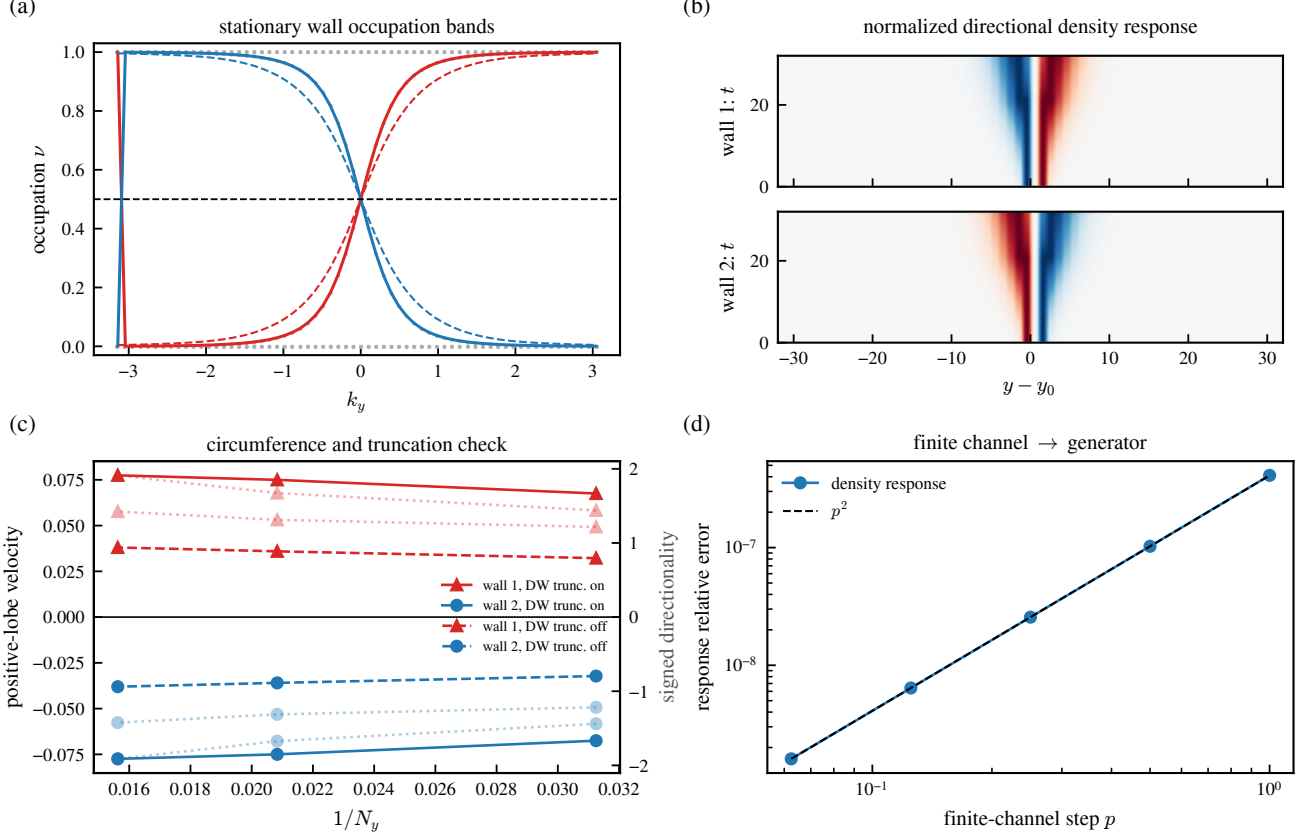


FIG. 24. Exact averaged-channel interface diagnostics. (a) The k_y -resolved occupation spectrum and wall-selected branches for $N_x = 20$, $N_y = 64$, $n_{\text{shell}} = \text{None}$, $n_a = 1/2$, and $(\alpha_{\text{in}}, \alpha_{\text{out}}) = (1, 30)$. Exact y -translation symmetry gives independent $2N_x \times 2N_x$ $V_{\pm}(k_y)$ blocks from Eq. (46); Eqs. (49) and (A29) are solved and the resulting $C_{\text{ss}}(k_y)$ is diagonalized at each k_y . There is no sliding average over y . Solid and dashed curves denote `dw_truncation=True` and false, respectively; both are momentum resolved because both are unwindowed. The colored branches use the three-column wall weight, 25% eligibility gate, half-occupation cost, fallback, and near-zero-momentum fit of Eq. (50), using $|k_y| \leq \max(2.52\pi/N_y, 0.22)$. (b) Signed central-difference density responses to local $\pm 10^{-3}$ phase kicks at the two walls, with each nonzero time slice normalized by its own maximum absolute amplitude only for display. The underlying response used in fits is not slice-normalized; this panel is the $N_y = 64$, unwindowed, `dw_truncation=True` arm through $t = N_y/2$. (c) Positive-lobe velocities from Eq. (51) and signed directionality versus $1/N_y$ for $N_y = 32, 48, 64$, $n_{\text{shell}} = \text{None}$, and both domain-wall truncation settings; each velocity fit uses $2 \leq t \leq 0.375N_y$. (d) Relative response errors at fixed $t = N_y/2$ for the $N_y = 64$ exact gain/loss affine channels at $p = 1, 1/2, 1/4, 1/8, 1/16$; their slope tests second-order convergence to the continuous generator. All panels are deterministic and therefore have no trajectory error bars. The occupation slope is not a group velocity, and the response velocity is not a quantized conductance.

VIII. NUMERICAL LEDGER AND SOURCE MAP

Tables I and II are the compact lookup tables for future work. Source entries are deliberately given as useful repository subfolders rather than unwieldy full file paths; the exact files are recorded in `figure_manifest.json`.

IX. WHAT THE REPOSITORY ALREADY SAYS, AND WHAT REMAINS OPEN

Taken together, the legacy archive already supports a coherent but carefully qualified picture. The deterministic wall has the intended $c = 1$ topology and handedness. The adaptive ensemble has a near-unit bulk Chern response. Introducing a wall produces a strongly localized slow tangent sector, concentrates wrong-outcome activity there, delays max-mix purification there, and leaves an entropy curve close to the two-wall $1/3$ coefficient. Independent modular and probability-record observers see the same spatial organization: oriented packet motion in the former and roughly tenfold wall-enhanced uncertainty in the latter. Independent mass sweeps further place the dynamical crossover near the exact $\alpha = 2$ transition. The flattened parent sharpens the static benchmark: its translation-resolved spectrum exposes the counter-propagating wall branches directly, the uniform one-shell parent is independently gapped and Chern nontrivial, and the wall shell hierarchy warns that a good entropy coefficient can precede convergence of local correlator tails. The independent OW form-factor reconstruction adds a more microscopic support check: the actual three-by-three controller mode preserves the target-band overlap zero and winding while retaining more than 99% of its norm, without misidentifying that overlap obstruction as a Chern number of the compact spinor. The four-point mass sweep makes the bundle relation explicit: all four tested overlaps have $\sum_a \nu_a = -1$ on the $C = 1$ side and a finite overlap gap with zero winding on the $C = 0$ side. Spatial heat maps show where each signal lives, and trajectory histograms show that the near-universal behavior is typical rather than exact record by record. The size-normalized filling distribution also narrows with circumference in two independent $S = 100$ preparations. The independent $N_x = 16$ covariance snapshots reproduce the joint near- $c = 1$ /wall-correlator pattern at two shell depths. Their newly reduced subsystem charge fluctuations add the missing direct current-sector precursor: k_{eff} is near one and quantitatively tracks the matched entropy coefficient on all four available ensembles. The transfer-mode maps show how feedback reorganizes the conditioned reduced-slab sector. The newly recovered `DW=True`, untruncated perfect-correction sweep further shows that the explicit interface survives overlapping controller support and undergoes the same sharp purification crossover near $\alpha = 2$, although it purifies faster than the terminated slab below the transition. This is sub-

stantially more than a collection of isolated plots: it is a mutually compatible evidence network spanning state topology, entanglement, conditional stability, controller activity, purification, handedness, effective-parent spectra, quantum charge fluctuations, and classical records.

The missing step is equally clear. Most stochastic state-ments are assembled from different campaigns rather than measured jointly on the accepted $S = 100$ master trajectories. Consequently the archive does not yet prove the recordwise tangent-purification identity, a causal activity-gap relation, a fully controlled thermodynamic $c = 1$ extrapolation with its charge sector, frozen-record integer twist flow, physical Born susceptibility, or record-only CFT universality. The atlas should therefore be used as a benchmark and data index: it says exactly which signals are already present, where their arrays live, how broad their trajectory distributions are, and which stronger claims still require new experiments. In particular, ensemble conditioning remains an explicit sensitivity diagnostic rather than a license to discard inconvenient records.

Appendix A: Complete computation formalism for the atlas

This appendix states every reduction used in Figs. 1–24. Its purpose is conceptual and reproducibility-oriented: the truth notebook remains the executable source, whereas the equations below say precisely what that source computes. An overbar denotes the trajectory-first average in Eq. (16). Deterministic calculations carry no artificial sampling bars.

1. Gaussian restriction, entropy, contours, correlations, and charge

For a set of one-body indices A , let R_A be its restriction matrix and $C_A = R_A C R_A^\dagger$. Diagonalizing $C_A u_q = \lambda_q u_q$ gives the von Neumann entropy [3]

$$S_A = - \sum_q [\lambda_q \log \lambda_q + (1 - \lambda_q) \log(1 - \lambda_q)]. \quad (\text{A1})$$

Eigenvalues are clipped only at machine precision when evaluating logarithms. The site-resolved entropy and charge-variance contours used by the atlas are

$$\begin{aligned} s_A(\mathbf{r}) &= \sum_{\mu, q} |\langle \mathbf{r}, \mu | u_q \rangle|^2 h_2(\lambda_q), \\ f_A(\mathbf{r}) &= \sum_{\mu, q} |\langle \mathbf{r}, \mu | u_q \rangle|^2 \lambda_q (1 - \lambda_q). \end{aligned} \quad (\text{A2})$$

Summing over all sites in A returns S_A or $\text{Tr}[C_A(\mathbf{1}_A - C_A)]$. A one-wall value in Results 1, 5, and 12 is a restricted sum of a full-subsystem contour, not a separately restricted covariance.

The CFT fit in Eq. (10) is ordinary least squares on the captioned A_y window. Its displayed line may be

TABLE I. Summary of archived numerical findings, Results 1–12.

No.	result	strongest quantitative content	reusable source subfolder
1	Exact wall benchmark	$c = 1.000092$; correlator exponent 2.074; modular velocities $(-2.930, +4.536)$; unit signed twist flows; $N_x = 20$ accepted.	experiment_review/b0_exact_domain_wall/
2	Slow tangent sector	CPU wall/uniform contrast 0.101–0.112 versus 0.894–0.896; GPU 0.0197–0.0270 versus 0.722–0.768; wall weight 0.933–0.996.	topological_frustration_diagnostics/; colab_lyapunov/
3	Wall purification residual	At $2N_y$, $S = 0.244$ – 0.331 and $V_Q = 0.0665$ – 0.0973 ; 94.25% of a smaller-run residual lies on the wall columns.	colab_charge_fluctuations/; colab_small_system_testing/
4	Log-chord entropy	Common-depth $c_{\text{eff}} = 1.0672, 1.0528, 1.0476$ for $N_y = 30, 40, 50$; separate time-scaled series reaches 1.04098 at $N_y = 100$; all controlled fits have $R^2 > 0.99997$.	colab_large_entanglement_scaling_N20/
5	Tripartite information	Exact $c_{\text{TMI}} = 1.001806$ per wall and 1.002697 over both walls, with full/wall slope ratio 2.00178; the stochastic pilot preserves counting but gives biased $c_{\text{TMI}} = 1.49$ – 1.58 .	exact_DW_benchmark_notes/; sample_average_testing/
6	Uniform Chern response	Final means 0.999995, 0.999687, 0.999927 at $N_y = 30, 40, 50$; global minimum 0.992548.	colab_lyapunov/
7	Feedback activity	Wall activity enhanced by 4.47–4.84; independently measured tangent gap suppressed by 7.94–8.83; two sharp interface peaks.	topological_frustration_diagnostics/
8	α transition	Transfer and max-mix entropy/variance diagnostics all change sharply near the exact $\alpha_c = 2$; finite-sector loss is explicitly censored.	colab_no_feedback_alpha_sweep_transfer/; colab_charge_fluctuations/
9	Modular handedness	Exact modular velocities $(-2.930, +4.536)$; stochastic packet displacements have the orientation-dependent endpoint sign pattern at two shell choices.	experiment_review/b0_exact_domain_wall/; colab_small_system_testing/
10	Record uncertainty	$\hat{H}_{\text{chain}}/N_y = 2.989$ – 2.995 ; wall-neighborhood uncertainty 0.1013 versus 0.0101 nats per event outside.	colab_charge_fluctuations/
11	Flattened spectrum	Translation-resolved X, Y, Z parents show two in-gap branches; the X, Y zero-energy crossings are especially clean.	exact_DW_benchmark_notes/; notebooks/
12	Corrected flattened-parent audit	The uniform $n_{\text{shell}} = 1$ parent is gapped with $C_R = -0.999999829$ and independent $C_{\text{FHS}} = -1$ on the same occupied projector; the $\alpha = 3$ control is trivial. In the separate wall geometry, one shell gives entropy slopes 0.33398/0.16675, while the correlator exponent converges from 3.2033 to 2.0042.	notebooks/flattened_hamiltonian_analysis/

extended beyond that window for visual clarity; only gray-window points enter the fit. The intercept absorbs the cutoff-dependent constant. A one-wall contour slope is multiplied by six and a two-wall full-width slope by three. The exact correlator target follows from a chiral complex fermion of scaling dimension $h = 1/2$:

$$|\langle c^\dagger(y)c(0) \rangle|^2 \propto \ell(r_y)^{-2}. \quad (\text{A3})$$

Results 1, 12, and 18 fit $\log |C_{ij}|^2 = a - \beta \log \ell(r_y)$ only on the displayed distance window.

The full counting statistics of number in A are encoded by the Gaussian determinant formula [4]

$$\begin{aligned} \chi_A(\theta) &= \langle e^{i\theta Q_A} \rangle \\ &= \det[\mathbf{1}_A - C_A + e^{i\theta} C_A], \\ \kappa_n &= (-i\partial_\theta)^n \log \chi_A|_{\theta=0}. \end{aligned} \quad (\text{A4})$$

In particular,

$$\kappa_1 = \text{Tr } C_A, \quad \kappa_2 = \text{Tr}[C_A(\mathbf{1}_A - C_A)]. \quad (\text{A5})$$

Result 21 evaluates κ_2 for a full- x interval separately in each trajectory and averages origins within that trajectory. With $d = 2N_x$, $\bar{n} = N_y^{-1} \text{Tr } C$, and $K_r = N_y^{-1} \sum_y \|C_{y,y+r}\|_F^2$, direct block algebra gives

$$\bar{F}_\xi^q(A_y) = A_y(\bar{n} - K_0) - \sum_{r=1}^{A_y-1} (A_y - r)(K_r + K_{-r}), \quad (\text{A6})$$

which is checked against explicit restricted matrices. Its two-wall current-algebra fit is Eq. (32). Result 17 instead

uses the classical record variance in Eq. (26); this is not κ_2 . Results 3 and 20 use the full-layer $S(C)$ and $V_Q(C)$ of Eq. (11) to quantify mixed-state purification.

2. Tripartite information: sign, cross ratio, CFT reduction, and errors

Let A, B, C be adjacent periodic- y intervals of lengths L_A, L_B, L_C . Every entropy subsystem initially spans the full x direction, and S_Y below is its ordinary von Neumann entropy. The two full-width information combinations are

$$\begin{aligned} I_2(A : C) &= S_A + S_C - S_{AC}, \\ I_3(A : B : C) &= S_A + S_B + S_C - S_{AB} \\ &\quad - S_{AC} - S_{BC} + S_{ABC}. \end{aligned} \quad (\text{A7})$$

Their full-width difference is exactly the conditional mutual information,

$$\begin{aligned} \Delta I^{(\text{full})} &\equiv I_2 - I_3 \\ &= S_{AB} + S_{BC} - S_B - S_{ABC} \\ &= I(A : C | B) \geq 0. \end{aligned} \quad (\text{A8})$$

Thus $I_3 - I_2 = -\Delta I^{(\text{full})}$. Its nonnegativity is strong subadditivity.

For a transverse reporting window $\mathcal{X}$, the atlas instead sums an entropy contour computed from that same full- x

TABLE II. Summary of archived numerical findings, Results 13–24.

No.	result	strongest quantitative content	reusable source subfolder
13	Spatial evidence maps	Exact Chern, slow-vector, entropy-contour, and charge-variance maps independently resolve the two-wall geometry.	exact_DW_benchmark_notes/ colab_lyapunov/ colab_small_system_testing/ colab_charge_fluctuations/
14	Modular density maps	Cycle-50 modular snapshots at $t_{\text{mod}} = 0, 8, 16, 32$ display the spatial propagation underlying the endpoint sign reduction.	colab_small_system_testing/
15	Trajectory distributions	c_{eff} and C are typically near one, but low-Chern records and broad, right-skewed max-mix V_Q tails remain.	colab_charge_fluctuations/
16	Cleaning sensitivity	Charge-, wall-, and Chern-conditioned c_{eff} shifts are not uniform across size or retained fraction; 10% retains only ten records.	colab_charge_fluctuations/ analysis_outputs/
17	Filling-distribution narrowing	At $C = 2N_y$, σ_ν^2 falls $0.29684 \rightarrow 0.16534$ (pure state) and $0.28533 \rightarrow 0.17023$ (max mix) from $N_y = 30 \rightarrow 50$, reductions of 44.3% and 40.3%.	colab_charge_fluctuations/
18	Independent state-data cross-check	$c_{\text{eff}} = 1.032\text{--}1.086$ and fixed-window wall exponents $2.092\text{--}2.191$ at $n_{\text{shell}} = 1, 2$; nonwall controls are $4.11\text{--}6.37$.	colab_small_system_testing/ colab_charge_fluctuations/ analysis_outputs/correlator_scaling_diagnostics/
19	Conditioned transfer-mode localization	At $N_y = 40$, $\alpha = 1$, perfect correction raises one-column edge weight $0.45187 \rightarrow 0.87933$ and lowers mean edge distance $1.9944 \rightarrow 0.5745$ cells.	colab_no_feedback_alpha_sweep_transfer/
20	Untruncated overlapping interface	With $\text{DW}=\text{True}$, $\alpha_{\text{out}} = 30$, and support truncation disabled, 270 perfect-correction trajectories show a sharp purification crossover near $\alpha_{\text{in}} = 2$; at $N_y = 32$, S_{tot} falls $0.26286 \rightarrow 6.06 \times 10^{-5}$ from $\alpha_{\text{in}} = 1.5 \rightarrow 1.9$.	colab_charge_fluctuations/
21	Critical subsystem charge fluctuations	Cycle-50 trajectory fits give $k_{\text{eff}} = 1.033\text{--}1.085$ across $N_y = 30, 40$ and $n_{\text{shell}} = 1, 2$; matched entropy coefficients are $c_{\text{eff}} = 1.032\text{--}1.086$, and every mean charge curve has $R^2 > 0.99996$.	colab_small_system_testing/gpu_data/pure_state_covariance_snapshots/
22	Finite-shell OW obstruction	The $n_{\text{shell}} = 1$ three-by-three mask retains 99.2155% of the OW norm, shifts the overlap zero only from 0.5 to 0.4922, and preserves $\nu_n = -1$, while the compact spinor has $C_{\psi_n} = 0$.	form_factor_analysis/
23	Overlap-winding mass sweep	Unwindowed and $n_{\text{shell}} = 0, 1, 2$ modes all have $\nu = -1$ at $\alpha = 1, 1.5$ where $C = +1$; at $\alpha = 3, 30$ all four overlaps are gapped and have $\nu = 0$ where $C = 0$.	form_factor_analysis/
24	Exact mean-channel interface	At $N_y = 64$ the canonical untruncated arm has paired half-occupation wall branches with slopes ± 0.949605 ; its two walls have signed positive-lobe velocities ± 0.077455 , while the finite-channel response converges to the continuous generator with order 2.0008 at fixed time.	mean_channel_lindblad_cpu_campaign/ experiment_review/ legacy_evidence_figure_atlas/data/

restriction:

$$S^{(\mathcal{X})}(Y) = \sum_{x \in \mathcal{X}} \sum_{y \in Y} s_Y(x, y), \quad (\text{A9})$$

$$\Delta I^{(\mathcal{X})} = S_{AB}^{(\mathcal{X})} + S_{BC}^{(\mathcal{X})} - S_B^{(\mathcal{X})} - S_{ABC}^{(\mathcal{X})}. \quad (\text{A10})$$

Although the contour contributions add to the full result when the windows partition the transverse direction, $S^{(\mathcal{X})}$ is not generally the entropy of a tensor factor. Consequently $\Delta I^{(\mathcal{X})}$ is an algebraic spatial contribution, not a conditional mutual information, and strong subadditivity does not assign its sign. The Result-5 wall curves happen to be positive and are displayed in the same positive orientation as $\Delta I^{(\text{full})}$.

For endpoints on a circle, define

$$x = \frac{\ell(L_A)\ell(L_C)}{\ell(L_A + L_B)\ell(L_B + L_C)}. \quad (\text{A11})$$

The elementary sine identity

$$\begin{aligned} \sin(a+b)\sin(b+c) - \sin a \sin c \\ = \sin b \sin(a+b+c) \end{aligned} \quad (\text{A12})$$

implies

$$1 - x = \frac{\ell(L_B)\ell(L_A + L_B + L_C)}{\ell(L_A + L_B)\ell(L_B + L_C)}. \quad (\text{A13})$$

For either the full width or a declared contour window q , suppose the relevant single-interval quantity obeys $S^{(q)}(L) = \kappa_q \log \ell(L) + s_q$. Substitution into its four-term algebraic combination cancels all constants:

$$\Delta I^{(q)} = \kappa_q \log \left[\frac{\ell(L_A + L_B)\ell(L_B + L_C)}{\ell(L_B)\ell(L_A + L_B + L_C)} \right] = \kappa_q \log \frac{1}{1-x}. \quad (\text{A14})$$

Here $\kappa_{\mathcal{X}} = c/6$ is the one-wall contour ansatz, whereas a full-width interval cuts both walls and has $\kappa_{\text{full}} = c/3$. Therefore $c_{\text{TMI}} = 6m$ and $3m$, respectively, and the two-wall/one-wall slope ratio is two. For the contour result this statement assumes the one-wall CFT form for each contour sum; it is not a consequence of strong subadditivity. For the full-width result, the reduction does not require setting I_2 to zero; rather, Eq. (A8) shows which combination survives. If the separated outer intervals additionally satisfy $I_2(A : C) \simeq 0$, as in the intended wall geometry, then $I_3 \simeq -\Delta I$ and the same prediction may equivalently be written $I_3 \simeq \kappa \log(1-x)$. This optional equivalence refers to the full-width ΔI and does not turn a post-contour contribution into a conditional mutual information.

For stochastic Result 5 and each plotted reduction $q \in \{\text{full}, \mathcal{X}\}$, the order at each abscissa z is

$$\begin{aligned} \Delta I_{\xi}^{(q)}(z) &= \frac{1}{N_{y_0}} \sum_{y_0} \Delta I_{\xi, y_0}^{(q)}(z), \\ \overline{\Delta I^{(q)}}(z) &= \frac{1}{S} \sum_{\xi=1}^S \Delta I_{\xi}^{(q)}(z). \end{aligned} \quad (\text{A15})$$

The plotted pointwise uncertainty is

$$\text{SEM}_q(z) = \sqrt{\frac{1}{S(S-1)} \sum_{\xi=1}^S \left[\Delta I_{\xi}^{(q)}(z) - \overline{\Delta I^{(q)}}(z) \right]^2}. \quad (\text{A16})$$

The parenthetical error on a quoted mean-curve slope is the least-squares regression standard error. The fixed-seed bootstrap resamples whole trajectories, preserves every trajectory's correlated z vector, recomputes the mean curve, and refits; it is a sensitivity diagnostic rather than a replacement for either pointwise SEM or regression error. Exact curves have regression uncertainty but no sampling SEM.

3. Real-space topology, flattened parents, and overlap winding

For the disk tripartition used in Results 1, 6, 12, and 15, let A, B, C be three counterclockwise angular sectors and let $P = \Gamma$ be the occupied projector in the operator ordering of Eq. (6); an archived production matrix is converted by $P = C^*$. Let P_{XY} denote its block from Y to X . The estimator is the real-space Chern formula [5]

$$\mathcal{C}_R = 12\pi i [\text{Tr}(P_{CA}P_{AB}P_{BC}) - \text{Tr}(P_{AC}P_{CB}P_{BA})]. \quad (\text{A17})$$

The disk center and radius are part of the protocol. A disk that crosses both interfaces does not define a uniform bulk invariant, which is why the corresponding old domain-wall scalar is excluded from Result 12. For a spatial map, the Bianco–Resta local marker [6] is evaluated as

$$\mathcal{C}(\mathbf{r}) = \sum_{\mu} \langle \mathbf{r}, \mu | 2\pi i (PXPYP - PYPXP) | \mathbf{r}, \mu \rangle. \quad (\text{A18})$$

Here X and Y are the diagonal one-body position operators in unit-cell coordinates; the finite-window convention and any bulk averaging are fixed by the archived calculation. The archived Result 13 render applies $\tanh$ only as a bounded color transformation.

For a translation-invariant occupied Bloch vector $|u(\mathbf{k})\rangle$, the Fukui–Hatsugai–Suzuki (FHS) links [7] are

$$\begin{aligned} U_{\mu}(\mathbf{k}) &= \frac{\langle u(\mathbf{k}) | u(\mathbf{k} + \hat{\mu}) \rangle}{|\langle u(\mathbf{k}) | u(\mathbf{k} + \hat{\mu}) \rangle|}, \\ C_{\text{FHS}} &= \frac{1}{2\pi} \sum_{\mathbf{k}} \text{Arg} [U_x(\mathbf{k}) U_y(\mathbf{k} + \hat{x}) U_x^*(\mathbf{k} + \hat{y}) U_y^*(\mathbf{k})]. \end{aligned} \quad (\text{A19})$$

The momentum orientation is recorded with the result, because reversing it reverses the reported sign without changing the phase.

The static flattened parent in Eq. (14) is diagonalized at fixed particle number. For eigenvalues $\epsilon_1^{\uparrow} \leq \dots \leq \epsilon_N^{\uparrow}$, Ky Fan's principle gives

$$\min_{\substack{P=P^{\dagger}=P^2 \\ \text{Tr } P=N_{\text{occ}}}} \text{Tr}(PH_{\text{flat}}) = \sum_{a=1}^{N_{\text{occ}}} \epsilon_a^{\uparrow}. \quad (\text{A20})$$

Result 11 exploits exact y translation symmetry to diagonalize $H_{\text{flat}}(k_y)$; Result 12 separately tests the uniform parent's gap and topology. Neither calculation asserts that the conditioned finite-depth ensemble equals this ground state.

For Results 22 and 23, a real-space mask g_n produces the effective target-band overlap in Eq. (34). The gauge-invariant search field is

$$|f_n(\mathbf{k})|^2 = \langle \phi_n(\mathbf{k}) | P_- | \phi_n(\mathbf{k}) \rangle, \quad (\text{A21})$$

and local phase circulation is computed in the smooth projector-defined frame of Eq. (44). The punctured-torus proof in Eqs. (41)–(42) gives $\sum_a \nu_a = -C$ in the form-factor sweep's convention. This overlap index belongs to a section of the dual occupied-band bundle; it is not the Chern number of the compact spinor, which remains zero when smooth and nonvanishing.

4. Conditioned Gaussian updates, activity, tangent frames, and transfer modes

For a rank-one measured mode $P = |u\rangle\langle u|$, put $Q = \mathbf{1} - P$ and $p = \text{Tr}(PC) = \langle u | C | u \rangle$. The number-measurement outcomes $m = 1, 0$ occur with probabilities $p, 1 - p$ and update a number-conserving Gaussian covariance as

$$\begin{aligned} C_{m=1} &= P + QCQ - \frac{QCPCQ}{p}, \\ C_{m=0} &= QCQ + \frac{QCPCQ}{1-p}. \end{aligned} \quad (\text{A22})$$

The singular branches $p = 0$ or 1 are taken only when their Born probability is nonzero, so the corresponding quotient is never evaluated on an impossible outcome. The prescribed correction then acts by $C \mapsto U_m C U_m^\dagger$ on the enlarged physical-plus-ancilla covariance. Restricting back to the physical layer after the declared ancillary step yields $C_\xi(C_{\text{cyc}})$. Result 7's wrong-outcome activity is the saved premeasurement probability of, or sampled indicator for, the outcome opposite the prescribed target; it is averaged over the stated wall or bulk channel set. Separate activity and tangent ensembles are never paired retrospectively.

For the conditional tangent calculation, the ordered linearized frame maps form the finite-time cocycle underlying the multiplicative-ergodic construction of Oseledec [8]. They are stabilized by

$$J_t Q_{t-1} = Q_t R_t, \quad \lambda_a^{(T)} = \frac{1}{T} \sum_{t=1}^T \log(R_t)_{aa}, \quad (\text{A23})$$

with positive diagonal entries of R_t . The gap plotted in Result 2 is the campaign's declared separation of leading finite-time rates, and its profile sums the selected vector probability over orbital and periodic coordinates. Since the CPU and GPU campaigns normalize different horizons, only within-campaign wall-uniform contrasts and wall weights are compared.

The particle–Choi construction in Results 8 and 19 is instead reduced from a regularized cumulative Choi covariance Σ :

$$T_p = -(\mathbf{1} + \Sigma_{LL})^{-1} \Sigma_{LR}. \quad (\text{A24})$$

If $s_j(T_p)$ are its finite singular values at completed cycle $C_{\text{cyc}} > 0$, the archived transfer exponents and plotted finite gap are

$$\lambda_j^{(p)} = \frac{1}{C_{\text{cyc}}} \log s_j(T_p), \quad \Delta_p = \min_{j \text{ finite}} |\lambda_j^{(p)}|. \quad (\text{A25})$$

Equivalently, for a finite eigenvalue $a_j \in (-1, 1)$ of the Hermitian Σ_{LL} chart used by the solver, $\lambda_j^{(p)} = [\log(1 - a_j) - \log(1 + a_j)] / (2C_{\text{cyc}})$. Endpoint sectors $a_j = \pm 1$, corresponding to zero or pole singular values, are excluded from the finite minimum rather than plotted as infinite observations. Its finite eigenmodes are normalized in the saved reduced slab basis and mapped back to physical unit cells. For unit-cell probabilities p_i , the atlas reports

$$\begin{aligned} w_{\text{edge}} &= \sum_{i \in \text{edge}} p_i, \\ d_{\text{edge}} &= \sum_i p_i d(i, \text{edge}), \\ \text{PR} &= \left(\sum_i p_i^2 \right)^{-1}. \end{aligned} \quad (\text{A26})$$

Missing finite sectors are censoring, not infinite exponents. These transfer modes are not the QR/Oseledec vectors in Eq. (A23).

Finally, Result 10 applies Eq. (12) chronologically to saved premeasurement probabilities. The chain rule identifies it as the Born expectation of the conditional record entropy. Without the actual outcome word it cannot reconstruct the realized likelihood p_ξ or $-\log p_\xi$.

5. Modular, physical, twist, and dissipative chirality

For a restricted Gaussian state, the one-body modular generator is

$$K_A = \log[(\mathbf{1}_A - C_A)C_A^{-1}] = -2 \operatorname{arctanh} G_A. \quad (\text{A27})$$

A localized packet is evolved by $e^{-iK_A t_{\text{mod}}}$; its density is averaged over translated origins within each record and then over records. Result 14 shows that density before reduction, whereas Results 1 and 9 unwrap its periodic center D_w and fit Eq. (13). Physical benchmark propagation instead uses $e^{-iH_{\text{DW}} t_{\text{phys}}}$. The common signed-coordinate convention is increasing y , so opposite wall orientation reverses the sign. Since K_A is dimensionless and state dependent, $|v_{\text{mod}}|$ is not compared to $|v_{\text{phys}}|$.

The twist-flow diagnostic is a third construction. A boundary holonomy $e^{i\phi}$ is inserted in the periodic y direction, and the same branch is followed continuously in

ϕ . For restricted entanglement occupations $\nu_a(\phi)$, define

$$\begin{aligned} \varepsilon_a(\phi) &= \log \frac{1 - \nu_a(\phi)}{\nu_a(\phi)}, \\ \nu_{\text{flow}, w} &= \sum_{\substack{\text{tracked crossings} \\ a \text{ on } w}} \text{sgn } \Delta_\phi \varepsilon_a. \end{aligned} \quad (\text{A28})$$

It is an integer turn count of tracked entanglement levels through $\nu = 1/2$ and not a velocity or transported charge. Result 1 uses the deterministic benchmark's opposite wall flows; future conditioned-record applications must freeze both the outcome word and external schedule throughout the twist circle.

Result 24 uses neither Hamiltonian nor modular time. Its exact generator is Eqs. (47)–(49); its signed speed is the positive-response-center derivative in Eq. (51). The occupation slope $d\nu/dk_y$, the physical velocity, the modular velocity, the twist integer, and the mean-channel response velocity therefore diagnose handedness in five different spaces. Only their orientation reversal is compared.

6. Record-discarded channel, Lyapunov solution, and finite-channel limit

For $n_{\text{shell}} = \text{None}$, inverse Fourier transformation of the physical- y index of every gain/loss frame yields independent $2N_x \times 2N_x$ blocks $V_\pm(k_y)$. This is an exact block diagonalization, not a sliding average. Writing $D = n_a V_- + (1 - n_a) V_+$ gives Eq. (47). If $D(k_y) = W(k_y) \text{diag}(d_a) W^\dagger(k_y)$ and $B = W^\dagger(n_a V_-) W$, the stationary Lyapunov solution [9] is

$$[W^\dagger C_{\text{ss}} W]_{ab} = \frac{2B_{ab}}{d_a + d_b}, \quad (\text{A29})$$

for nonzero denominators, followed by Hermitian symmetrization and a residual check. The homogeneous propagator is $e^{-Dt/2}$, so

$$C(t) = C_{\text{ss}} + e^{-Dt/2} [C(0) - C_{\text{ss}}] e^{-Dt/2}. \quad (\text{A30})$$

These are standard covariance equations for number-conserving quadratic gain/loss Lindbladians [10, 11].

The finite success-step map is represented exactly as $C \mapsto ACA^\dagger + B$ and composed by binary affine exponentiation. Symmetric gain-half/loss-full/gain-half Strang composition at substep p converges at fixed physical time as

$$\frac{\|\delta n_p(t) - \delta n_{\text{cont}}(t)\|_2}{\|\delta n_{\text{cont}}(t)\|_2} = O(p^2). \quad (\text{A31})$$

The response to a local phase kick is central-differenced in ϵ and propagated with the homogeneous affine part; only the rank-four source vectors are materialized, not a full covariance-response history. This explains both the exactness and the storage limitation stated in Result 24.

7. Distributions, conditioning, maps, and uncertainty semantics

Ordinary stochastic bars use

$$\text{SEM}(O) = \frac{s_O}{\sqrt{S}}, \quad s_O^2 = \frac{1}{S-1} \sum_{\xi=1}^S (O_\xi - \bar{O})^2. \quad (\text{A32})$$

Regression bars are conditional on the displayed fit window and are not trajectory SEMs. Result 6(d) intentionally uses the trajectory standard deviation to display ensemble spread. Results 17 and related variance curves use a delete-one-trajectory jackknife:

$$\text{SE}_{\text{JK}}^2 = \frac{S-1}{S} \sum_{\xi=1}^S (\hat{\theta}_{(-\xi)} - \bar{\hat{\theta}}_{(-)})^2. \quad (\text{A33})$$

Whole-trajectory bootstraps resample records, never translated origins or interval lengths as if independent.

Result 15 uses fixed bin edges and ordinary filled, black-outlined histogram rectangles. Result 16 ranks one final-cycle scalar per trajectory, retains the declared fraction, and recomputes the unweighted subset mean and SEM; it is a sensitivity analysis, not a new ensemble. Heat maps show their actual discrete arrays. White meshes delineate physical unit cells only; smooth momentum fields use their native momentum sampling. A slice-normalized heatmap, such as Result 24(b), is identified explicitly and never used as the fitted numerical input.

Appendix B: Implementation cross-checks for the truth notebook

This appendix makes the numerical reductions independently legible without duplicating the source notebook. In every stochastic result, an overbar denotes the arithmetic mean over saved trajectories. Shaded bands and ordinary error bars denote the standard error $s/\sqrt{S}$ unless a caption explicitly says regression error, standard deviation, or jackknife error. Deterministic calculations carry no artificial sampling bars.

1. Gaussian-state observables: Results 1, 3–5, 12–18, 20, and 21

Every entropy calculation diagonalizes the restricted number covariance C_A and uses

$$S_A = - \sum_q [\lambda_q \log \lambda_q + (1 - \lambda_q) \log (1 - \lambda_q)]. \quad (\text{B1})$$

Results 1, 4, 12, 15, 16, and 18 regress this quantity against the log-chord coordinate of Eq. (10); $c_{\text{eff}} = 3m$ for a full periodic interval crossing two walls and $c_{\text{eff}} = 6m$ for a one-wall contour. Result 5 forms, for $q \in \{\text{full}, \mathcal{X}\}$,

$$\Delta I^{(q)} = S_{AB}^{(q)} + S_{BC}^{(q)} - S_B^{(q)} - S_{ABC}^{(q)} \quad (\text{B2})$$

and fits the declared conformal cross-ratio coordinate. For $q = \text{full}$ these are ordinary entropies and the result is the rigorous CMI. For $q = \mathcal{X}$ each term is the post-contour sum of Eq. (A9), so the result is a one-wall algebraic contribution. The $3m$ versus $6m$ conversion follows from the two-wall full entropy versus one-wall contour CFT coefficient, respectively.

Results 3 and 20 use the full-layer eigenvalues in $V_Q = \text{Tr}[C(\mathbf{1}-C)]$ and Eq. (11); Result 21 instead restricts C to a periodic interval and uses Eqs. (30)–(32). Entanglement and charge contours in Results 3 and 13 are obtained by distributing each restricted-covariance eigenmode’s scalar contribution over its site probability and summing over modes. Result 17 is different: it takes a sample variance of trajectory means, not the within-trajectory quantum variance.

For squared correlators in Results 1, 12, and 18, the notebook averages the declared matrix elements of C over translations and fits $\log |C_{ij}|^2 = a - \beta \log \ell(r_y)$ only on the displayed distance window. Result 14 constructs the Gaussian modular one-body generator

$$K_A = \log[(\mathbf{1}_A - C_A)C_A^{-1}], \quad (\text{B3})$$

evolves a localized one-body packet with $e^{-it_{\text{mod}} K_A}$, and then averages the density over the saved origins and trajectories; Result 9 reduces the same density to an unwrapped center of mass before fitting its initial displacement.

2. Topology, effective parents, and form factors: Results 1, 6, 11–13, 22, and 23

The real-space Chern values in Results 1, 6, 12, 13, and 15 use the archived tripartition estimator built from cyclic products of the occupied covariance between three spatial sectors; the atlas imports the saved scalar or field and never substitutes a Bott index. Result 11 diagonalizes each translation-resolved block $H_{\text{flat}}(k_y)$, while Result 12 fills its lowest fixed-charge eigenstates according to the Ky Fan construction in Eqs. (14)–(15). For the uniform parent it computes both the tripartition response and the FHS integer from normalized nearest-neighbor links of the occupied Bloch eigenvector. Domain-wall entropy and correlator reductions are kept separate because the corresponding real-space disk crosses the gapless interfaces.

Results 22 and 23 Fourier transform the analytic two-band projector to obtain the finite coefficients $C_{\mathbf{r}}$ in Eq. (35). Retained norm is the sum of $\|C_{\mathbf{r}}|\tau\rangle\|^2$ over the requested real-space support. Equation (43) locates overlap zeros without choosing a Bloch gauge; Eq. (44) then computes their local phase indices in a smooth patch. The independent target-band FHS check multiplies normalized nearest-neighbor link overlaps around every momentum plaquette and sums their principal phases. This separates three objects that must not be conflated: the target-band Chern number C , the overlap-section index $\sum_a \nu_a = -C$, and the Chern number of the smooth compact spinor, which is zero when that spinor is nonvanishing.

3. Conditional dynamics, activity, and transfer reductions: Results 2, 7, 8, 10, and 19

Result 2 propagates a tangent basis through the ordered linearized conditional maps and performs stabilized QR factorizations. If R_t is the positive-diagonal triangular factor, the finite-time exponents are accumulated as

$$\lambda_a^{(T)} = T^{-1} \sum_t \log(R_t)_{aa}; \quad (\text{B4})$$

the plotted slow gap is the declared separation of the leading conditional rates, and the transverse profile sums the selected singular-vector probability over orbitals and the periodic coordinate. Result 7 averages the saved wrong-outcome indicators at each channel and then sums either over wall neighborhoods or the complete cycle. The activity and tangent ratios are propagated from separate ensembles and are not treated as a recordwise covariance.

Result 8 parses the finite particle–Choi transfer rates of Eq. (A25) and separately loads the max-mix entropy and variance mass sweep; missing finite transfer sectors are censored, not converted to infinite rates. Result 19 normalizes each saved transfer eigenvector, maps reduced-basis entries back to physical cells, and evaluates edge weight, mean edge distance, and participation ratio $1/\sum_i p_i^2$. Result 10 applies the binary entropy $h_2(s_j)$ to every saved premeasurement Born probability and sums chronologically as in Eq. (12); it does not reconstruct unsaved outcome-bit likelihoods.

4. Distributions, conditioning, and maps: Results 13–17

Result 15 applies fixed bin edges to one scalar per trajectory and draws ordinary filled, black-outlined histogram bars; no kernel density estimate or staircase surrogate is used. Result 16 ranks the same final-cycle feature table by each declared metric in Eq. (25), retains a fixed leading fraction, and recomputes the mean and standard error of c_{eff} without reweighting. Result 17 recomputes Eq. (26) at every cycle and estimates its uncertainty by deleting one record at a time. Heat maps in Results 10, 13, 14, 18, 19, and 22 show the actual discrete cell or momentum array; white cell meshes are used only for physical unit-cell arrays, whereas smooth momentum fields retain their native continuum sampling.

Appendix C: Result-and-panel equation ledger

Tables III and IV are the shortest route from any panel to its mathematical definition. They supplement the geometry, protocol, averaging, uncertainty, and limitation statements in each caption.

TABLE III. Formalism map for Results 1–12. Panel letters follow the atlas figures; *det.* means that sampling error bars do not apply.

No.	panel quantities	defining equations	uncertainty and principal qualification
1	(a) entropy; (b) wall correlator; (c) physical/modular speeds; (d) width gate	(2), (10), (A3), (13)	Regression errors; exact target, not adaptive preparation.
2	(a,b) finite-time gaps; (c,d) selected-vector density/weight	(A23), (16)	Trajectory SEM; CPU/GPU clocks differ.
3	(a) entropy; (b) intrinsic variance; (c) mass control; (d) entropy contour	(11), (A2)	Trajectory SEM except mean-only contour.
4	(a,c) log-chord fit/residual; (b,d) c_{eff}	(A1), (9), (10)	Point SEM versus regression SE stated separately.
5	(a,b) full CMI/one-wall contour ΔI ; (c) c_{TMI} ; (d) wall-count ratio	(A7)–(A16)	Full width is rigorous CMI; wall is an algebraic contour contribution. Exact regression SE; stochastic SEM, regression SE, and bootstrap separated.
6	(a–d) tripartition Chern trajectories and distributions	(6), (A17)	SEM except panel (d) SD; one finite-radius estimator.
7	(a–c) wrong-outcome activity; (d) activity/gap ratios	(A22), (A23)	SEM/propagated SEM; activity and tangent suites are independent.
8	(a,b) finite Choi exponents; purification	(c,d) (A24), (A25), (11)	Finite-sample SEM; open marks are censoring.
9	(a,c,d) modular displacement; (b) signed physical/modular fits	(A27), (13)	Exact regression SE or trajectory-first SEM after within-record origin averaging; clocks differ.
10	(a–d) chain entropy and its spatial density	(12), (16)	Trajectory SEM; no realized-bit self-information.
11	(a–c) k_y -resolved flattened-parent spectra	(14), (A20)	Det.; archived render, no exported eigenvalue rows.
12	(a) Chern; (b) entropy; (c) correlator; (d) parent gap	(A17), (A19), (A3), (A20)	Mostly det.; static parent is not the trajectory ensemble.

TABLE IV. Formalism map for Results 13–24.

No.	panel quantities	defining equations	uncertainty and principal qualification
13	(a) local Chern; (b) slow vector; (c) entropy contour; (d) charge contour	(A18), (A23), (A2)	Different precursor ensembles; no recordwise equality.
14	(a–d) modular density snapshots	(A27), (16)	Origins are within-record samples; not real-time transport.
15	(a–d) trajectory histograms	(25), (10), (A17), (11)	One scalar per record; pure/max-mix ensembles are unpaired.
16	(a–c) rank-conditioned c_{eff}	(25), (A32)	Post hoc sensitivity analysis, not ensemble redefinition.
17	(a–c) record filling variance	(26), (A33)	Population variance with jackknife SE; not quantum variance.
18	(a) entropy; (b–d) correlator exponents/window	(A1), (10), (A3)	Regression SE and trajectory SEM separated; two sizes only.
19	(a,b) edge statistics; (c,d) transfer-mode maps	(A24)–(A26)	Trajectory SEM; reduced Choi mode, not tangent vector.
20	(a,b) entropy/variance; (c,d) construction ratios	(28), (29), (11)	Trajectory SEM; interface retained in both arms.
21	(a) interval variance; (b–d) k_{eff} summaries	(A4)–(A6), (32)	Trajectory-first fits; matched entropy comparison.
22	(a) norm/scales; (b,c) overlap; (d) overlap/spinor indices	(36), (37), (34), (A21), (42)	Det.; one trial spinor and mask family.
23	(a–d) overlap zeros, gaps, and winding	(39)–(45)	Det.; four masses, critical point excluded.
24	(a) occupations; (b,c) response; (d) channel convergence	(46)–(50), (51), (A29)–(A31)	Det.; averaged mixed channel, not conditioned trajectory.

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